



**Fully-auto chemiluminescence
immunoassay analyzer**

MAGLUMI[®]
XE

Operating Instructions

Dear users! Thank you for using our Fully-auto chemiluminescence immunoassay analyzer!

To make sure you using the analyzer safely and skillfully, and improve your working efficiency, please read the instructions carefully before operating the analyzer.

Please properly keep the instructions after reading, and placed it in a readily accessible place in order to obtain easily at any time.

If you have any questions regarding your Fully-auto chemiluminescence immunoassay analyzer, please contact your local representative.



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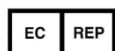
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Device	Catalogue Number
MAGLUMI X3	010101003301

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Product-related information

Product Name: Fully-auto chemiluminescence immunoassay analyzer

Model: MAGLUMI X3

Structure and components: The analyzer is composed of the main machine, accessories and software, wherein the main machine includes the material provision module, the liquid path module, the temperature control module, the mechanical drive module, the optical path detection module and the circuit control module; and the accessories include a desktop computer suite and cables.

Intended Use: Used in conjunction with adapter reagents for qualitative and/or quantitative analysis of the analytes in the human sample.

About Operating Instructions

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Applicable Scope of Software: 1.1.2.1 or compatible version

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Cautions

This section contains all important safety-related and proper use information and regulations.

Please read the Operating Instructions before using the analyzer.

Instructions

The MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer and reagents are strictly limited to professional in-vitro-diagnosis use. Please use reagents and consumables manufactured by Shenzhen New Industries Biomedical Engineering Co., Ltd. only, so as to prevent measurement errors or product failures.

This Operating Instructions provides a guide on the usage of the MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer, aiming to help users understand the structural principles, operation and usage, routine maintenance, troubleshooting and other aspects of the MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer. Please operate the analyzer in accordance with the instructions herein.

Alert description

Symbol	Text	Description
	Warning	Warning reminds the user that it could lead to personal injury, damage to the instrument, error in data or risk of biological infection under this condition.
	Note	Note reminds the user of an important message worth attention.

Safety precautions

Please read the Instructions carefully before operating the analyzer for safe use of the system. Any operation that violates the safety precautions could lead to personal injury or damage to the analyzer.

This system has been manufactured in compliance with the requirements for the safe production of electronic analyzers and medical analyzers. There are corresponding requirements of laws and regulations for the installation and operation of this system, therefore the installer and operator are obliged to comply with the related laws and regulations.



Warning:

1. The user should regularly check and maintain the analyzer by strictly following the instructions, so as to prevent analyzer faults and hazards to personal safety.
 2. To ensure the safety and reliability of the analyzer, its installation and maintenance can only be carried out by our authorized technical service engineers and personnel or with their approval. All analyzer parts must be checked and provided by our company or our authorized distributors.
-

1. Prevention of personal injury from moving parts

Please comply with the following precautions to prevent the moving parts from causing personal injury when the analyzer is working.



Warning:

1. Please do not touch moving parts on the analyzer, or their path of movement, when the analyzer is working. Please do not place any obstacle in the path of movement as this may cause personal injury or damage to the analyzer.
2. Follow the specified procedures when loading liquid or solid consumables into the analyzer and properly remove the cover, sealing film or other packaging from the consumables, so as to prevent personal injury or damage to the analyzer.

2. Prevention of electrical hazards

Please comply with the following precautions to prevent electric shock.



Warning:

1. Confirm that the input voltage complies with the analyzer requirements.
2. Unauthorized maintenance personnel should not open the front or rear cover of the analyzer while the analyzer is powered on.
3. Reagents, samples or other fluids entering the analyzer may cause analyzer faults or electric shock. Please turn off the power immediately and contact SNIBE's service department.
4. If you need to open the front or rear cover to replace parts, please cut off power to the analyzer before proceeding.
5. Improper grounding may lead to electric shock and damage to the analyzer.
6. Do not touch or discharge static electricity from parts with electrostatic protection warning labels.

3. Prevention of fire hazards

The use of organic solutions can lead to a risk of fires. Please comply with the following precautions.



Warning:

1. Do not use organic solutions in tests.
2. The analyzer is not designed to be explosion-proof. Do not use organic solutions with ignition points below 65 °C near the analyzer. Otherwise, this may cause a fire or explosion.

4. Prevention of biochemical danger

Please comply with the following precautions to effectively prevent biochemical danger.



Warning:

1. Please carefully read the reagent and consumable SDS before use to understand the safety instructions and preventive actions.
2. Avoid direct hand or clothes contact with reagents and consumables. If your hands or clothes are accidentally exposed, wash them with soap and water immediately. If something gets into your eyes accidentally, wash them with a

- large amount of water immediately and consult an ophthalmologist.
3. Improper use of samples may cause infection. Do not touch the sample, mixture and waste liquid with your hand or other body parts. Do wear gloves, a gauze mask and work clothes during operation to prevent infection, and wear protective glasses when necessary.
 4. Some reagents and system liquid may damage the skin. Use reagents and system liquid carefully to prevent your hands and clothes from being directly exposed to the reagents and system liquid. If your hands or clothes are accidentally exposed, wash them with soap and water immediately. If something gets into your eyes accidentally, wash them with a large amount of water immediately and consult an ophthalmologist.
 5. If a small amount of reagent or sample spatters on the analyzer during use, use a clean cotton cloth with alcohol to wipe the analyzer clean before operating it. If there is a large amount of reagent or sample on the analyzer, stop the analyzer immediately and contact an engineer authorized by our company to deal with it promptly.
 6. When the analyzer needs to be moved over a long distance, it should be completely disinfected to prevent the spread of potential infection sources.
-

5. Protection against laser hazard

Please comply with the following precautions to prevent laser burns caused by the barcode reader.



Warning:

The laser produced by the barcode reader will directly irradiate the human retina, causing damage to eyes. Do not look directly into the laser beam of the barcode reader. Please wear goggles if you need to check the laser beam of a charged barcode reader.

6. Disposal of waste liquid

To prevent environmental contamination and personal injury, please comply with the following precautions in the disposal of waste liquid.



Warning:

1. Some substances in the QC solution, calibration solution and waste liquid are controlled by pollution regulations and discharge standards. Please comply with local discharge standards and consult with the relevant reagent manufacturer or distributor.
 2. Waste liquid from infectious disease patients' samples should be discharged to an infectious disease treatment facility of the hospital.
-

7. Disposal of waste analyzers

Please comply with the following requirements to dispose of the waste analyzers.

**Warning:**

Some substances from the waste analyzer are controlled by pollution regulations. Please comply with local related regulatory requirements during disposal.

Precautions for use

Please read the following precautions for use carefully, in order to use the analyzer correctly and effectively.

1. General precautions

The purpose of this analyzer and general precautions should be defined before you use the analyzer. If the analyzer is not used according to the methods stipulated in the instructions, the protection provided by it might be damaged.

Note:

1. When making clinical decisions based on analysis results, please also consider the clinical symptoms or other lab test results.
2. The Operating Instructions may be revised without prior notice. The user should consult the customer representative based on the condition.
3. This analyzer is limited to medical inspection professionals' use only and requires basic analyzer operation abilities, experimental skills and laboratory knowledge.
4. Do not touch the display, mouse or keyboard of the computer when you have chemicals on your hands.
5. Please do not fold or press on the drainage pipe. Otherwise, waste liquid may spill from other places due to obstructed drainage, leading to damage to the analyzer in some serious cases.
6. The analyzer produces heat during operation, which is dissipated from the rear. The work environment should have good ventilation to ensure good heat dissipation and a ventilation device should be used when necessary. However, avoid having the airflow blow directly to the analyzer. Otherwise, the reliability of test results might be compromised.
7. Each component of the analyzer should be adjusted before initial use to make sure the parameters of each component are accurate.
8. There should be no bubbles in the starter or system liquid. Otherwise, the reliability of test results cannot be guaranteed.
9. Make sure the starter bottle of the immunology module is correctly connected. Do not use mixed starter. Otherwise, the reliability of test results cannot be guaranteed.
10. Please use the supporting cuvettes designated by our company in order to ensure safe operation of the analyzer and accuracy of test results.
11. Please do not use expired system liquid in order to ensure safe operation of the analyzer and stability of test results.



12. Start up the analyzer at least 30 minutes before use to stabilize the measurement system.
 13. Check whether there are sufficient consumables for testing before starting any tests (system liquid, starter, and cuvettes among others).
 14. Before starting a test, perform at least one BGW test and confirm that the BGW test result is within the normal range. Otherwise, the reliability of test results cannot be guaranteed.
 15. There should be no foam on the surface of a liquid sample when it is injected. Otherwise, this could lead to an error when sample is injected. The reagents must not be changed or moved before the reagents are fully injected.
 16. QC testing must be conducted when the analyzer is used for sample analysis. Otherwise, the reliability of test results cannot be guaranteed.
 17. The analyzer is equipped with an alarm light and a loudspeaker. When the analyzer has a fault warning, the alarm light and loudspeaker will send out an alarm signal. At such times, please rectify the fault in time to ensure normal operation of the analyzer and the accuracy of test results. Refer to the troubleshooting methods in Section 7 of the instructions: Common faults and troubleshooting.
 18. Please use the system tubing cleaning solution designated by SNIBE in order to carry out cleaning and maintenance of pipes.
 19. The analyzer surfaces and interior should be appropriately disinfected if any hazardous substance leaks or spills onto or into the analyzer. Do not use cleaning agents or disinfectants that may chemically react with materials contained in the analyzer's parts or components. Consult the manufacturer or agent for any questions regarding the compatibility of a disinfectant or cleaning agent with the materials contained in the analyzer or parts thereof.
 20. It is recommended that the technical service engineers authorized by Shenzhen New Industries Biomedical Engineering Co., Ltd. perform a calibration on the analyzer on an annual basis to ensure the equipment safety and normal operation.
 21. If any serious incident has occurred in relation to the device, please report to Shenzhen New Industries Biomedical Engineering Co., Ltd. (Snibe) or our authorized representative and the competent authority of the Member State in which you are established.
-

2. Operation environment



Note:

Correctly install the analyzer in accordance with the installation environment specified in the Operating Instructions. Installation and use beyond the specified conditions may produce unreliable results and cause damage to the analyzer.

3. System maintenance



Warning:

1. Carry out regular maintenance of the analyzer based on the relevant parts of these Operating Instructions. Improper maintenance may affect the accuracy and precision of test results, and may even lead to analyzer faults or personal injury.
 2. The surface of the analyzer may collect dust if the analyzer is not used for long periods. Use a clean cotton cloth which has been soaked in water and wrung out to gently wipe the surface. Take necessary precautions to prevent water from dropping inside the analyzer.
 3. Please turn off the power before maintenance (excluding hotline maintenance). Otherwise, this could lead to analyzer faults or personal injury.
 4. The analyzer may be stained with samples from potentially infected patients. Do wear gloves and work clothes to prevent infection during maintenance.
 5. The analyzer itself does not contain any parts that the user can service. Do not try to open the cover case and remove any part. Contact one of SNIBE's authorized technical service engineers if you need assistance.
 6. Please contact an authorized technical service engineer to check whether the overheat protection or liquid level protective device or system is fully operational.
-

4. Samples, reagents and QC solution



Warning:

1. Substances such as drugs, anticoagulants and preservatives that exist in the sample may interfere with test results.
 2. Samples affected by hemolysis, lipemia and jaundice will affect test results.
 3. Make sure there are no clots in the sample. Otherwise, the pipettor will be blocked, which will seriously affect the test results.
 4. Take correct sample storage measures. Improper sample storage measures may change the component structure of samples, leading to incorrect test results.
 5. Do not leave samples open for long periods, to prevent them from evaporating. The evaporation of samples may lead to incorrect test results.
 6. Improper storage of reagents or QC solution could lead to incorrect test results and sub-optimal system performance, even if they are within their validity period. Reagents and QC solution should be used and stored in accordance with the instructions for use of the relevant reagent manufacturer.
 7. The calibration analysis should be conducted after changing the reagent. Correctness of test results may be compromised if calibration analysis is not conducted.
-

5. Data backup



Note:

The system possesses the function of automatically storing data on the computer hard disk. However, the data on the computer hard disk cannot be recovered if the data is deleted or the hard disk is damaged due to other reasons. Please regularly back up the test results and analyzer parameters to another medium, such as a CD-ROM.

6. Electromagnetic compatibility



Note:

1. The MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer complies with the emission and disturbance immunity requirements stipulated in IEC 61326-2-6:2012.
 2. The user has the responsibility to guarantee the electromagnetic compatibility environment for the analyzer to enable the analyzer's proper operation.
 3. It is recommended to evaluate the electromagnetic environment before using the analyzer.
-



Warning:

1. The MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer has been designed and tested as a type A device in accordance with IEC/CISPR 11:2010. The analyzer may cause radio interference in a home environment. Therefore, please implement protective measures in such scenarios.
 2. It is forbidden to use the analyzer near strong radiation sources (such as unshielded RF sources). Otherwise, these may interfere in the normal operation of the analyzer.
-

Chapter 1 Measuring principle

The operation principles of the analyzer include the test principle, analysis method and calibration method. This section will introduce these in detail.

1.1 Lab test steps

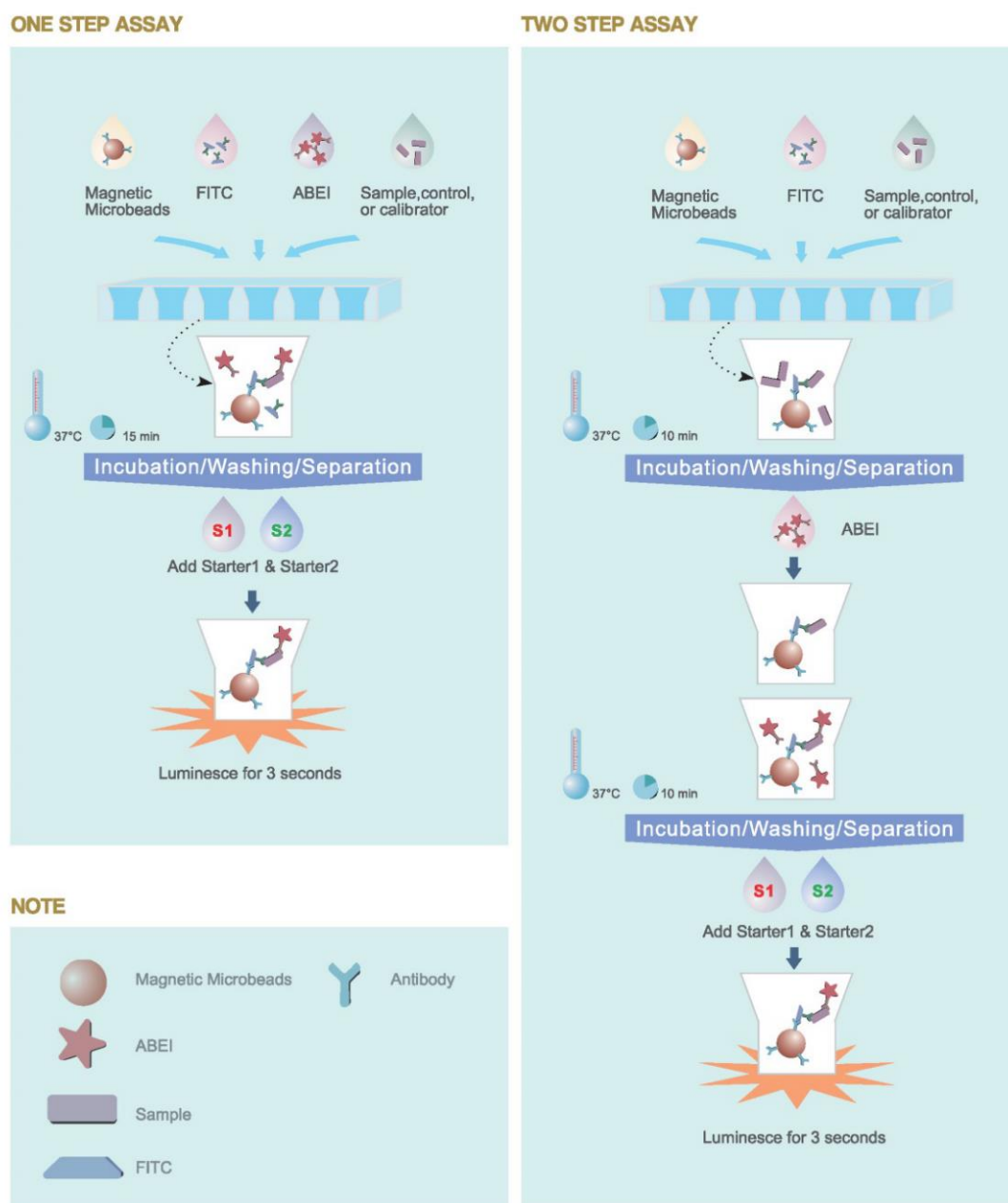


Fig. 1.1-1 Test steps

1.2 Measuring principle

The PMT in the analyzer is used to detect the light generated from the chemiluminescence reaction. The linear measuring range of light wavelengths detected by the PMT is 300–650 nm and the peak wavelength of light generated by chemiluminescence reactions is 420 nm. The light generated by the chemiluminescence reaction strikes the PMT where it is converted into an electrical signal. This signal is collected, processed and converted by the peripheral circuits into a digital value that reflects the magnitude of the light intensity.

We use relative light units (RLUs) as the quantitative unit of raw data in order to eliminate differences among individual PMTs and ensure the test result consistency among different analyzers.

After the sample and reagent are added to the cuvette, they undergo three steps of mixing, washing and separation, and then the cuvette enters the chamber. Starter S1 is injected into the cuvette, then after a 2.5 second delay, starter S2 is injected into the same cuvette to stimulate a chemiluminescence reaction. Detection of the optical signal starts 0.1 seconds after the chemiluminescence reaction occurs to obtain the optical signal within a period of 3.0 seconds.

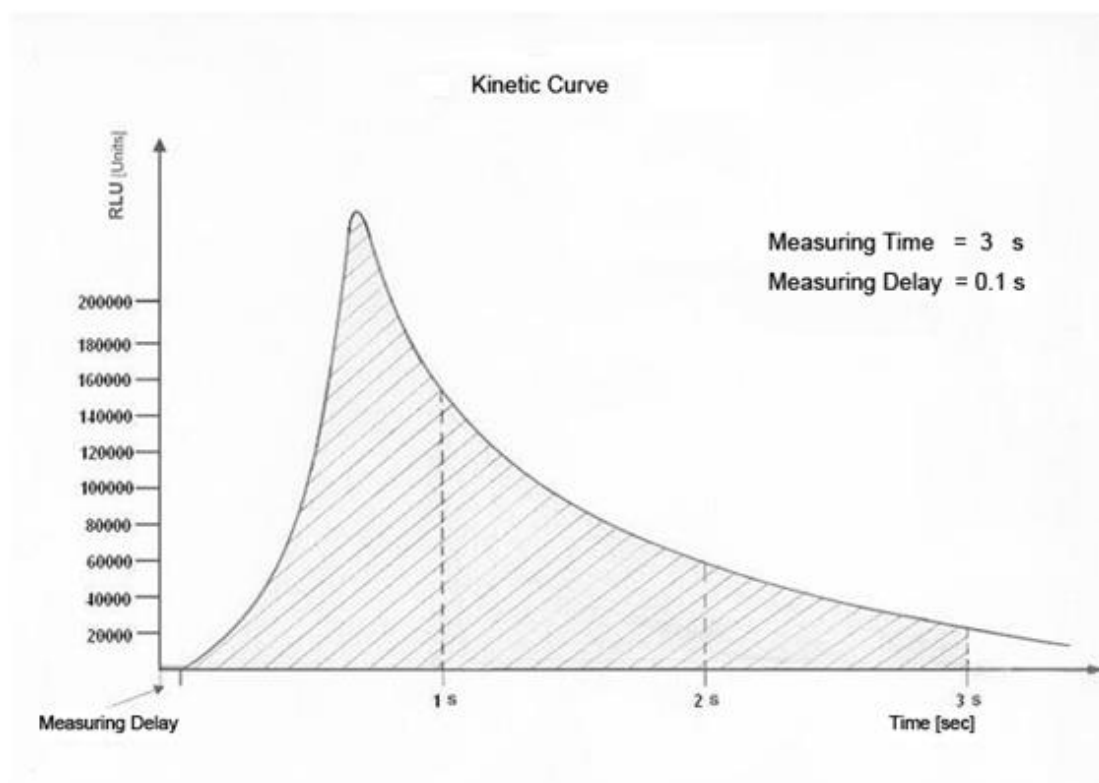


Fig. 1.2-1 Chemiluminescence kinetics curve

1.3 Calibration method

The master curve should be corrected due to the difference between the actual work environment of the analyzer and the lab environment, to generate a work curve that is consistent with the actual work

environment.

Brief description:

- The master curve is determined using ten standard points.
- Compare the relative light units of two calibration points obtained from the calibration with the relative light unit of the corresponding concentration in the master curve.
- The difference in relative light intensity between the two calibration points and the relative light intensity of the corresponding concentration in the master curve is calculated and a linear inference is made based on the recalculated relative light intensity (y-axis) and concentration (x-axis).
- For other points in the master curve, the difference of their relative light units is calculated with the assistance of the compensation curve, and the relative light intensity (y-axis) and concentration (x-axis) are recalculated.
- This recalculated curve is the valid work curve (see Fig. 1.3-1: Calibration principle).

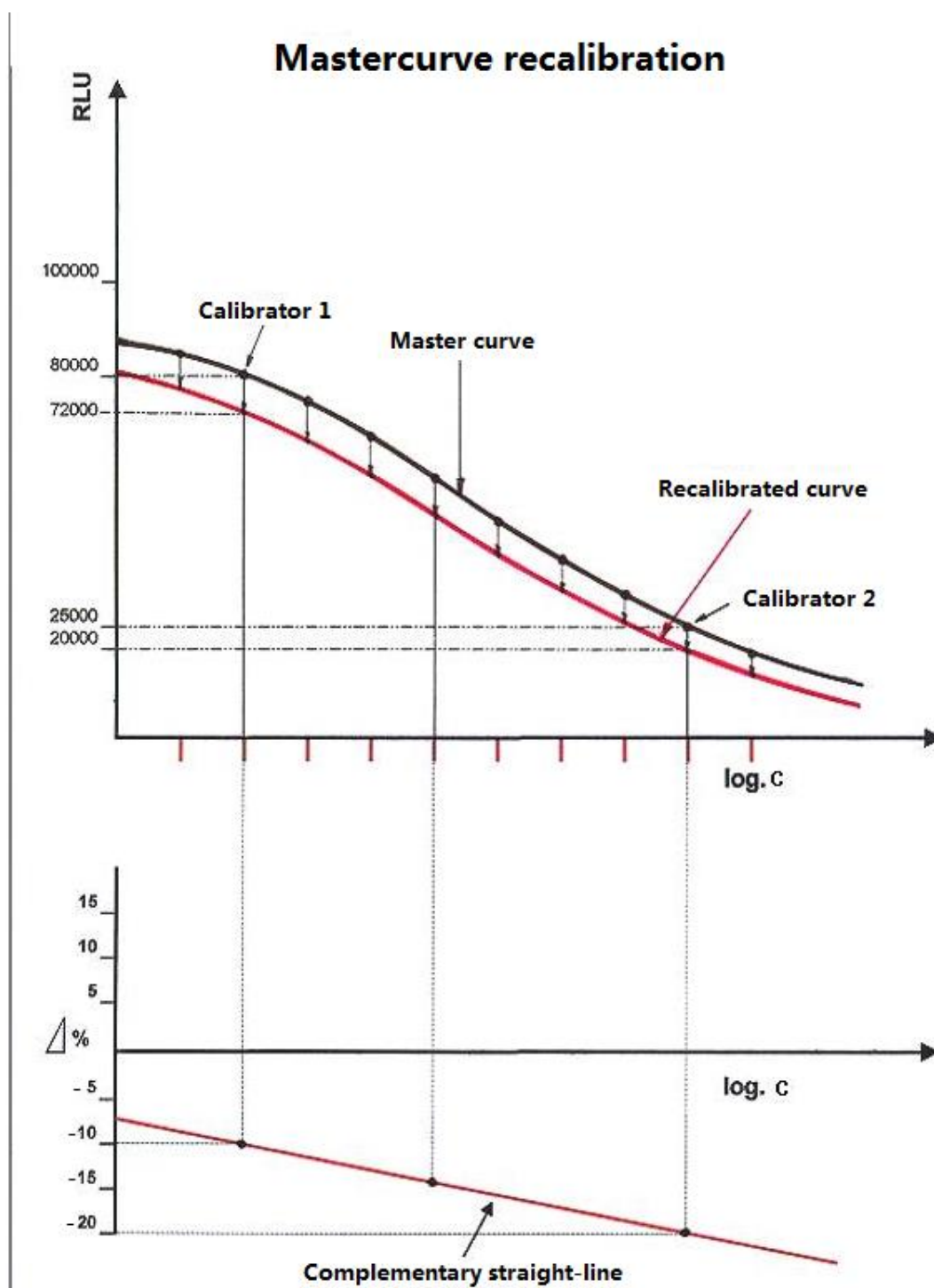


Fig. 1.3-1 Calibration principle

Chapter 2 Analyzer overview

2.1 Overview

The MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer and the supporting diagnostic reagent series constitute a magnetic separation, ABEI-labeled, and accurate direct chemiluminescence micro immunoassay system. The system is used for qualitative or quantitative analysis of the analytes in human samples. The analyzer performs automatic sample pipetting, reagent loading, incubation, washing, measurements, and result calculations, which reduces test errors and improves the accuracy and precision of test results.

2.2 Specifications of analyzer

Table 2.2-1 Specifications of analyzer

Item		Specification
Basic characteristics	Test speed	200 tests/hour
	Sample type	Serum, plasma, urine
	Barcode type	Code128, Code39, Code93, Codabar, 2/5 Interleaved
	QC	Monthly QC analysis
Analyzer characteristics	Sample volume	10–200 μ L
	Sample reagent area	A total of 72 sample positions and 20 reagent positions on six sample racks
	SampleArm	Liquid level detection, measure-and-track, clot detection, anti-bubble interference, collision detection, automatic washing
	Reagent volume	10–450 μ L
	Temperature of sample reagent area	Reagent area temperature is 10 $^{\circ}$ C $\pm$ 3 $^{\circ}$ C
	Cuvette quantity	A total of 364 can be loaded
	Reaction temperature	37.0 $^{\circ}$ C $\pm$ 0.3 $^{\circ}$ C, with the fluctuating no more than 0.2 $^{\circ}$ C
	Mixing method	Mechanical oscillation
External interface	Interface	Ethernet
Work environment	Temperature	10–30 $^{\circ}$ C
	Relative humidity	$\leq$ 70%
	Atmospheric pressure	85.0–106.0 kPa
	Other	Keep away from electromagnetic field interference sources

Analyzer overview

Storage environment	Temperature	-20–55 ℃
	Relative humidity	≤93%
	Atmospheric pressure	50.0–106.0 kPa
	Other	Room free of strong sunlight and corrosive gas, with good ventilation
Safety classification	Anti-electric shock grade	Type I
	Overvoltage type	Type II
	Pollution grade	Grade 2
Total	REF ID	010101003301
	Weight	161 kg
	Outer dimensions Length * width * height	90 cm*75 cm*78 cm
	External packing size Length * width * height	104.8 cm*89.8 cm*97.7 cm
	Power supply	AC 100-240V,50/60Hz
	Power consumption (VA)	600VA

2.3 Overall composition of analyzer

The analyzer is composed of the main machine, accessories and software, wherein the main machine is composed of the material preparation module, the liquid module, the temperature control module, the mechanical drive module, the optical path detection module and the circuit control module; and the accessories include a desktop computer suite and cables.

- The material preparation module includes the cuvette storage module, the sample reagent area module, the starter module, and the cuvette dustbin module.
- The liquid module includes the pipetting liquid module, the washing liquid module, the optical path detection module, and the cooling liquid module.
- The temperature control module includes the incubator heating module (which can incubate 80 cuvettes at the same time), the sample reagent area cooling module, and the PMT temperature control module.
- The mechanical drive module includes the cuvette storage, the sample disk, the SampleArm, the washer, the chamber, the claw, and the sample reagent area shaker.
- The optical path detection module is composed of the chamber module, the PMT module, and the main control circuit module.

- The circuit control module is composed of the power source module, main control board, wiring harness, and different types of sensors, motors, etc.

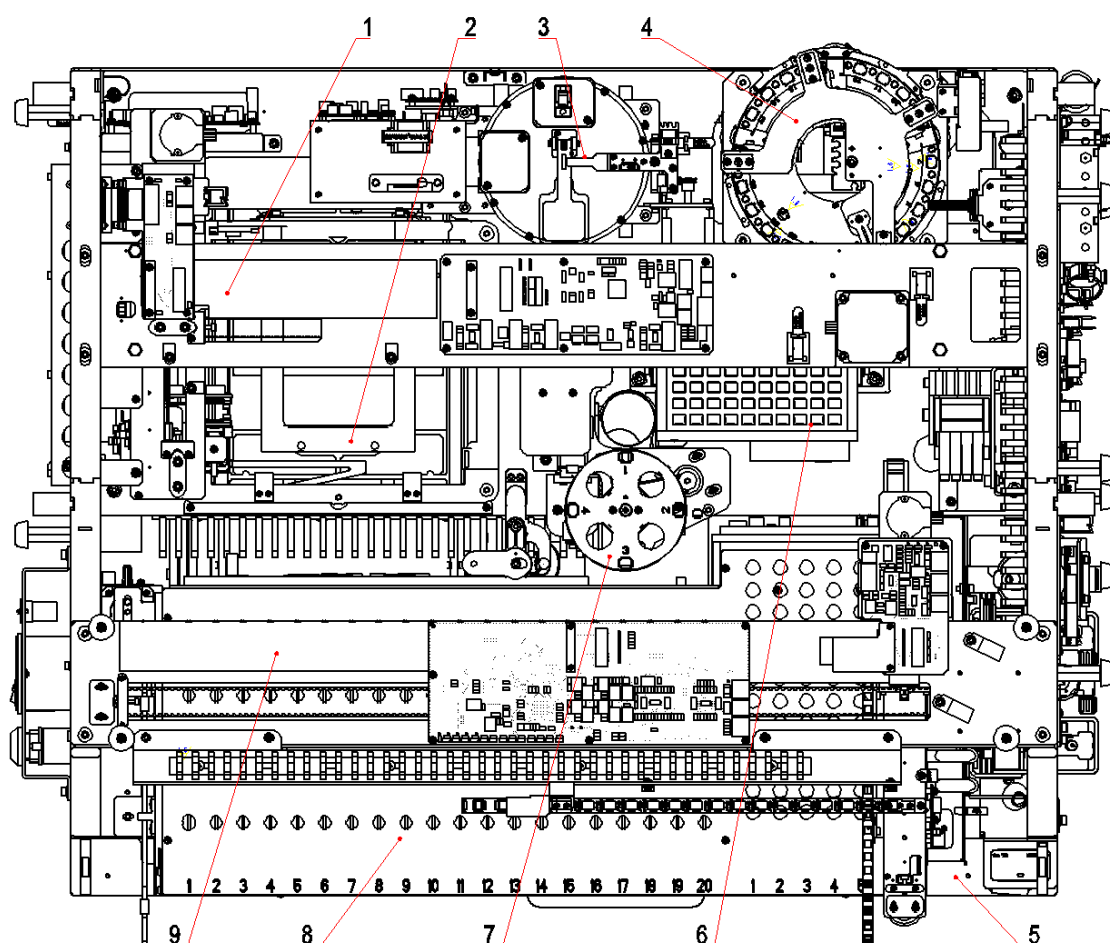


Fig. 2.3-1 Internal structure of analyzer

- | | | |
|-----------------|-------------------------|---------------|
| (1) Claw | (2) Cuvette storage | (3) Chamber |
| (4) Washer | (5) Rack component | (6) Incubator |
| (7) Sample disk | (8) Sample reagent area | (9) SampleArm |

2.4 Main components of analyzer

A rundown of the specific components of each module of the MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer is as follows.

2.4.1 Cuvette storage



Warning:

Do not touch the new cuvette box usage area of the cuvette storage while the analyzer is operating. Otherwise, this could lead to personal injury or damage to the analyzer.

(1) Components

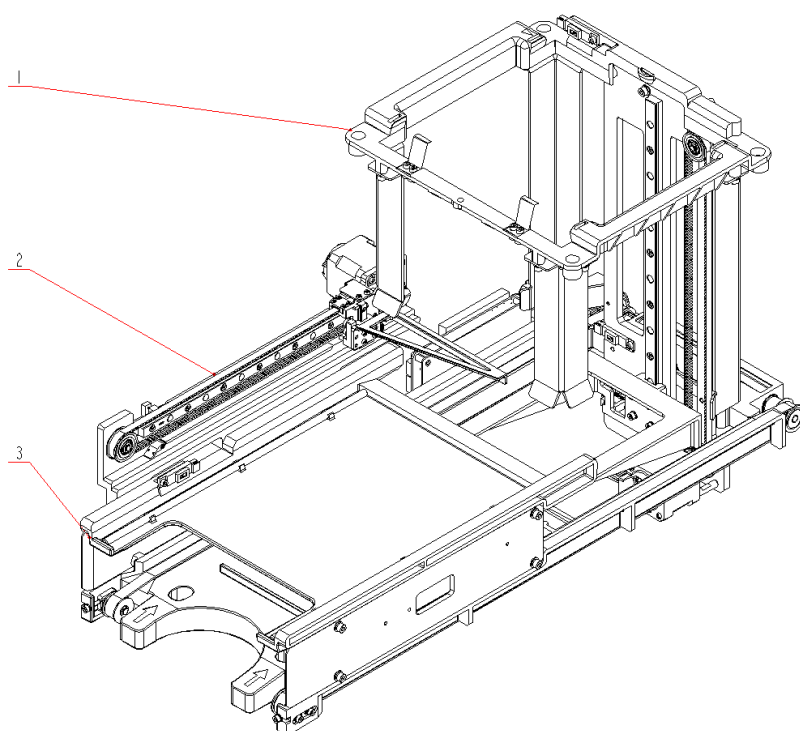


Fig. 2.4-1 Cuvette storage line drawing

1. Lifter

2. Waste box recoverer

3. New box transfer

(2) Function

The cuvette storage transfers new cuvette boxes containing cuvettes to the usage area and recovers waste cuvettes.

The cuvette storage can be divided into three layers. The bottom layer stores the new boxes transferred by the new box transfer. The middle layer stores the waste boxes transferred by the waste box recoverer. The top layer stores the boxes currently in use transferred by the lifter. It can store two boxes of cuvettes.

2.4.2 Sample reagent area

(1) Components

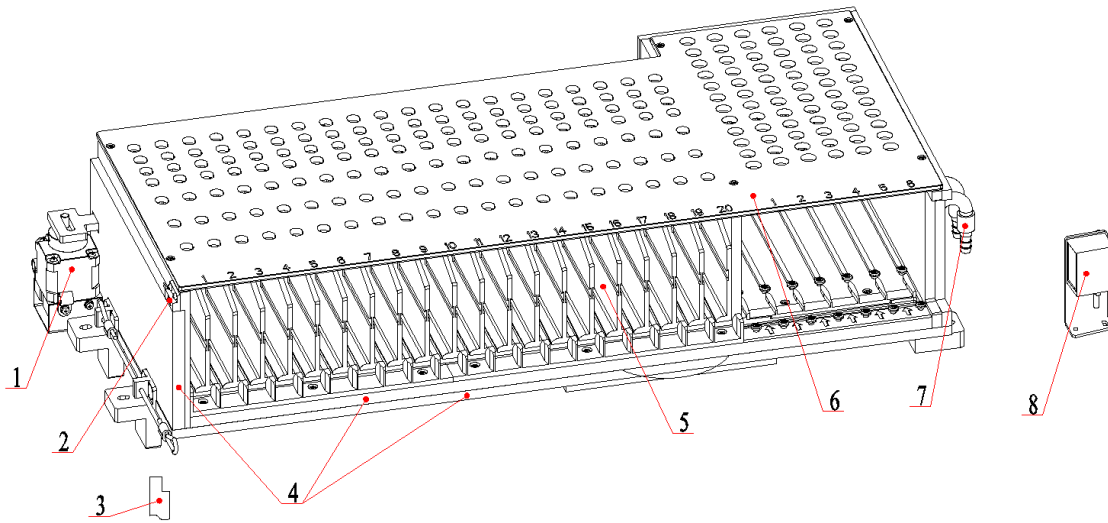


Fig. 2.4-2 Sample reagent area line drawing

- | | | | |
|---------------------|--------------------------|---------------------|----------------------|
| 1. Shaker component | 2. Door detection sensor | 3. RFID card reader | 4. Insulation cotton |
| 5. Area component | 6. Plate | 7. Radiator | 8. Barcode reader |

(2) Function

Places the reagent kit and the sample rack. The pipettor aspirates reagents and samples through the holes in the cover plate. The reagent area of the sample reagent area is cooled by water medium, and the temperature in the area is lowered down by heat conduction of the area metal to refrigerate the 20 boxes of reagents in it. If you need to turn off the refrigeration system, turn off the main power supply.

2.4.3 SampleArm



Warning:

Do not touch the SampleArm while the analyzer is operating. Otherwise, this could lead to personal injury or damage to the analyzer.

(1) Components

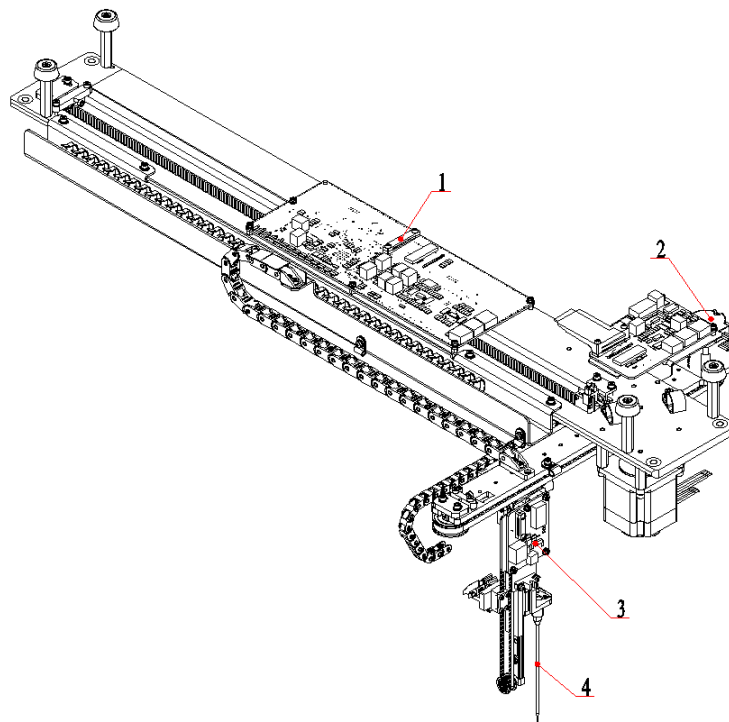


Fig. 2.4-3 SampleArm line drawing

1. X-axis

2. Y-axis

3. Z-axis

4. Pipettor

(2) Function

Transfers reagents from reagent kits in the sample reagent area and samples in sample tubes on the sample rack to cuvettes. It has liquid level detection, measure-and-track, clot detection, anti-bubble interference, collision detection, and automatic washing functions.

2.4.4 Sample disk

**Warning:**

Do not touch the sample disk while the analyzer is operating. Otherwise, this could lead to personal injury or damage to the analyzer.

(1) Components

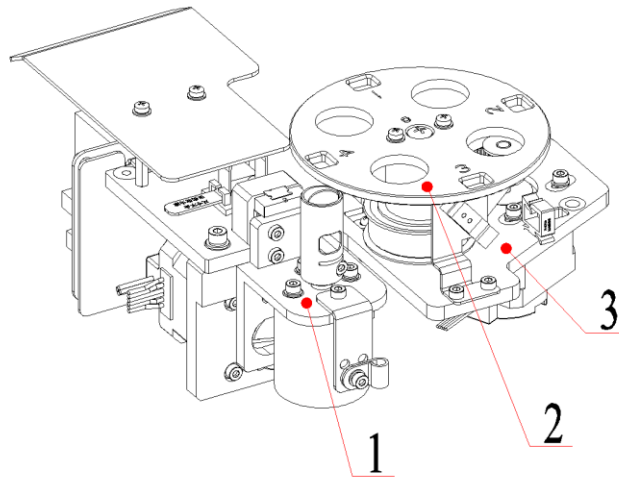


Figure 2.4-4 Sample disk line drawing

1. Lifter

2. Mixer disk

3. Turntable

(2) Function

Transports empty cuvettes to the corresponding work locations for sample and reagent loading, and mixes sample-loaded or reagent-loaded cuvettes.

2.4.5 Washer



Warning:

Do not touch the washer while the analyzer is operating. Otherwise, this could lead to personal injury or damage to the analyzer.

(1) Components

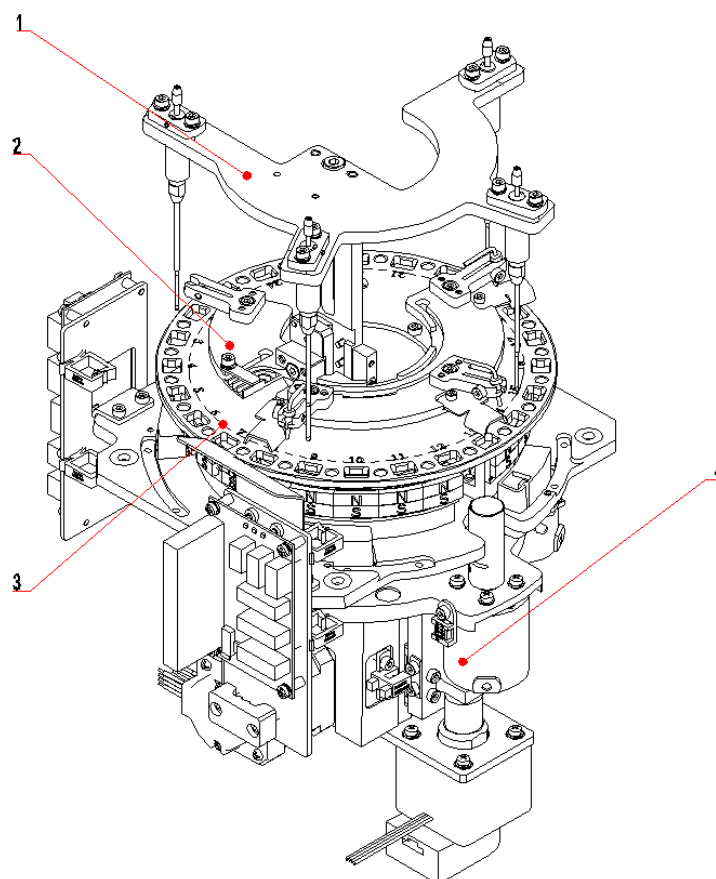


Figure 2.4-5 Washer line drawing

- | | | | |
|-----------------------|-------------------------|----------------|-----------------|
| 1. Waste liquid probe | 2. Washing liquid probe | 3. Washer disk | 4. Washer mixer |
|-----------------------|-------------------------|----------------|-----------------|

(2) Function

The main function of the washer is to remove impurities that are not bound to the solid-phase antigen of the magnetic spheres in the reaction liquid, by means of tertiary washing. The washer disk is responsible for transporting cuvettes to the magnetic field or various work locations. The four groups of waste liquid probes are responsible for aspirating waste liquid. The three groups of washing probes are responsible for injecting washing liquid. The washer mixer is responsible for mixing the mixture in the cuvettes.

2.4.6 Chamber

**Warning:**

Do not touch the chamber components while the analyzer is operating. Otherwise, this could lead to personal injury or damage to the analyzer.

(1) Components

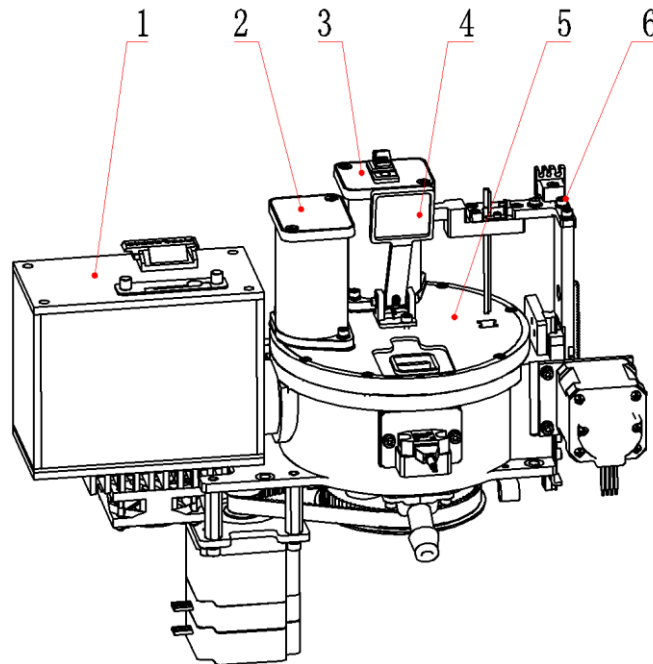


Fig. 2.4-6 Chamber line drawing

- | | | |
|-------------------|---------------------|---------------------------|
| 1. PMT module | 2. Starter 2 nozzle | 3. Starter 1 nozzle |
| 4. Shading module | 5. Chamber disk | 6. Waste liquid extractor |

(2) Function

In the chamber, starters 1 and 2 are injected into the cuvette that has been washed and shaded by the shading module. It then acts on the analyte to generate an optical signal. The optical signal is collected by the PMT and control circuit and processed, then converted into a light intensity value. The waste liquid extractor is responsible for extracting the waste liquid in cuvettes that have been tested.

2.4.7 Cuvette claw



Warning:

Do not touch the cuvette claws while the analyzer is operating. Otherwise, this could lead to personal injury or damage to the analyzer.

(1) Components

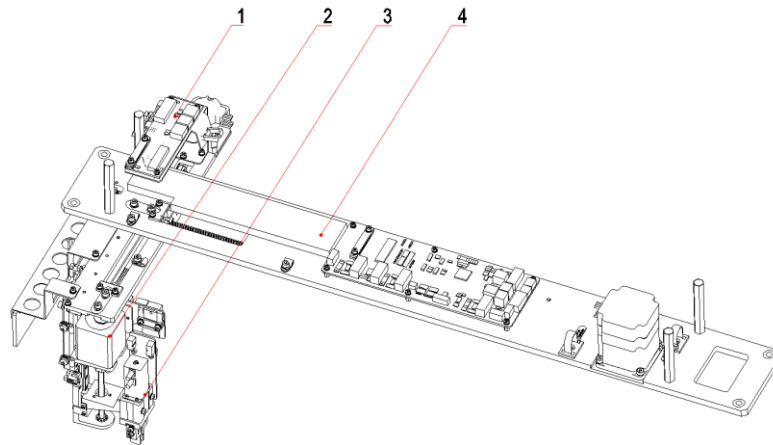


Fig. 2.4-7 Cuvette claw line drawing

- | | | | |
|-----------|-----------|------------|-----------|
| 1. Y-axis | 2. Z-axis | 3. Gripper | 4. X-axis |
|-----------|-----------|------------|-----------|

(2) Function

The claw transports cuvettes among various components of the analyzer.

2.5 Operation unit

The operation unit is a computer which has the analyzer operating software installed. It is composed of a display, computer, keyboard and mouse.

- Basic configuration of the computer: CPU frequency ≥ 3.0 GHz; hard disk ≥ 500 GB; RAM ≥ 4 GB; RS-232 interface; USB interface; RJ45 network interface; LCD; keyboard; and mouse.
- Supported OS: Windows 10.
- Security software: Common security software such as 360 Total Security, 360 Antivirus, Tencent PC Manager and Kingsoft Antivirus are supported.
- Data interface of software: wired network interface; data is stored in the XML format and in MySQL databases.
- User access: User management is available. Permission management based on two roles is supported: administrators and users.
- Network conditions: The software and the analyzer are connected via a wired network (TCP/IP), and the analyzer doesn't need to be connected to a local area network (LAN).

2.6 Analyzer labels

2.6.1 Module labels

"Protective grounding symbol" label

Protective grounding label

Located in the following positions:

- 1) One on the rack grounding block.
- 2) One on the left support of the reagent area.



"Cuvette bin" label

Cuvette bin for storing waste cuvettes.

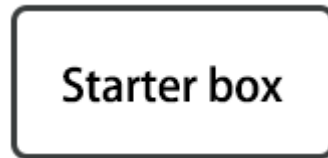
Located on the inside plate of the cuvette bin door.



"Starter box" label

Starter box for storing starter.

Located on the inside plate of the starter box door.



"S1" label

Store the starter 1.

Located on the inside plate of the starter box.



"S2" label

Store the starter 2.

Located on the inside plate of the starter box.



"Beware of laser" label

Remind users to be ware of the laser here and not to look straight at the beam.

Located in the laser beam area on the analyzer: the left door plate of the reagent area of the sample reagent area.



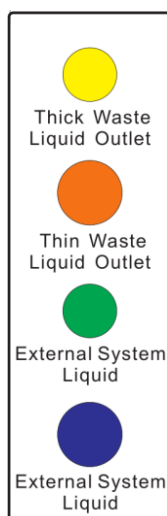
Laser window

Remind users of the laser window here.

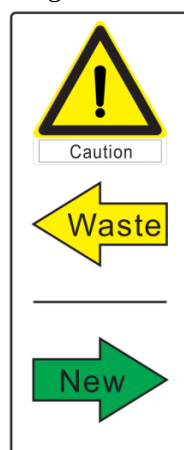
Located in the laser beam window on the analyzer: the right door plate of the sample area of the sample reagent area.

**External liquid interface label**

Connect to the external system liquid, thick waste liquid and light waste liquid devices.
Located on the left side of the external liquid interface.

**Tip labels of new cuvette box**

Prompt users to note the correct positions of loading new cuvettes and recovering waste cuvette boxes.
Located on the left door plate of the cuvette storage.

**Tip label of power switch**

Tip label of the network interface, main switch, and submain switch.
Located above the power switch.

LAN

Main Switch

Submain Switch

"Cuvette storage" label

Cuvette storage that stores cuvettes and recovers waste cuvette boxes.

Located on the inside plate of the cuvette storage door.

Cuvette storage

"Main switch of cover" label

Remind users to lock the analyzer during the analyzer's normal operation.

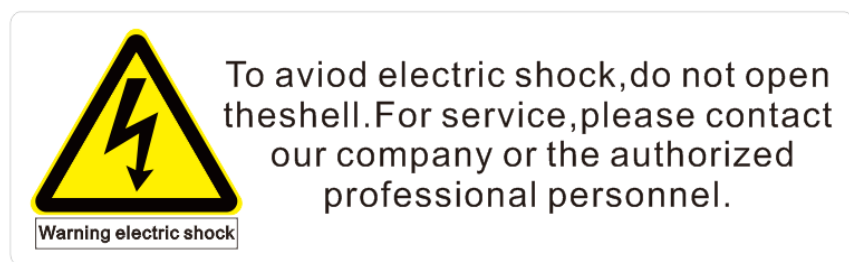
Located beneath the flip cover.



"Warning on opening casing" label

Remind users not to open the casing to prevent electric shock. If you need services, please contact our company or our company's authorized service technicians.

Located at the rear of the casing.



"Test counter" label

Record the number of tests.

Located inside the right part of the casing.

Test counter

"Beware of infection" label

This reminds the user to beware of potential biological infection in the particular area in question.

Located in the following positions:

- ① Located on the inside plate of the cuvette dustbin door.
- ② Located on the left side of the external liquid interface.
- ③ On the inner plate of the sample area door.

**"Thick Waste Liquid" label**

Remind users that thick waste liquid is stored in the tank.

Located on the thick waste liquid tank.

**"Thin Waste Liquid" label**

Remind users that Thin waste liquid is stored in the tank.

Located on the Thin waste liquid tank.

**"System liquid" label**

Remind users that system liquid is stored in the tank.

Located on the system liquid tank.



"No mixing" label

Remind users that starter 1 and starter 2 must not be mixed.

Located on the inside plate of the starter box door.



2.6.2 Symbols use in label

Symbol	Description
	Manufacturer
	Authorized representative in the European Community
	Date of manufacture
	Catalogue number
	In vitro diagnostic medical device
	Serial number
	Note: Refer to attached documents
	Importer

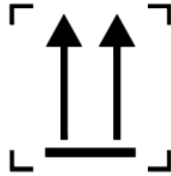
The definition for the following WEEE tag applies to EU member states only. WEEE tagged item indicates that this item should not be processed as domestic waste. By ensuring the proper disposal of this device, you help avoid the potential impacts of hazardous substances on the environment and human health. For more information, please contact the distributor from whom you purchased this product.



Up

This symbol means the transport package should be kept upright during transportation.

Located in the upper center of the packaging box sides.

**Keep dry**

This symbol means the transport package should be kept away from rain.

Located in the upper center of the packaging box sides.

**Fragile**

This symbol indicates that the content in the transport package is fragile and you should handle it with care when moving.

Located in the upper center of the packaging box sides.

**Do not roll**

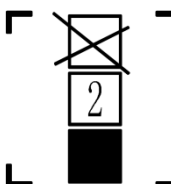
This symbol indicates that the transport package should not be rolled when moving.

Located in the upper center of the packaging box sides.

**Stacking layer limit**

This symbol means this package can be stacked two layers high at the maximum.

Located in the upper center of the packaging box sides.



2.7 Materials provided

See the packing list.

2.8 Accessories required but not provided

Catalogue Number	Name
130105000101	Reaction Cup
130299005M	Wash Concentrate
130299007M	System Tubing Cleaning Solution
130299004M	Starter 1+2
130299006M	Light Check
130210000201	Waste Bag

Chapter 3 Analyzer installation

3.1 Storage and transportation requirements of analyzer

3.1.1 Storage requirements

The packed MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer should be stored in a well-ventilated environment, free of corrosive gas, at -20–55 °C, with relative humidity not greater than 93% and atmospheric pressure of 50.0–106.0 kPa.

3.1.2 Transportation requirements

The MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer should be packaged for shipment in accordance with the requirements in the order contract. It should be kept away from violent impact, moisture or sunlight.

3.2 Installation requirements

3.2.1 Space requirements

The installation of the analyzer should meet the following requirements to guarantee the space necessary for operation, maintenance and good repair:

- The distance between the left side of the analyzer and the wall should be no less than 50cm.
- The distance between the right side of the analyzer and the wall should be no less than 50cm.
- The distance between the front of the analyzer and other devices should be no less than 125cm.
- Make sure you provide space for installation of the waste liquid drainage device.
- There should be ample space surrounding the power outlet to facilitate plugging and unplugging the power cord. Do not place the analyzer in a position which makes it difficult to disconnect the device.

3.2.2 Environment requirements

- Ambient temperature during use: 10 – 30 °C
- Relative humidity: ≤70%
- Atmospheric pressure: 85.0–106.0kPa

- The environment should be free of dust, mechanical vibration, loud noise sources or power supply interference and have good ventilation.
- Do not place the equipment close to a brushed electric motor, a flashing fluorescent light or electrical devices that turn on and off frequently.
- Avoid direct sunlight or placing the equipment in front of a heat or air source. The ground should be flat and able to bear a weight of at least 161kg.

3.2.3 Power supply requirements

- The analyzer is suitable for an AC 100-240V, 50/60Hz power supply.
- Rated power of analyzer: 600VA.
- Fuse model: 250V-10A—fast fusing.
- High-temperature protector: 2455R-70820258 (specification: OPEN: 53 °C±3 °C; CLOSE: 45 °C+4 °C/-5 °C; 12Vdc, 15A), 9700K51-1638 (specification: OPEN: 75 °C±5 °C; CLOSE: 50 °C±15 °C; 230Vac, 13A).
- To ensure the reliable operation of the analyzer, more than 20% spare capacity should be maintained for the distribution box.
- The analyzer should be connected to a well grounded power supply for safety.
- For safety purposes, an over-current protection device with a rated current of 15 A should be provided between the power supply of the building and the power supply of the device.



Warning

If the service environment or power supply cannot meet the above requirements, this may affect the analyzer's performance or cause damage to the analyzer and even endanger personal safety.

3.3 Out of box inspection

3.3.1 Unpacking steps

Please inspect the analyzer package upon receipt to check for physical damage. In case of damage, please contact Shenzhen New Industries Biomedical Engineering Co., Ltd. or your local agent. Open the package according to the following steps once you are sure there is no external damage:

- Keep the package upright so that the arrow label on the package is upward.
- Open the package to check all the components are complete based on the packing list. If you find anything missing, please contact Shenzhen New Industries Biomedical Engineering Co., Ltd. or your local agent.
- Carefully inspect the appearance of the analyzer for any damage. If you find any, please contact Shenzhen New Industries Biomedical Engineering Co., Ltd. or your local agent.

3.3.2 Moving and positioning

- For short-distance movement, raise the analyzer directly by pulling the rope handle on the analyzer. When the handle is used, it is recommended that you perform the operation on the left and right sides of the analyzer. Do not raise the handle against the edge of the analyzer to avoid injury to hands.
- The analyzer should be kept upright during the whole moving process and should not be tilted or laid on its side. It should be moved gently and slowly.
- Avoid vibration during moving. The analyzer should be inspected and debugged after moving and before operation.
- The analyzer should be firmly placed and do not wobble, and the height of the feet should be adjusted so that the analyzer is level.

3.4 Installation of analyzer

The analyzer should be installed by professionals from our company. It is not recommended to disassemble or install the analyzer other than for routine system maintenance.

3.4.1 Electrical connection

- Connect the monitor, keyboard and mouse to the appropriate connectors on the computer.
- The touch screen cable of the display is connected to a USB port on the back of the computer.
- Connect the power cables of the computer case and monitor to the corresponding ports.
- Connect the analyzer's power cord to a power connector that meets the specified requirements.
- The analyzer's network interface and the computer's network interface are connected using a network cable.

3.4.2 System liquid connection

- The analyzer's external system liquid tank, the system liquid interface, and the system liquid level sensor interface are located on the front right of the analyzer;
- The analyzer has two system liquid pipes. Insert the end with a system liquid pipe connector of each of the two pipes to the corresponding system liquid interface on the analyzer, and insert the end with a system liquid level sensor line to the corresponding liquid level sensor interface on the analyzer. Then install the other end of the tank cover with a system liquid level sensor and a system liquid pipe to the tank labeled "System Liquid".

3.4.3 Waste liquid connection

- The analyzer's external waste liquid tank, the waste liquid interface, and the waste liquid level sensor are located on the front right of the analyzer;

- The analyzer has two waste liquid pipes. Insert the end with a waste liquid pipe connector of each of the two pipes to the corresponding waste liquid interface on the analyzer, and insert the end with waste liquid level sensor to the corresponding liquid level sensor interface on the analyzer.
- Then install the other end of the tank cover with a waste liquid level sensor and waste liquid pipe to the tank labeled "Waste Liquid".

3.5 System startup and status confirmation

After the analyzer is installed, ensure that the power and communication lines of the analyzer and computer, as well as the system liquid and waste liquid pipes have been connected properly and firmly. Then power on and start the system.

1. Power on the MAGLUMI X3 Fully-auto chemiluminescence immunoassay analyzer, and turn on the main and auxiliary switches on the left side of the analyzer.
2. After logging in to the Windows operating system, start and log in to the user software.
3. The analyzer will enter an initialized status after the user logs in to the user software and each component of the analyzer completes initialization.
4. Load consumables such as cuvettes, system liquid, and starters.
5. Confirm that each status displayed in the software is in the normal range, and check whether there are sufficient solid and liquid consumables.
6. System test verification. Perform five LC tests and 10 BGW tests (see "4.2.3 Confirming analyzer status" for the steps and expected results). When the system test results meet the requirements, the system test is passed and the installation of the analyzer is successful.

Chapter 4 Basic operation

This section mainly introduces the basic operation process of this system. Users can handle basic routine operations after learning the content in this section.

4.1 Routine operation process

Table 4.1-1 Routine operation process

Operation steps	Procedure
1. Pre-start checks (1) Check the power supply (2) Check the analyzer	Check whether the power supply is normal. Check the appearance and status of the analyzer as necessary.
2. Connect the power supply	Power on the analyzer, and turn on the main and auxiliary switches of the analyzer in succession.
3. Log in to the user software	Enter the username and password of the operator in the software's login dialog.
4. Check consumables	Confirm there are sufficient system liquid, starter 1, starter 2, and cuvettes for the present sample test.
5. Confirm analyzer status (1) Status confirmation (2) System test confirmation	In the software [Status] interface, confirm the temperature, voltage, humidity, consumables and waste status. Wait until all statuses display as green before testing. Perform system test operations and confirm whether the measurement values of LC tests and BGW tests are within the allowed range.
6. Confirm test conditions (1) Confirm the assay parameters (2) Profile settings	Confirm the test assay parameters. Set the profile based on the actual conditions.
7. Prepare reagent (1) Confirm the remaining volume of reagent (2) Time to shake the magnetic microspheres	Confirm that the number of remaining reagent tests for each assay is sufficient to complete the sample test. Confirm that the magnetic microspheres of the reagent have been shaken for at least 30 minutes.
8. Calibration of test assay QC assay registration	Confirm that the reagent has been calibrated and run the procedure to calibrate reagents with no valid calibration. Perform QC procedures on reagents
9. Sample registration	Register routine sample. Register STAT sample. Register diluted sample.
10. Start test	Click the <Start> button to test.

11. Test added sample	Register the added sample during the course of the test. (If the added sample is an STAT sample, it needs to be registered as an STAT sample. If it is a diluted sample, it needs to be registered as a diluted sample.) Click the <Start> button to test.
12. Confirm the test result	Query, recalculate, remeasure, delete and print the test results.
13. Procedures at the end Exit operating software	Exit the operating software.
14. Shut down	Power off the analyzer and the computer.
15. Operations at the end	Remove the reagent kits and samples from the sample reagent area; empty the waste liquid tank and the cuvette dustbin.

4.2 Preparation for analysis

Complete the preparation for routine sample analysis via the following steps.

4.2.1 Pre-start checks

Inspect the following items before starting up to ensure the system works normally after starting up.

1. Check the power supply to guarantee normal power supply.
2. Check the communication line and power line of the analyzer and computer, to confirm they are connected and not loose.
3. Check whether the system liquid, starter 1 and starter 2 are properly connected and the volume of liquid is sufficient.
4. Check whether there are sufficient cuvettes.
5. Check whether the waste liquid has been cleared.
6. Check whether the cuvette dustbin and the waste cuvette box are cleared.



Warning:

Beware of risk of biological infection. Do wear gloves and work clothes during the above inspection work to protect from infection and wear protective glasses when necessary.

4.2.2 Connecting power supply and logging in to software

1. Power on the analyzer, and turn on the main and auxiliary switches of the analyzer in succession.
2. Double-click the shortcut of the user software on the desktop after logging in to the Windows operating system to start the user software. A login dialog will pop up in the interface after starting up. Enter the username and password, then click the <Login> button to enter the user software interface.
3. The analyzer will enter the initialization status after you log in to the user software and each

component of the analyzer completes the initialization. After waiting for all statuses to be displayed in green, you can start testing.

4.2.3 Confirming analyzer status

4.2.3.1 Confirming status

In the software **[Status]** interface, confirm the status of Consumables, Environmental Parameters and Waste in sequence. Normal status is displayed in green.

If the Consumables item displays as abnormal, follow the "Loading consumables" section of these Operating Instructions.

After the analyzer is initialized, you need to wait for 30 minutes to ensure that the Environmental Parameters are stable.

If the Waste item is abnormal, follow the steps in the "Waste disposal" section of this instructions.



Note:

After each component of the analyzer completes initialization, wait until all the status indicators are green before you start the sample test. If the analyzer fails to enter the normal status after a long time, please contact our technical service department.

4.2.3.2 System test confirmation

System test is divided into the BGW and the LC parts.

Before performing the system test, put the LC solution in the corresponding positions of the sample reagent area. See "5.6.2 Loading LC solution" for details.

Click the **[System Test]** button on the main menu to go to the **[System Test]** interface. Click the **<Add>** button to open the **[System Test Registration]** interface. Enter the numbers of BGW tests and LC tests, and then click the **<Register>** button to register the system test. A **[Query]** interface appears. Confirm the system test information, and then click the **<OK>** button to start the system test. Check the system test results after the test ends.

Run five LC tests and 10 BGW tests. When the system test results meet the following requirements, it is deemed that the system test has been passed:

- | | | |
|-------|---------------|-----------|
| ● BGW | 100—1200 | CV ≤ 10%; |
| ● LC | 450000—650000 | CV ≤ 3%; |



Note:

1. The tests can be performed only when the BGW and LC test results and CV values satisfy the requirements. It is not recommended to continue the sample testing if the values are abnormal (abnormal data is marked in red in the list of system test results). Otherwise, it may undermine the reliability of the test

results.

2. The routine LC test is to be carried out once a week. When the analyzer parts are replaced, an LC is also required. It is not necessary to load the LC test liquid when an LC test is not being performed.
 3. A BGW test should be performed every day before sample testing to ensure the accuracy of results.
-

4.2.4 Replacing consumables

When the analyzer has insufficient or no cuvettes, system liquid, or starters, an alarm will be prompted. The analyzer status returns to normal after the consumables are loaded.

4.2.4.1 Loading cuvettes

- Loading cuvettes

New cuvette boxes and waste cuvette boxes are stored in the cuvette storage. The cuvette storage is located on the left side of the analyzer. Open the lower door of the analyzer, and put the new cuvette boxes on the new cuvette box transporting area at the bottom to load new cuvette boxes or remove waste cuvette boxes.

Procedure:

1. Log in to the user software. Once the analyzer is initialized, check the quantity of available new cuvettes and the quantity of waste cuvettes in the software.
2. Remove the cuvette packaging and prepare to test the required number of new cuvette boxes.
3. Open the lower door of the analyzer. If any waste cuvette boxes are found in the waste cuvette box recovery area, remove them before loading new cuvette boxes.
4. Put a new cuvette box on the new cuvette box transporting area at the bottom layer of the cuvette storage. The new cuvette box will be transported to the new cuvette box usage area at the top layer to complete loading.

Note:

1. Pay attention to the quantity of cuvettes remaining before and during tests.
 2. Make sure you remove any waste cuvette boxes before putting new ones in. Otherwise, you won't be able to replace the cuvette boxes in the usage area.
 3. If the analyzer is not used for one week, clear the cuvette storage before use to ensure that the cuvettes are clean.
 4. The cuvette box and the cuvette storage are designed to prevent errors. Please insert them in the correct orientation.
-



4.2.4.2 Filling system liquid and starter

- Filling system liquid

The system liquid is placed at the lower part of the front right of the analyzer and connected to the analyzer using a hose. The remaining amount of system liquid can be viewed in the [Home] and [Status] interfaces.

Procedure:

1. Unscrew the empty system liquid tank's cover which has a liquid level probe and a hose. Seal the cover with a clean sealed bag and take out an empty system liquid tank;
2. Take the tank cover out of the sealed bag and screw it to a system liquid loaded tank. Make sure that the connection is safe with no liquid leakage.



Note:

- 1) The system liquid must be prepared in advance. Make sure no impurities exist in the liquid tank.
- 2) Pay attention to the amount of system liquid before and during tests.

- Filling starter

Procedure:

1. Unscrew the empty starter bottle's cap that has a liquid level probe and a hose. Seal the cap with a clean sealed bag and take out an empty starter bottle;
2. Take the bottle cap out of the sealed bag and screw it to a new starter bottle. Make sure that the connection is safe with no liquid leakage;
3. Enter "30" as the number of primings in the **[Priming]** interface and click the **<Start>** button to start priming;
4. Perform the BGW test in the [System Test] interface.



Note:

1. Pay attention to the amount of starter before and during the test!
2. The starter 1 and 2 bottles and tubes are equipped with signs. Be careful not to install them in reverse!

4.2.5 Loading reagent

Scan the reagent RFID on the left side of the analyzer and insert the reagent into the reagent area position of the sample reagent area. In the **[Reagent]** interface of the software, confirm the following information before you start the calibration, QC test and sample test of the reagent:

- The reagent must be within its period of validity.
- The number of reagents remaining should be sufficient to complete the sample tests.

- The magnetic microspheres of the reagent are to be shaken for 30 minutes.

4.2.5.1 Reagent kit structure

A reagent kit has the following basic structure:

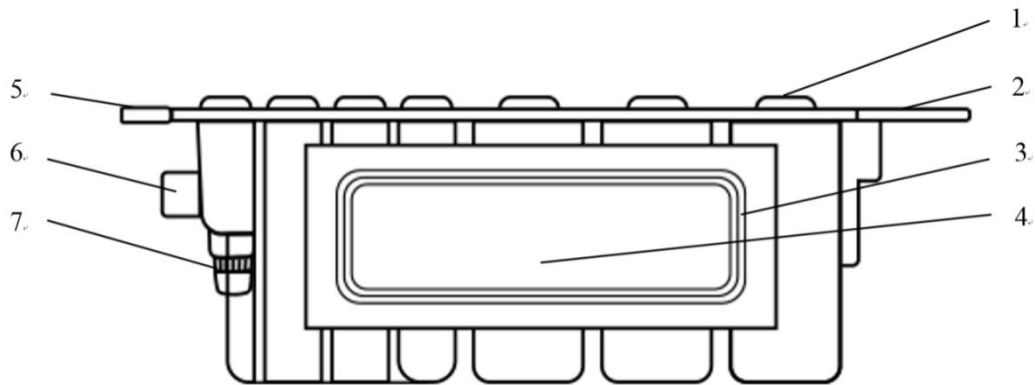


Figure 4.2-1 Reagent kit structure

1. Sealing film: Seals reagents. The analyzer reagent probe enters the reagent kit via the sealing film to aspirate reagent.
2. Handle: Used to load and remove the reagent kit.
3. RFID tag: Records the reagent information of the reagent kit.
4. ID label: Indicating the reagent composition, name, period of validity, and other information of the reagent kit.
5. Fixing clip: Fixes the reagent kit inserted into the reagent area.
6. ID blocker: Blocks the optocoupler when the reagent kit is inserted into the reagent area to identify the kit.
7. Magnetic microsphere mixing gear: Cooperates with the reagent area shaker to mix the magnetic microspheres.

4.2.5.2 Loading the reagent kit

1. Preparing the reagent kit

Make sure that the reagent kit is stored in the correct orientation (the sealing film faces up) and do not invert or shake the reagent kit.

- 1) Remove the reagent kit packaging box from the refrigerator.
- 2) Remove the reagent kit from the packaging box.
- 3) Check for leaking of the sealing film or other parts of the reagent kit. If there is liquid leakage, please contact the agent in good time.

- 4) Check the liquid in each component of the reagent kit for bubbles. If air bubbles are present, use a pipette to remove the bubbles and verify that the bubbles are completely removed before use.
- 5) Gently rotate the magnetic microsphere mixing gears of the reagent kit and ensure that the gears can rotate freely.
- 6) Carefully peel off the reagent kit sealing film.
- 7) Remove any liquid from the surface of the reagent kit to avoid cross-contamination.

Note:



- 1) Remove the sealing film from the reagent kit before using the new reagent.
- 2) The reagent should be prepared, used and stored in strict accordance with the reagent instructions. Do not allow bubbles to form in reagents. This could affect the accuracy of reagent intake and affect test results.
- 3) Reagents from different reagent kits must not be mixed, as this may affect the reliability of test results.

2. Loading the reagent kit

Before you insert a reagent kit into the reagent area, make sure the analyzer and software are operating properly and that the analyzer has completed initialization.

- 1) Open the sample reagent area door.
- 2) Remove unused reagent kits in accordance with the indicator colors of various channels in the reagent area.
- 3) Hold the handle of the reagent kit and approach the RFID tag on the reagent kit to the sensing area (and pause there for about two seconds). The buzzer will beep, indicating that RFID is recognized successfully.
- 4) Insert the reagent kit straight to the bottom along a vacant reagent channel until the analyzer gives off a "beep" sound, indicating that the reagent kit has been properly loaded.
- 5) Reagent related data is read by the software. The corresponding position of the **[Reagent]** interface is lit and the reagent name is displayed.
- 6) If there is a failure when the data is read, repeat the above steps.
- 7) After the reagent information is read, close the sample reagent area door.

Reagent kit statuses indicated by reagent area indicator colors:

Indicator color	Reagent kit status
No color	Idle, meaning a reagent kit can be loaded.
Steady green	Waiting. The rack has a reagent kit loaded.
Flashing green	The reagent kit in the rack has completed the test and can be pulled out.
Steady yellow	Running. The reagent kit is in use.
Flashing yellow	Error. For example, a reagent kit in use is removed.

3. Confirm the remaining volume of reagent

After the RFID e-tag of the reagent is scanned in the reagent area and the reagent kit is inserted properly, you need to confirm that the number of remaining tests displayed in the reagent area is sufficient for this sample test. If the reagent is not sufficient to finish this sample test, load a new reagent kit in the reagent area.

4. Shaking time of reagent magnetic microspheres

When a reagent kit is inserted, the reagent kit's assay name, the number of remaining tests of the reagent, the calibration status and the magnetic microsphere shaking time will be displayed in the corresponding reagent channel in the [Reagent] dialog. See "5.4.1 Reagent information" for details.

4.2.6 Loading samples

4.2.6.1 Sample rack structure

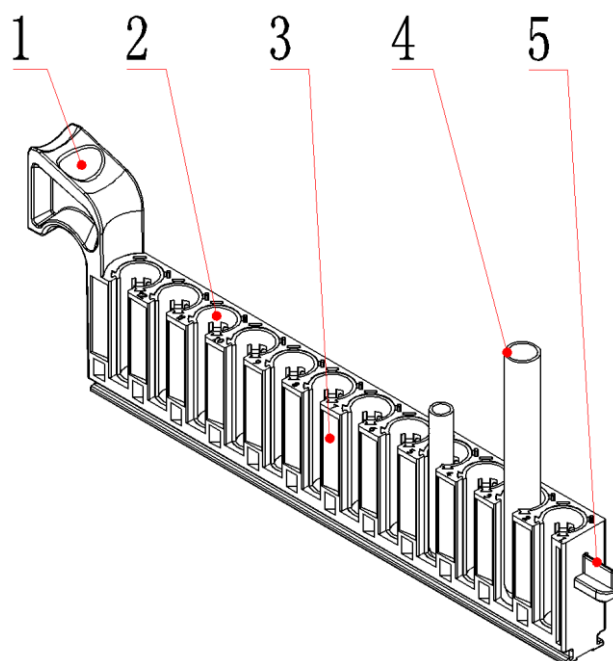


Fig. 4.2-2 Sample rack line drawing

1. Handle	2. Test tube holder	3. Sample rack ID label
4. Test tube	5. Optocoupler sensor chip	

4.2.6.2 Loading samples

1. Preparing samples

The following conditions must be met before a sample is tested:

- 1) Samples that have hemolyzed, are affected by lipemia, or are infected with microbes cannot be used for testing.
- 2) Make sure the sample is free of air bubbles before testing.

After conditions for the sample are met, you can insert the sample tube into the sample rack as follows.

- 1) The sample tube must meet the specifications.
- 2) Carefully insert the sample tube into the sample rack.
- 3) If the sample tube has a barcode affixed, the barcode should be oriented towards the opening of the sample rack for scanning by the barcode reader.

Note:

1. Wear gloves before working on samples so as to avoid infection.
 2. Make sure the sample tube cap is open. Otherwise, the pipettor may be damaged.
 3. When there is insufficient or no sample liquid, the test will generate a concentration result, but the result is flagged as "L". Therefore, when the analyzer warns that "The SampleArm cannot detect the liquid level", you should perform another test.
 4. For safety reasons, the sample must meet the requirements and operating conditions in order to avoid the generation of air bubbles and blood clots.
 5. Do not rotate sample tubes with a barcode. Rotating sample tubes may cause damage to the barcode.
 6. After inserting the sample rack with the sample tube into the sample platform, do not randomly change the position of the sample tube so that test results are not mixed up.
-



2. Loading sample rack

Please follow the steps below to load the sample rack:

- 1) Open the sample reagent area door.
- 2) Remove unused sample racks based on the indicator color of each sample channel in the sample area.
- 3) Select an empty sample channel and push a sample rack into the channel slowly to make sure the barcode of the rack can be read properly.

- 4) Push the sample rack vertically and stably until the front blocker of the rack enters the sensor at the end of the sample area channel and produces a "click" sound.
- 5) When the barcode fails to be recognized, please re-load the sample rack to scan the barcode.

Indicator color	Status of sample rack
No color	Idle, meaning a sample rack can be inserted.
Steady green	Waiting
Flashing green	Sample has been loaded and the sample rack can be removed.
Steady yellow	In operation, the sample rack is loading
Flashing yellow	Error. For example, a sample rack in use is removed.



Note:

Before the sample rack enters the sample platform, make sure that the analyzer and software are in normal working order, otherwise the sample rack will not be recognized and subsequent operations will be invalid.

4.2.7 Validating analysis conditions

Before running a test, first confirm whether the reagent parameters and reference range of the test assay are set correctly. Confirm the reagent for the assay has been calibrated and the calibration has not expired. Confirm whether the profile, dilution and QC material are configured. (For the specific steps to configure the reagent parameters, profile, dilution and QC material, please refer to "5.8.1 Parameter definitions".)

4.3 Test analysis

After the preparatory work is completed, routine sample analysis can be performed.

4.3.1 Calibration assay registration

Click the **<Reagent>** button in the main menu to access the **[Reagent]** interface. Check the reagents that need to be calibrated and then click the **<Cali. Register>** button to finish registering the calibration assays. Click the **<Start>** button on the operation bar to run the calibration test.

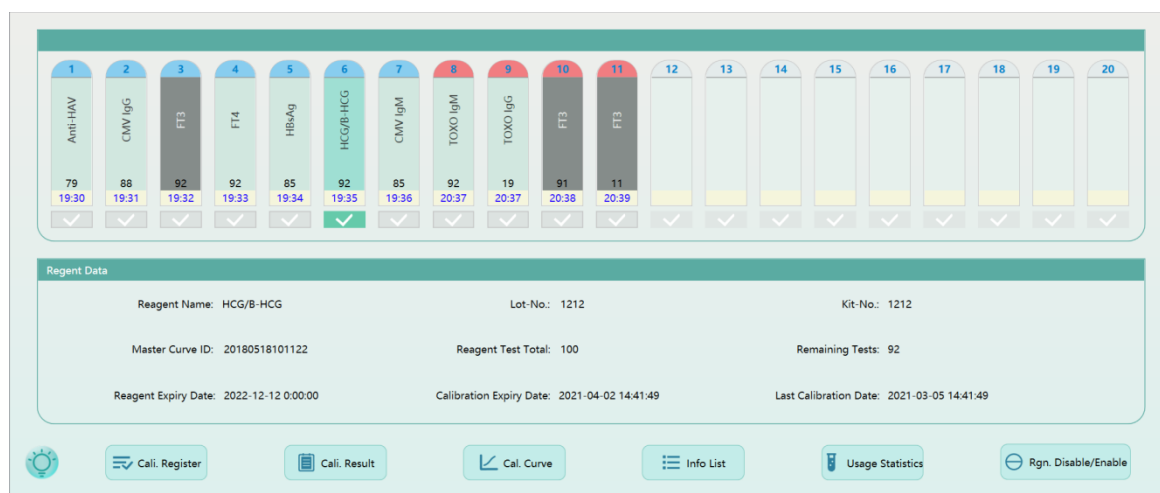


Fig. 4.3-1 Reagent interface

After the calibration test is complete, double-click the corresponding assay reagent or selected a reagent and click the <Cal. Curve> button to go to the [Reagent Details] interface. View the curve information and accept or reject the new curve.

Select the "Validate" option and click the <OK> button to validate the new calibration data. Select the "Reject" option and click the <OK> button to discard the new calibration data. You can go to the [Reagent] interface to re-register the reagent and run the calibration operation.

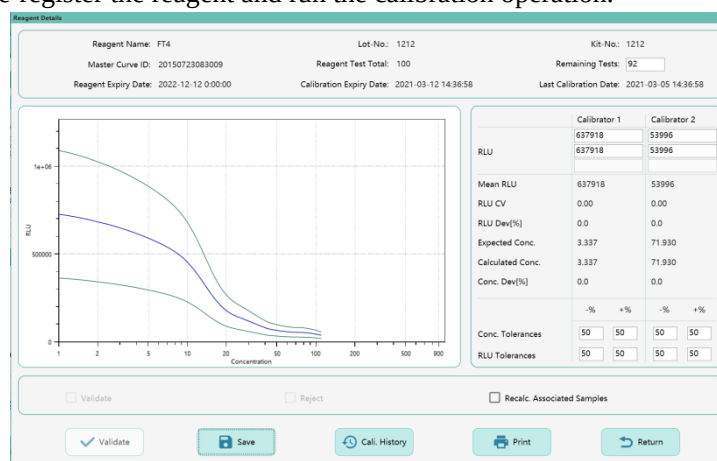


Fig. 4.3-2 Reagent calibration details

4.3.2 QC assay registration

Click the <Work List> button on the main menu to go to the [Work List] interface. Load the sample rack that contains the QC material properly, select the position where the QC material is placed in the sample position of [Sample Area], and click the <QC> button to go to the [QC Register] interface. Select the QC material in the list and select the QC assay, then click the <Register> button to register the QC assay.

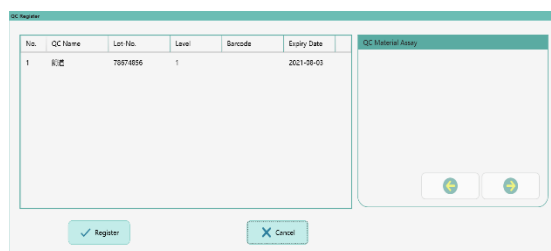


Fig. 4.3-3 QC registration

4.3.3 Sample registration

Click the **<Work List>** button on the main menu to access the **[Work List]** interface.

Load the sample to test onto the sample rack (if the sample has a barcode, keep the barcode facing the opening of the sample rack), and then insert the sample rack into the sample area properly. The scanner scans the sample when the sample rack is being inserted to the sample area, and the software automatically displays the sample rack ID and barcode scanned.

Select the sample rack to edit, enter the sample ID (not needed if there is a barcode), select the desired assay in the **Assay Selection** area, or use the profile function to quickly select multiple test assays to register the samples.

If the sample is for the STAT purpose, click the position of the corresponding STAT sample tube and enter the sample ID (not needed if there is a barcode). Then select the assay for STAT and click the **<STAT>** button to prioritize the STAT sample for tests.

If the sample needs to be diluted, click the position of the sample to dilute on the sample rack in the software, and enter the sample ID (not needed if there is a barcode). Select the assay to dilute, click the **<Dilute>** button and select the dilution ratio.

To use the LIS feature, you only need to properly configure and connect the LIS server. After the sample ID is entered, the analyzer will automatically retrieve the sample test assays from the LIS server.

After the sample is registered, you can click the **<Work List>** button to view the statuses of all registered tests and delete or modify the number of repetitions or export data. Alternatively, you can click the **<Browse>** button to view edited assays for each sample or the number of registered assays and the number of remaining tests of the reagent.

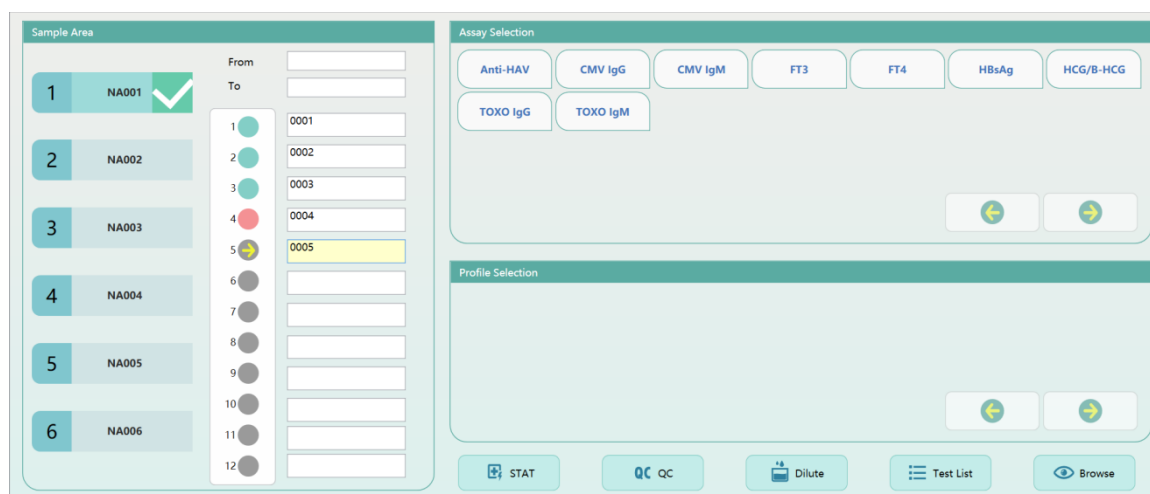


Fig. 4.3-4 Work list

4.3.4 Starting tests

After the sample registration is complete, click the **<Start>** button on the operation bar to send the test command and the analyzer will start the test.



Note:

- 1) Do not open the analyzer's cover while the analyzer is operating.
 - 2) Avoid touching the pipettor, sample disk and other moving components and their motion paths when the analyzer is operating.
-

4.3.5 Testing added samples

1. Additional test assays for registered samples

If you need to add assays during the testing process, you should register the sample first. Select the desired sample rack in the "Sample Area" in the [Work List] interface and the desired sample to add an assay to and then register the sample. See "4.3.3 Sample registration" for details of the registration method.

2. Adding test assays for unregistered samples

If you need to add a sample during the test, place the sample to add onto the sample rack and load the rack to the sample area properly, and then register the sample. See "4.3.3 Sample registration" for details of the registration method.

If no reagent for the additional assay is in place in the reagent area, you need to load the corresponding reagent into the reagent area. If the reagent for the additional assay is in place in the reagent area, after the registration is completed, click the **<Start>** button in the operation bar to complete continuous loading.

4.4 Monitoring during a test

The number of samples, number of tests, consumables, temperature, and other statuses of the analyzer can be monitored in real time during a test via the main interface. If a major fault occurs, click the **<Stop>** button to stop the test. See "5.7 [Status] function description" for status details.

4.5 Processing results

Click the **<Result>** button on the main menu to access the **[Result]** interface. This interface displays result information on all the samples for the day.

You can double-click any result record or click the **<Details>** button after clicking a result record to check the details of the test result.

You can check the checkbox before each result record, or select multiple test results simultaneously in the "Sample ID" checkbox and then click a function button below the result list to delete, remeasure, put

online, or recalculate the select results or perform other operations.

Click the <Search> button at the bottom of the **[Result]** interface to go to the **[Result Search]** interface. Then enter the conditions in the search criteria field, and click the <OK> button. The information of test results in history queries will be shown in the test result list.

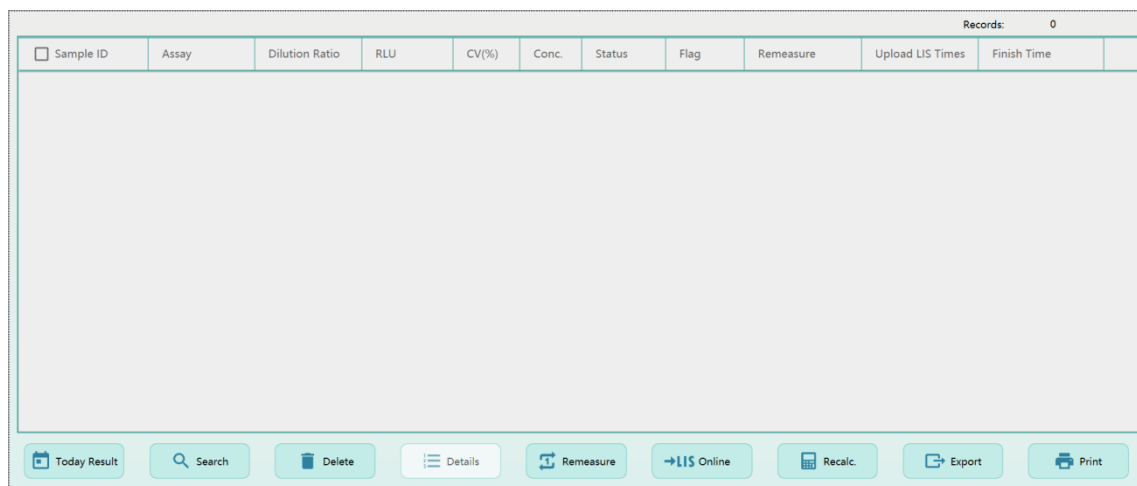


Fig. 4.5-1 Test results

4.6 End of analysis

When all the tests have finished, you can exit the user software and log out of the Windows operating system before turning off the power supply of each part.

4.6.1 Turning off the power supply

When all the tests are completed and the tests for the day are finished, you can exit the software and power off the printer, computer, analyzer, or other devices.



Note:

If the analyzer won't be used for a long time, please turn off the main and auxiliary switches of the analyzer.

4.6.2 Operations after power-off

Check the following items in preparation for the next test.

1. Check for leftover cuvettes on any horizontal surfaces.
2. Remove any samples from the sample area.
3. Remove any reagents from the reagent area.
4. Empty the waste liquid tank.
5. Empty the cuvette dustbin and clear waste cuvette boxes.

6. Check the plane part of the analyzer and whether the starter boxes have blots or stains. If yes, wipe the blots and strains off with a clean soft cloth.

4.6.3 Reagent kit removal

1. Open the sample reagent area door.
2. Hold the handle of the reagent kit and take the reagent kit out of the reagent area. If the reagent kit is empty, dispose of it properly. If the reagent kit is not empty, use an extra sealing film to cover the holes on the kit to prevent liquid evaporation.
3. Place the reagent kit in a refrigerator and ensure that it is correctly oriented.

4.6.4 Sample rack removal and maintenance

4.6.4.1 Removing the sample rack

1. Open the sample reagent area door.
2. Hold the sample rack handle and pull the sample rack out of the sample area stably until the rack is completely out of the sample area. If the sample tubes are empty, dispose of them in accordance with relevant regulations.
3. If the sample tubes are not empty and may be used in the future, seal and store the samples in accordance with the relevant regulations of the lab.

4.6.4.2 Maintaining the sample rack

In special cases, the sample rack requires maintenance.

- **Case 1:** After direct contact with biochemical substances.
- **Case 2:** The sample tubes cannot be secured in the sample rack.

Case 1 is handled as follows.

1. Move all samples on the contaminated sample rack to another clean sample rack.
2. Remove the sample rack and dry it with a clean, dry towel. Spray ethanol at contaminated areas to disinfect the rack.

Case 2 is handled as follows.

1. Take a pointed tool, insert its tip to under the metal tab of the inner retainer of the sample rack slot, as shown in Figure 4.6-1, and then gently lift the retainer.
2. Insert a test tube that meets the specifications to check whether it can be fixed securely.
3. If it is not possible to secure it, repeat steps 1 and 2 until it can be secured.



Fig. 4.6-1 Lifting the retainer

4.6.5 Waste disposal

4.6.5.1 Disposal of waste cuvettes

- Disposal of waste cuvettes

Waste cuvettes are placed in the cuvette dustbin in the lower middle of the analyzer. The software automatically counts the waste cuvettes. You can view their status in the **[Home]** and **[Status]** interfaces.

Draw out the cuvette dustbin, replace the waste bag in the cuvette dustbin with a new one, and close the cuvette dustbin. The discarded waste bag should be sealed with a cover and properly disposed of.



Warning:

- 1) After the dustbin is emptied, make sure you reset the number of waste cuvettes. Otherwise, the warning message will remain.
 - 2) Since residual material in cuvettes may cause contamination, they must be placed in a waste bag. Dispose of waste cuvettes and waste liquid in accordance with local laws and regulations.
-

Note: The dimensions of the cuvette dustbin are: 482mm x 143mm x 140mm (length x width x height). You can purchase suitable garbage bags in accordance with local regulations, or purchase according to 2.8.

- Disposal of waste cuvette boxes

Waste cuvette boxes are placed in the waste cuvette box recovery area at the middle layer of the cuvette storage. See "4.2.4.1 Loading cuvettes" for details of the removal method. Waste cuvette boxes are

free of biological contamination and should be disposed of following the conventional garbage disposal methods.

4.6.5.2 Disposal of waste liquid

The thick and thin waste liquid outlets are located in the lower right section on the front of the analyzer, with marks. The thick waste liquid comes from the SampleArm and contains magnetic microspheres, samples, reagents, and system liquid. The sample has a high concentration and is highly polluting. The thin waste liquid comes from the waste liquid of the chamber, the washer, and the reagent area and contains magnetic microspheres, starters, system liquid, and condensate water, but does not contain sample, and is less polluting.

Waste liquid can be stored in dedicated waste liquid tanks, or you can dispose of the waste liquid in accordance with local laws and regulations. You can check the status in the **[Home]** and **[Status]** interfaces. When the waste is marked as "Full", please dispose of it in time.



Warning:

- 1) Please wear gloves when disposing of waste!
 - 2) Dispose of waste liquid in accordance with local laws and regulations.
 - 3) See "6.4 Monthly maintenance" for details of regular washing of the waste liquid tanks.
-

Chapter 5 Software functions

5.1 Introduction to software interface

Double-click the user software icon on the desktop after you log in to the Windows operating system. A login dialog will pop up in the interface after the software starts up. Enter your username and password, then click the <Login> button to access the main interface of the user software. The structure of each part of the software interface is introduced below.

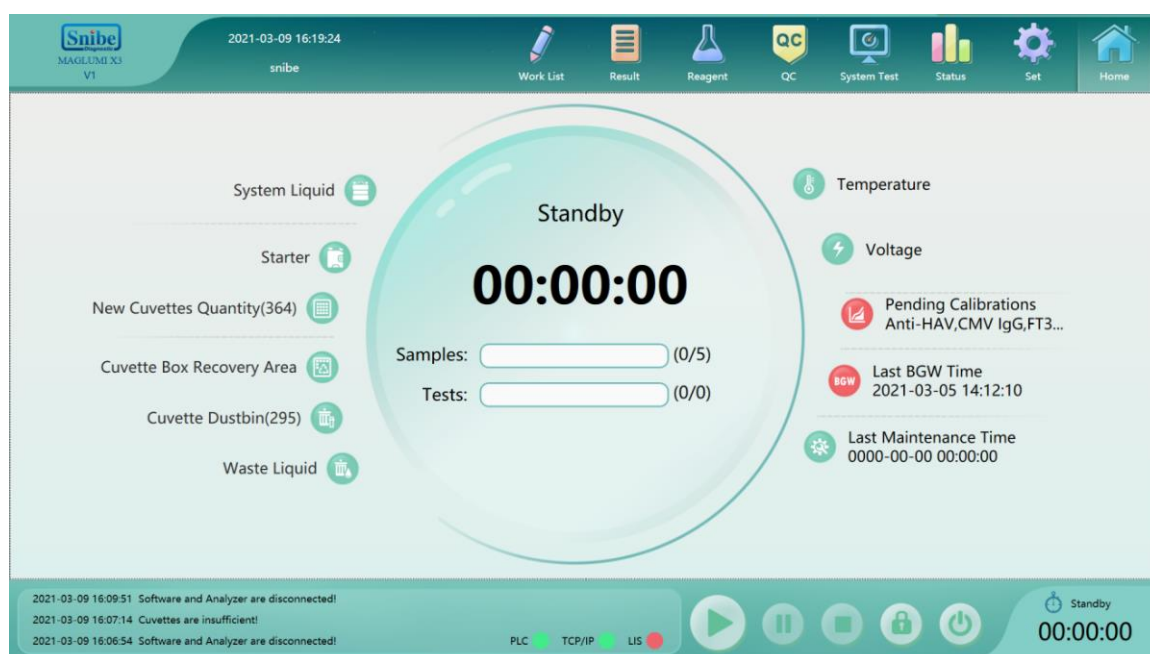


Fig. 5.1-1 User software interface

5.1.1 Main menu

The main menu shows eight menu buttons.



Fig. 5.1-2 Main menu of user software

Descriptions of button functions are as follows.

Button	Function
Work List	Register samples and QC tests, run checks and other procedures.
Result	Check or delete sample test results, perform remeasuring or other operations.

Reagent	Check reagent information, register reagent calibration, check calibration results or perform other operations.
QC	View QC results, QC charts and perform other operations.
System Tests	View system tests, system test results and perform other operations.
Status	Check analyzer temperature, voltage, consumables, etc.
Set	Set assay info, profile info, networking, language, etc.
Home	Display the status of the current part of the analyzer, test countdowns, etc.

5.1.2 Interfaces

The interfaces are used to show the details of each main function of the main menu and each interface is named after a main function of the main menu. The following figure shows the **[Work List]** interface.

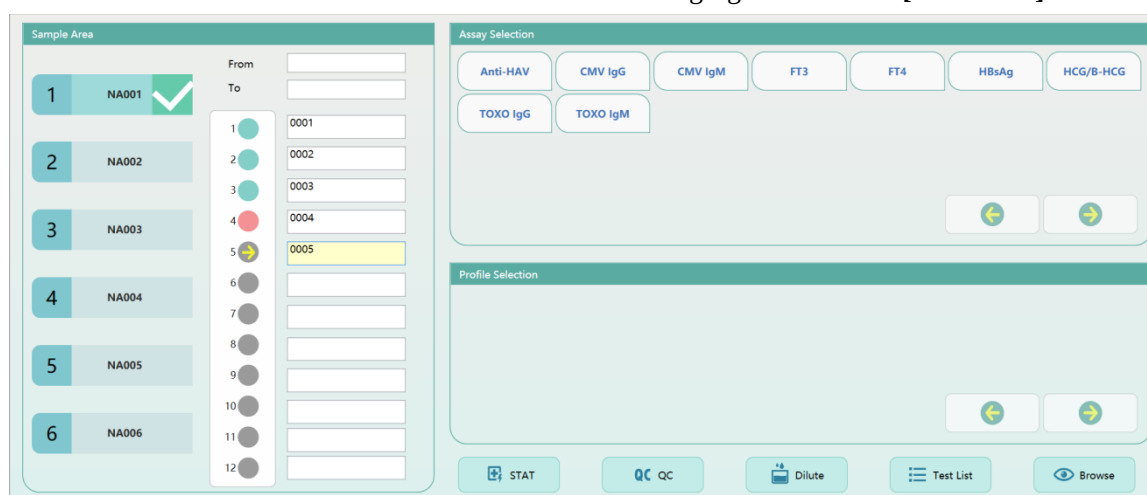


Fig. 5.1-3 [Work List] interface

5.1.3 Submenus

Click the **<Set>** button on the main menu to access a function submenu. The submenu is named after the main menu button. The following figure shows the **[Set]** menu.

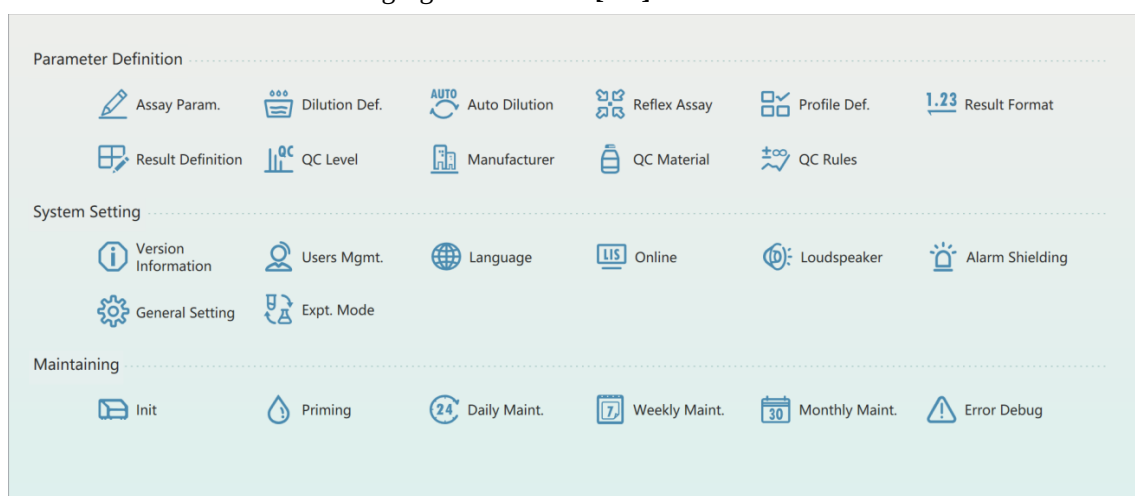


Fig. 5.1-4 [Set] menu

5.1.4 Operation bar

The operation bar includes multiple shortcut buttons and status displays.

1. A shortcut button is used to quickly run a certain function.

- : <Warning Message> button. After the analyzer gives an alarm, this button alerts the user of the nature of the error. The details can be viewed by clicking this button.
- : <Start> button. After sample registration, calibration registration and QC registration are completed, click this button and the software will start to send test command to the analyzer to start tests.
- : <Pause Cycle> button. If you need to pause a test during the analyzer's test process, click the button and the analyzer will pause the action.



Note:

Do not keep the system in the cycle paused status for a long time. Otherwise, it may affect the result of some tests.

- : <Stop> button. If you need to stop a test when the test is in process, click this button and the software will send a stop command to the analyzer to stop the analyzer immediately.



Note:

It is not recommended for the user to use this function, unless required under special conditions, such as the analyzer developing a fault. All the tests that did not finish the test process become invalid after they are stopped.

- : <Lock> button. Click this button and the software will display the lock screen interface, where you can switch user or lock the software. If you retype your username and password, you can log in to the software again. When the user software is locked, communication between the user software and the analyzer carries on as normal and the analyzer does not stop testing. The interface displays the current system time, the test countdown, the analyzer status, the number of samples, the number of tests, and error messages. Click the alarm light icon in the lower right corner to turn off the alarm loudspeaker.

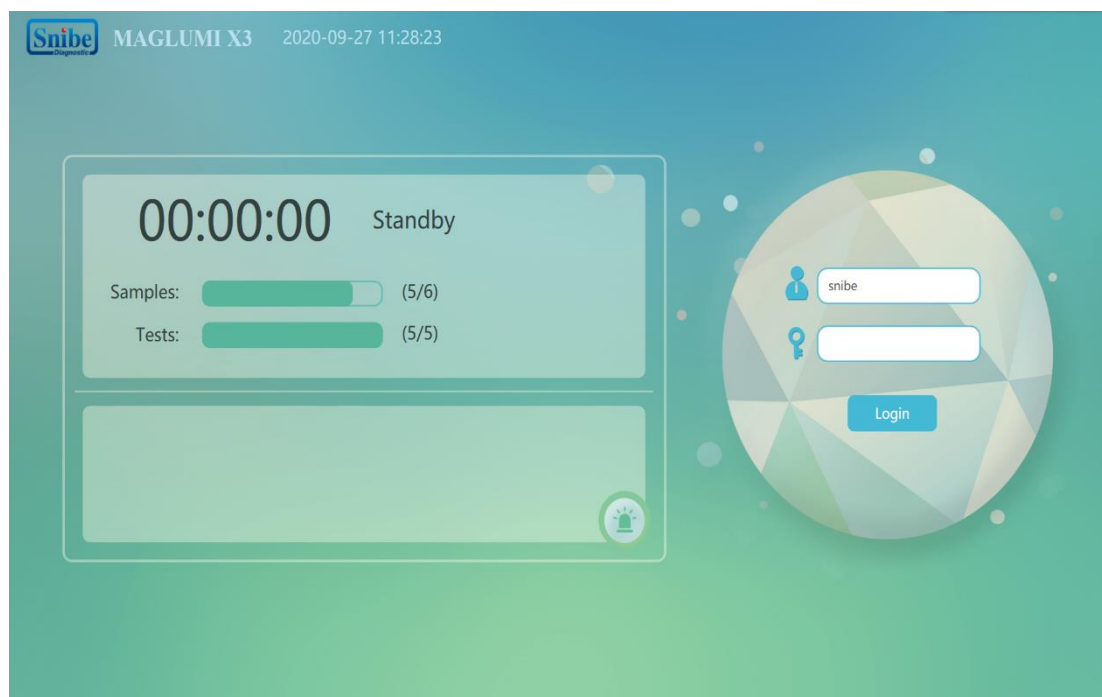


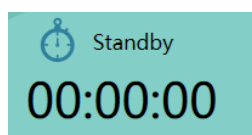
Fig. 5.1-5 Lock screen interface

- : <Shutdown> button. Click this button to exit the operating software.

2. The status display is used to quickly browse the current status of some analyzers.



- LIS status indicator (LIS connection status). The indicator lights up green when the LIS server is successfully connected and lights up red when the connection fails.
- PLC status indicator (lower computer connection status). The indicator lights up green when the lower computer is successfully connected and lights up red when the connection fails.
- TCP/IP status indicator (computer network port status). The indicator lights up green when the network port is enabled successfully and lights up red when the network port fails.



- Analyzer status: Analyzer statuses include "Connecting", "Init", "Standby", "Testing", "Maintaining", "Pause Cycle", "Continue", "Stop", "Analyzer Fault", and "System Test".
- Test countdown: Displays the countdown until the end of the testing process by the analyzer.

5.1.5 Dialogs

Dialogs are boxes which pop up after a certain button in the interface is clicked. One dialog box can open one or multiple subdialogs. Dialogs include function dialogs and information confirmation dialogs.

5.1.6 Button

A menu, interface, or dialog box contains one or more buttons that are used to trigger a function. These include three commonly seen buttons, namely <OK>, <Cancel> and <Return>. <OK> means to confirm completion of the operation in the dialog prompt; <Cancel> means to discard the operation in the dialog prompt; <Return> means to directly exit the dialog without performing any operation on the information in the dialog prompt.

5.2 [Work List] function description

Click the <Work List> button on the main menu to access the [Work List] interface. The [Work List] function is used for registering tests, checking the test list, and other operations.

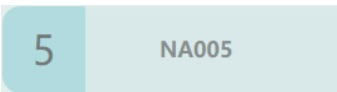
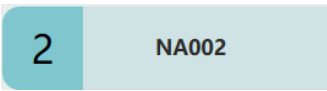
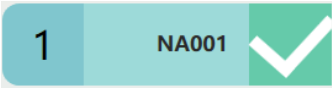
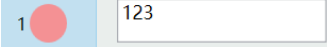
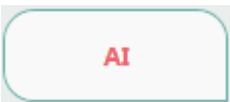
5.2.1 Sample registration

The screenshot displays the 'Work List' interface, which is divided into several sections:

- Sample Area:** A list of sample slots numbered 1 to 12. Slot 1 is selected and marked with a green checkmark. Slot 5 is highlighted in yellow. To the right of the sample list is a 'From' and 'To' range selector and a column of colored circles (green, red, yellow, grey) corresponding to the sample slots.
- Assay Selection:** A section containing buttons for various assays: Anti-HAV, CMV IgG, CMV IgM, FT3, FT4, HBsAg, HCG/B-HCG, TOXO IgG, and TOXO IgM. Navigation arrows are present at the bottom right of this section.
- Profile Selection:** A large empty box for selecting a profile, with navigation arrows at the bottom right.
- Bottom Bar:** A row of five buttons: STAT, QC, Dilute, Test List, and Browse.

Fig. 5.2-1 Work list

5.2.1.1 Legend description

Region	Legend	Description
Sample area		Sample rack is not loaded
		A routine sample rack is loaded
		Sample rack is selected
		Routine sample
		STAT sample
Assay selection area		The reagent area contains a reagent kit for this assay and the number remaining is greater than 0
		The reagent area contains a reagent kit for this assay and the number remaining is 0
		The reagent area does not contain a reagent kit for this assay

5.2.1.2 Registration methods

1. Registration of single sample

Select the sample rack to be edited and then enter the sample IDs one by one at the sample tube position (unless there is a barcode, in which case these steps are not needed). Select a sample ID, select the test assay in the Assay Selection Area, or use the package function to quickly select multiple test assays to finish registering the sample. If the sample is for the STAT purpose, click the position of the corresponding STAT sample tube and enter the sample ID (not needed if there is a barcode). Then select the assay for STAT and click the <STAT> button to prioritize the STAT sample for tests.

If the sample needs to be diluted, first select the assay to dilute, then click the <Dilute> button to select the dilution ratio.

To use the LIS feature, you only need to correctly configure and connect the LIS server. After the sample ID is entered, the analyzer will automatically retrieve the sample test assays from the LIS server.

2. Registration of batch samples

Select one or more sample racks in "Sample Area", then select the entire sample rack and enter the starting and ending sample IDs. Then press the Enter key to batch register the sample IDs on multiple sample racks. Then select an assay or use the profile function to select assays to register the samples.

See "1. Registration of single sample" for details of applying for test assays using the LIS function.

Batch registration is not supported for STAT and dilution samples.

5.2.2 Work list

Click the <Test List> button in the [Work List] interface to bring up the [Test List] interface. The functions of the interface include: Displaying the registered sample test list, modifying test replication, deleting already registered sample tests, and exporting the sample test list information to an Excel file.

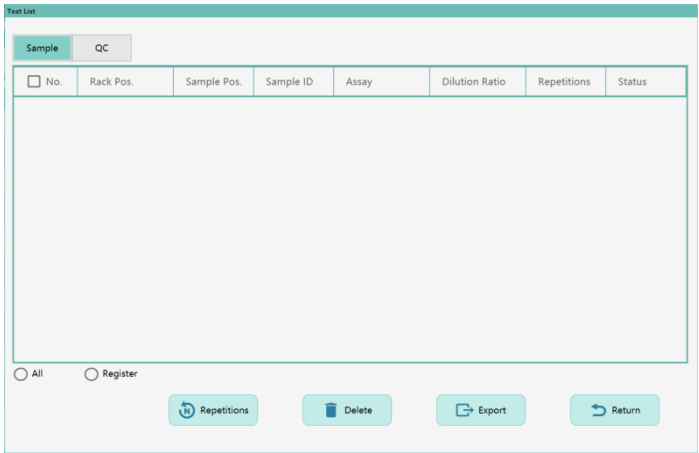


Fig. 5.2-2 Test list

1. Modify replications

You can click the "+" or "-" in the "Repetitions" column of the list to modify the replications of a single test. If you need to batch modify the replications for part of the registered assays, check the tests to modify repetitions of, click the <Repetitions> button to go to the [Modify Replications] interface, and modify the replications with "+" or "-" or manually input the replication value in the "Modify Value" column of the list.

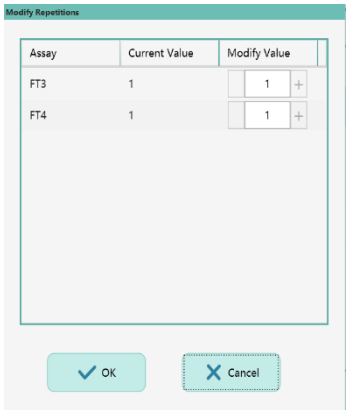


Fig. 5.2-3 Modifying replications

2. Delete a test

Check the sample test to delete and click the **<Delete>** button to delete the test. The deletion operation is only valid for registered tests.

3. Export a test

Select the sample test to export and click the **<Export>** button to export the test information to an Excel file.

5.2.3 Browsing

In the **[Work List]** interface, click the **<Browse>** button to access the **[Browse Sample Registration Info]** interface. In the dialog, you can browse all the test assays registered for each sample, the number of registrations for each assay, and the number of remaining tests for the reagent.

5.3 [Result] function description

Click the **<Result>** button to access the **[Result]** interface. [Result] functions are used to view, search, delete, view details of, remeasure, put online, recalculate, export and print results. This section mainly introduces functions of the **[Result]** menu one by one.

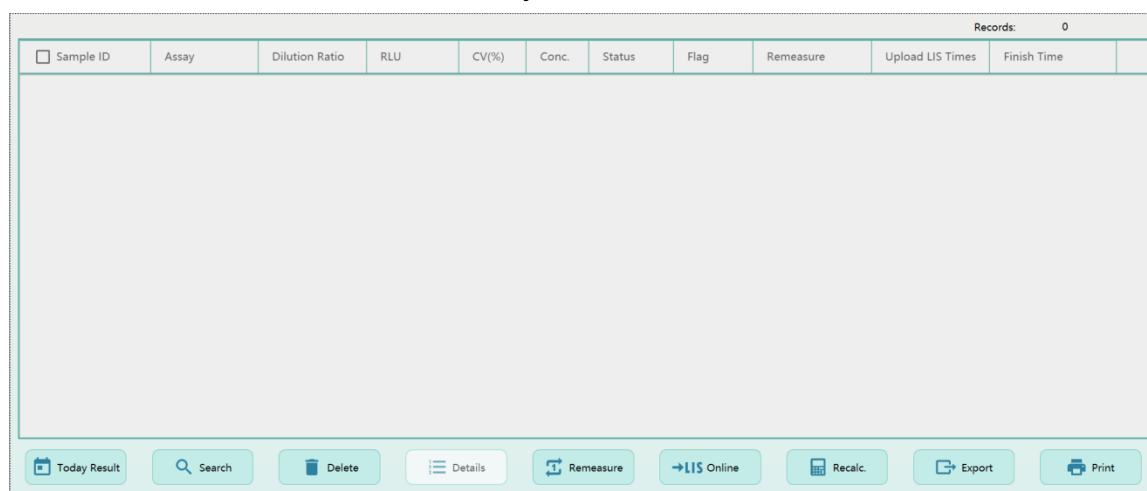


Fig. 5.3-1 Test results

5.3.1 Search results

In the **[Result]** interface, click the **<Search>** button and configure the search conditions in the **[Result Search]** interface to display test results that meet the criteria. Simply click **<Today Result>** to display all the test results for the current day. "Records" in the upper right corner displays the total records in the current list.

5.3.2 Deleting results

In the list of test results in the **[Result]** interface, select the test results you want to delete. Click the **<Delete>** button to finish deleting.



Note:

Please proceed with caution as deleted lab test data cannot be recovered.

5.3.3 Result details

In the test results list of the **[Result]** interface, select one test result and double-click this test result or click the **<Details>** button to open the **[Result Details]** interface to view the result details list.

Result Details

Result Information

Sample ID: 0001 Position: NAO01-1

Assay: Anti-HAV Dilution Ratio:

Conc.: 7.84 Conc. CV: 0

RLU: 29600 RLU CV: 0

Status: Finish Register Time: 2020-09-27 11:01:23

Reagent Information

Lot-No.: 1218 Kit-No.: 1218

Master Curve ID: 20181207101112 Unit: ng/mL

Normal Range: 0.69 ~ 2.15

Measuring Rang: 0.06 ~ 10

No.	RLU	Conc.	Status	Flag	Finish Cycle	Finish Time
1	29600	7.84	Finish	C ; >	193	2020-09-27 11:06:48

Flag

Calibration expired [C]

Above Normal Range [>]

Export

Save

Convert to QC

Return

Fig. 5.3-2 Result details

Click the **<Convert to QC>** button to open the **[Sample Convert to QC]** interface, select the QC lot-no. to be converted in Lot-No., and click **<Save>**. The software will transfer the assay results to the current day's results for the corresponding QC lot.

Fig. 5.3-3 Sample convert to QC

Click the **<Export>** button to export the result details to an Excel file.

5.3.4 Remeasuring results

In the list of test results in the **[Result]** interface, select the test results and click the **<Remeasure>** button. When the remeasuring is successfully registered, a message will pop up: "Registration is successful." In the newly added samples for remeasuring in the list of test results, review the samples for remeasuring which are flagged with an asterisk.



Note:

1. Remeasuring can only be carried out on samples that have been tested on the same day.
2. The sample to remeasure must be stored in the sample area and not be pulled out.
3. The assay to remeasure must be consistent with the original assay. If the test conditions change, you need to re-register the assay.

5.3.5 Sending results

After the LIS function is enabled and the LIS server is successfully connected, check the test results to send in the **[Result]** interface, and click the **<Online>** button to upload the test results to the LIS server.

5.3.6 Recalculate

If you need to recalculate the tested concentration results based on a new calibration curve, check the tests to recalculate in the test result list in the **[Result]** interface, and click the **<Recalc.>** button to run the recalculation operation. The test result list will display the test result values after the result for the corresponding assay is recalculated.

**Note:**

When the concentration is changed by the recalculate function, it cannot be restored to its original value. Please proceed with caution!

5.3.7 Exporting results

In the **[Result]** interface, select the test results of the assays to be exported in the test results list and click the **<Export>** button to export the test results to an Excel file.

5.3.8 Printing results

In the **[Test Results]** interface, select the test results to be exported in the test results list and click the **<Print>** button to open the **[Print Preview]** interface. You can view a print preview. After confirmation, click the print icon in the upper left corner of the dialog box to print the selected test results.

The screenshot shows a 'Print Preview' window with a title bar and a toolbar. The main content area displays a 'Sample Result Report' table. The table has columns for Sample ID, Assay, RLU, CV(%), Conc., Flag, and Finish Time. There are two rows of data. The top row shows Sample ID 0001, Assay Anti-HAV, RLU 29600, CV(%) 0, Conc. 7.84 ng/mL, Flag C ; >, and Finish Time 2020-09-27 11:41:41. The bottom row shows Sample ID 0001, Assay CMV IgG, RLU 29600, CV(%) 0, Conc. (blank), Flag (blank), and Finish Time 2020-09-27 11:39:58. A 'Return' button is located at the bottom right of the window.

Sample ID	Assay	RLU	CV(%)	Conc.	Flag	Finish Time
0001	Anti-HAV	29600	0	7.84 ng/mL	C ; >	2020-09-27 11:41:41
0001	CMV IgG	29600	0			2020-09-27 11:39:58

Fig. 5.3-4 Print preview report

5.4 [Reagent] function description

Click the **<Reagent>** button on the main menu to enter the **[Reagent]** interface. The **[Reagent]** functions are used to view reagent information and calibration information, register calibration, and view calibration results and reagent consumption. This section mainly introduces each function of **[Reagent]** one by one.

5.4.1 Reagent information

In the [Reagent] interface, you can view information such as whether the reagent kit is inserted, as well as status, name, magnetic sphere mixing time and reagent data of the reagent area.

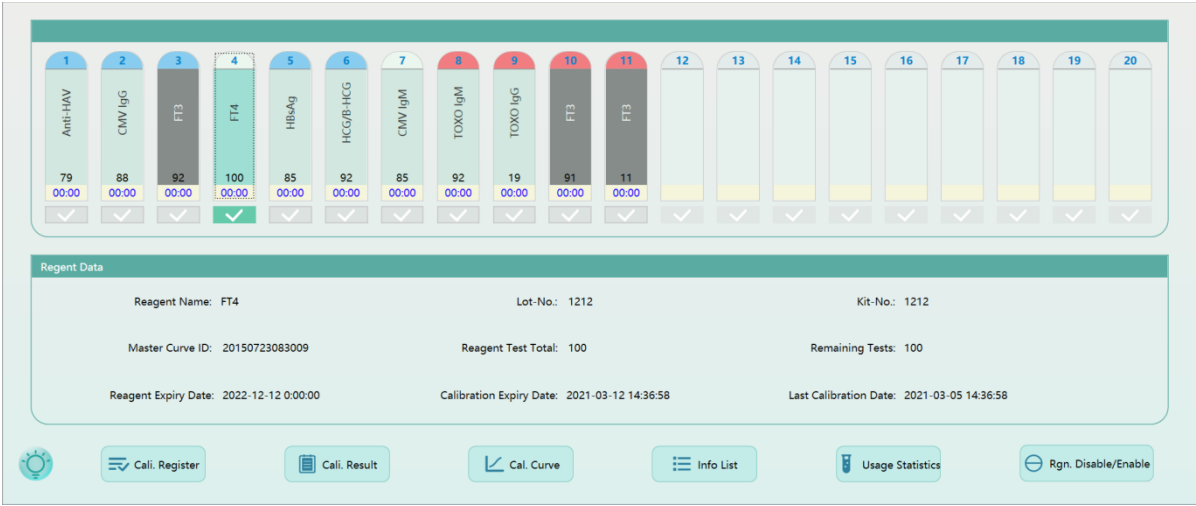
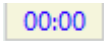


Fig. 5.4-1 [Reagent] interface

5.4.1.1 Reagent color definitions



Legend	Description
	The number "1" means that the reagent is in the no. 1 reagent position in the reagent area, the background color represents the current calibration status as "there is a valid calibration," and the calibration status color has the following meanings:
	: Has valid calibration status
	: No valid calibration
	: Has new calibration
	: Calibrating
	: Calibration expired

	"FT3" represents the current reagent name (ERROR is displayed when an insertion error occurs); "99" represents the number of remaining tests of the current reagent; "🔒" means the current reagent is in use and that the reagent kit cannot be removed; the background color represents the current status as "Reagent is not expired and not selected". The reagent status colors have the following meanings.  : Reagent expired  : Position has no reagent inserted  : Reagent is not expired and not selected  : Position is selected
	"00:00" represents the countdown for the current reagent magnetic microsphere mixing.
	A check indicates that the current reagent is selected

5.4.1.2 Definition of time status

The magnetic microsphere mixing time defaults to 30 minutes. If the countdown timer is less than 30 minutes and you click <Start> in the operation bar, it will prompt you that the magnetic microspheres are not fully mixed. The test can only be started if the user clicks the <OK> button.

5.4.1.3 Reagent usage sequence

There is more than one assay reagent kit loaded in the reagent area and the order of use is as follows:

- 1) Use the reagent in the ascending order of lot-no.
- 2) Use the reagent with a smaller number of remaining tests.
- 3) Use the reagent from left to right.

5.4.1.4 Reagent data

You can view the basic information of selected reagent in the "Reagent Data" area.

5.4.2 Calibration registration

In the [Reagent] interface, select the reagents that need to be calibrated (multiple selection allowed) and then click the <Cali. Register> button to finish registering the calibration assays. For details, see "4.3.1 [Calibration assay registration] function description."

5.4.3 Calibration results

On the [Reagent] interface, click the <Cali. Result> button to bring up the [Cali. Result] interface. The upper list shows the calibration results for all reagents. Select a calibration result and the lower list shows the corresponding details. The operation mode is similar to that for test results. For details, refer to "5.3 [Result] function description."

Fig. 5.4-2 Calibration results

5.4.4 Reagent details

In the [Reagent] interface, double-click a reagent to go to the [Reagent Details] interface. The dialog box displays basic information, curve information, calibration information, and related operations of the reagent.

Fig. 5.4-3 Reagent details

The following describes the meaning of the parameters of the calibration info:

Content	Description
Calibrator 1 Calibrator 2	Test results of two calibrations.
RLU	Measured RLU of calibration, up to three replicate tubes.
Mean RLU	The mean of the three measured RLU values.
RLU CV	The coefficient of variation of the three measured RLU values.
RLU Dev (%)	The percentage difference between the mean of the three measured RLU values and the factory standard value.
Expected concentration	The calibrated concentration set value.
Calculated concentration	Press <Cal. Curve> to calculate the concentration of the most recent calibration curve or a new calibration curve.
Concentration Dev(%)	The percentage difference between the expected concentration and the actual calibrated value.
Range of concentration tolerances	The calibrated measured concentration allows for a range of variation.
RLU Tolerances	The allowed variation range of the measured calibration RLU values.

5.4.4.1 Calibration result operations

After the reagent calibration test has finished, open the **[Reagent Details]** interface to view the calibration information and the work curve information. If the result is abnormal, the corresponding calibration info parameters are displayed in red. The user can judge from the actual situation and perform the following operations:

1. If the calibration result is acceptable, check "Validate" and click **<OK>**. The calibration result will be accepted by the system.
2. If the calibration result is not acceptable, check "Reject" and click **<OK>**. The calibration result will be rejected and deleted by the system and you need to perform the calibration again.
3. If you check "Recalc. Associated Samples" and then click **<OK>**, the latest calibration curve will be used to recalculate the concentration using all the sample results of this assay (only applicable to the sample results and QC results for the current day).
4. Click the **<Print>** button to print the current calibration information.
5. You can modify the Remaining Tests, Conc. Tolerances, and RLU Tolerances, with the maximum Remaining Tests value being 100.

5.4.4.2 Calibration history

Click the **<Cali. History>** button in the **[Reagent Details]** interface to open the **[Calibration History]** interface.

Calibration History								
Reagent Name: CMV IgM			Lot-No.: 20191011927					
Master Curve ID: 20180716143945			Kit-No.: 1212					
Date	Cali 1 Mean	Cali 2 Mean	Cali 1 CV	Cali 2 CV	Cali 1 Dev	Cali 2 Dev	Status	Kit-No.
2021-03-05 14:43:43	36052	242726	0	0	0	0	Valid	1212

Fig. 5.4-4 Calibration history

Select a record in the calibration history and click **<Recover Cali.>** to overwrite the calibration data of the current reagent with the selected calibration data. Click the **<Delete>** button to delete the record in the calibration history.

Double-click or select a record in the calibration history and click **<Details>** to access the **[Calibration History Details]** interface to view details of the calibration.

Click the **<Print>** button to print the current calibration history.

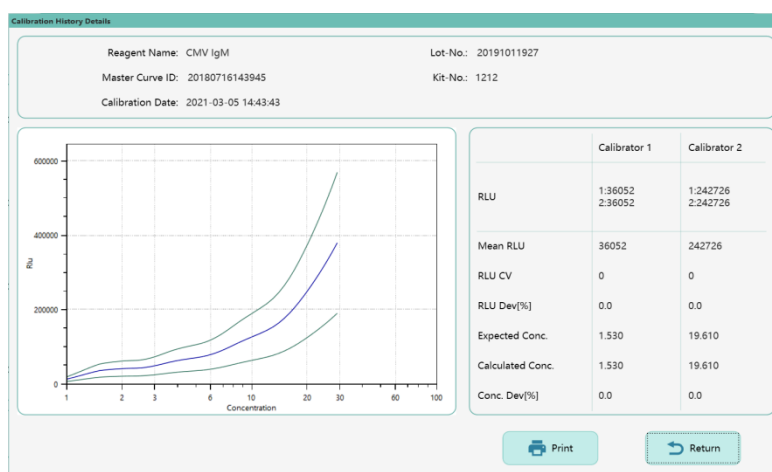


Fig. 5.4-5 Calibration history details

5.4.5 Reagent info list

In the **[Reagent]** interface, click the **<Reagent Info List>** button to open the **[Reagent Info List]** interface. A list of all of the reagents ever loaded into the analyzer can be viewed.

You can enter the assay, lot-no., kit-no., or check "Loaded", and then click the **<Search>** button to filter the reagent information that meets the requirements.

You can select one or more reagent kit entries and click the **<Delete>** button to delete the selected reagent entries, click the **<Export>** button to export the selected entries to an Excel file, and click the **<Print>** button to print the selected reagent entries.

Reagent Info List

Assay Lot-No. Kit-No. ☐ Loaded

☐ No.	Assay	Lot-No.	Kit-No.	Remaining Tests	Unsealing Days	Expiry Date

Fig. 5.4-6 Reagent info list

5.4.6 Reagent consumption statistics

On the **[Reagent]** interface, click the **<Reagent Consumption Statistics>** button to open the **[Reagent Consumption Statistics]** interface. You can view the statistics for reagent consumption for assays on the analyzer over a period of time.

Select the start and end times, then click the **<Search>** button and the actual consumption statistics list shows the consumption statistics for the assays over this period of time.

Select one or more information records, then click **<Export>** to export the selected information to an Excel file. Click **<Print>** to print out the selected reagent consumption statistics.

Reagent Consumption Statistics

Start End

☐ No.	Reagent Name	Reagent Consumption

Fig. 5.4-7 Reagent consumption statistics

5.4.7 Reagent Disable/Enable

In the **[Reagent]** interface, select a reagent kit and click the **<Reagent Disable/Enable>** button to disable or enable the reagent kit for use.

Select the reagent kit to disable, and then click the <Reagent Disable/Enable> button. The reagent kit will show a disabled icon "".

Select a disabled reagent kit and click the <Reagent Disable/Enable> button. The disabled icon on the reagent kit will disappear, indicating that this reagent kit allows to be used.

5.5 [QC] function description

Click the <QC> button on the main menu to access the [QC] interface. [QC] functions are used to manipulate QC results and view lot QC charts and monthly QC charts. This section mainly introduces each function of the [QC] interface one by one.

5.5.1 QC results

Click the <**QC Result**> tab on the [**QC**] interface to access the [**QC Result**] interface. This is used to search, delete, view details of, put online, recalculate, export and print QC results. The operation mode is similar to that for test results. For details, refer to "5.3 [**Result**] function description."



Fig. 5.5-1 QC results

5.5.2 QC analysis

Click the <QC Analysis> tab in the [QC] interface to go to the [QC Analysis] interface to view and print QC charts.

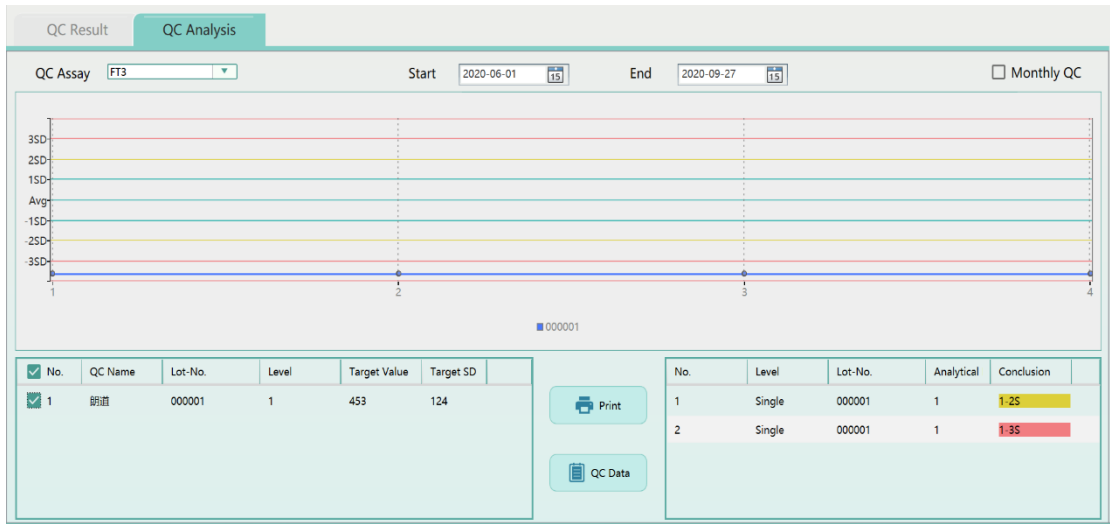


Fig. 5.5-2 QC Chart

5.5.2.1 Viewing QC charts

- Lot QC charts
Enter the search criteria at the top of [QC Analysis], and the search results will be displayed in the QC result list in the lower-left area. Check a data entry in the list, and the QC chart at the top will show the corresponding data curve based on the lot-no.
- Monthly QC charts
If you need to make a single month judgment on the QC results, you need to check the Monthly QC option when entering the search conditions, then you can judge the QC results of a single month and generate the corresponding QC chart. For the remaining operations, please refer to [Lot QC charts] operations above.

5.5.2.2 QC results operations

In the lower left QC results list, select a QC message, then click the **<QC Result>** button. The **[Ignore QC Data]** interface pops up and the QC result details corresponding to the QC lot-no. are displayed in the list. Check the QC results you want to ignore in the list and click the **<Ignore>** button, then the results selected to ignore in the QC chart and the QC conclusion list will be recalculated and displayed.

Select a QC message in the lower left QC results list and the corresponding QC conclusion can be viewed in the QC conclusion list at the bottom right. The QC conclusions are judged according to the user-predefined QC rules. See "5.8.1.11 QC rules" for details.

5.6 [System Test] function description

Click the **<System Test>** button on the main menu to access the **[System Test]** interface. [System

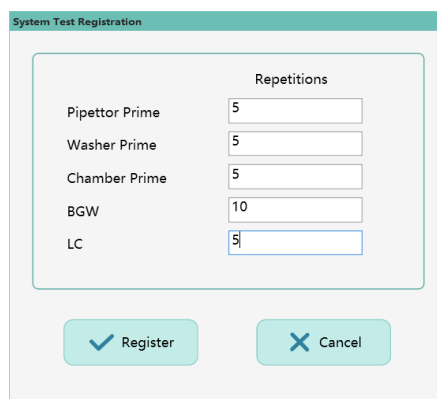
Test] functions are used to perform BGW and LC tests and view the BGW and LC test results. This section introduces the functions of the **[System Test]** interface.

5.6.1 System tests

5.6.1.1 Run a system test

In the **[System Test]** interface, click the **<Add>** button and the **[System Test Registration]** interface appears, as shown in the following figure.

In the **[System Test Registration]** interface, enter the numbers of BGW and LC tests and click the **<Register>** button to register the system test.



The 'System Test Registration' window contains a table with the following data:

	Repetitions
Pipettor Prime	5
Washer Prime	5
Chamber Prime	5
BGW	10
LC	5

At the bottom of the window are two buttons: 'Register' (with a checkmark icon) and 'Cancel' (with an 'X' icon).

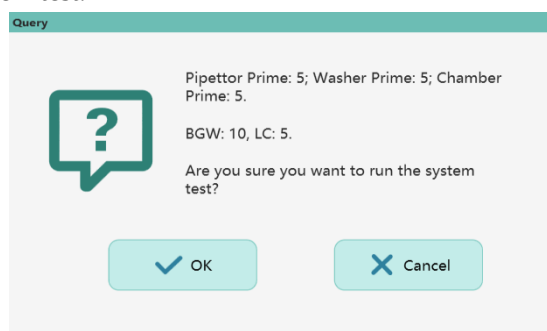
Fig. 5.6-1 System test registration



Note:

1. During a system test registration, you can register the pipettor, washer, and chamber, respectively. But you can't register a priming separately. Instead, you must include at least one system test to register a priming operation.
2. After you have registered priming successfully, the analyzer will first carry out the priming procedure and automatically start the system test procedure after the priming is completed.

After you click the **<Register>** button, the **[Query]** interface appears, showing the test details. If you click the **<OK>** button, the analyzer will perform the system test. If you click the **<Cancel>** button, the software will cancel the system test.



The 'Query' window displays the following information:

Pipettor Prime: 5; Washer Prime: 5; Chamber Prime: 5.

BGW: 10, LC: 5.

Are you sure you want to run the system test?

At the bottom are two buttons: 'OK' (with a checkmark icon) and 'Cancel' (with an 'X' icon).

Fig. 5.6-2 Query



Note:

1. Before performing an LC test, please load the LC solution to the corresponding positions in the reagent area. See "5.6.2 Loading LC solution" for details.
2. BGW and LC tests must be performed after the starter is replaced.

5.6.1.2 System test results

In the **[System Test]** interface, users can search, delete, view details of, export and print system test results. The operation mode is similar to that for test results. For details, refer to "5.3 **[Result]** function description."



Note:

Tests can be performed only when the system test results and CV values satisfy the requirements. It is not recommended to continue the sample testing if the values are abnormal (abnormal data is marked in red in the list of system test results). Otherwise, it may undermine the reliability of the test results.

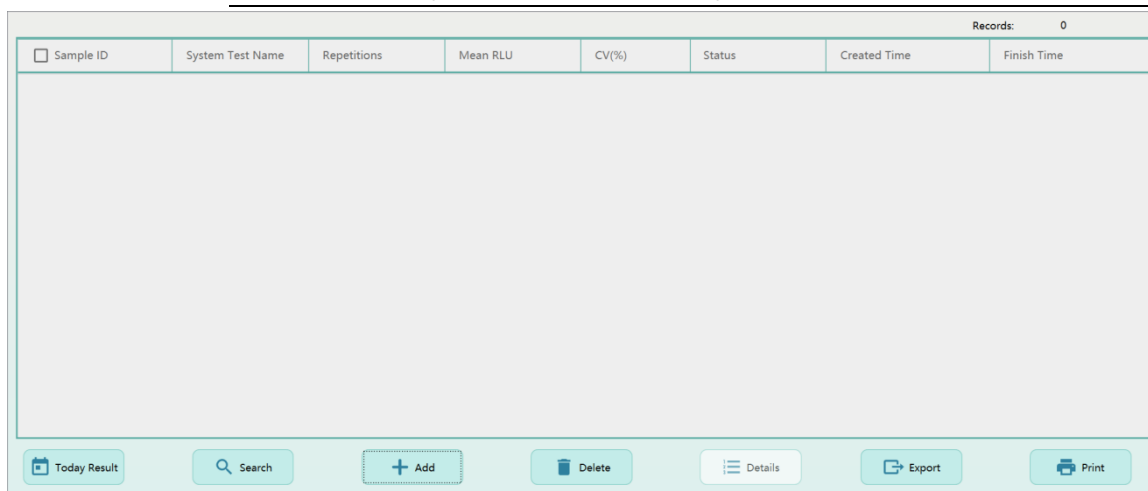


Figure 5.6-2-3 System test

5.6.2 Loading LC solution

The LC solution is used to test the pipetting accuracy performance of the pipettor of the analyzer. During an LC test, place the LC solution to the 4th position of the 20th channel of the reagent area.

Please strictly follow the instructions for preparing and storing the LC solution.

5.7 **[Status]** function description

Click the **<Status>** button on the main menu to enter the interface of **[Status]** and related functions, as shown in the following figure. This is used to display the status of each part of the analyzer.

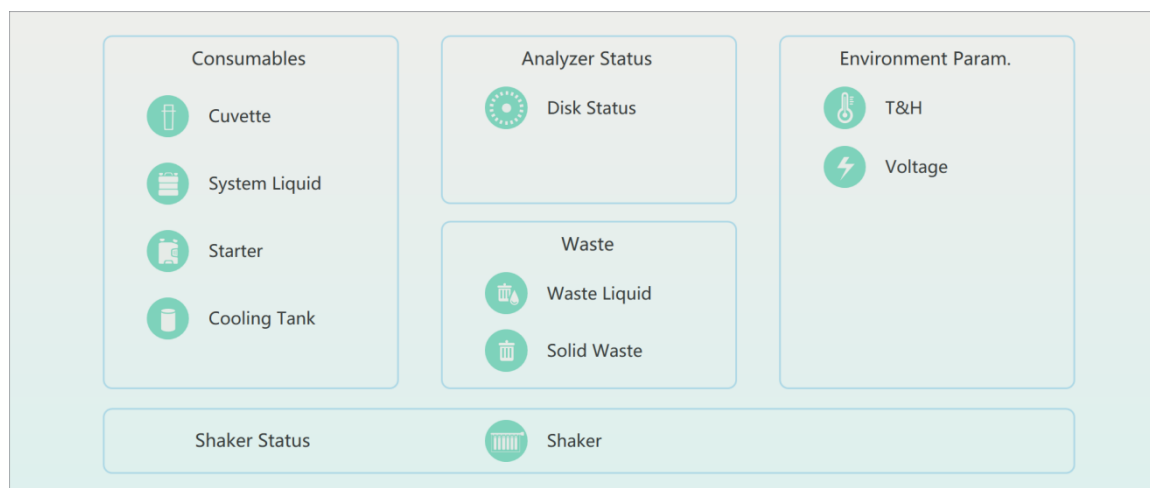


Fig. 5.7-1 Status

In the <Status> interface, you can view the statuses of the analyzer's consumables, environmental parameters, wastes and other components.

- When each status value is within the set range or the status is normal, the icon is displayed in green.
- When each status value reaches the warning range, the icon is displayed in yellow.
- When each status value exceeds the set range or the status is abnormal, the icon is displayed in red.

When a certain status represents an alert or an abnormality, the <Status> button icon flashes. To view the specific values, click on the corresponding icon to enter the details interface to view them. This section introduces each function of the [Status] menu one by one.

5.7.1 Consumables

In the consumables status area of the [Status] interface, you can view the statuses of cuvettes, system liquid, starters and cooling tank, as shown in the following figure. The following is an introduction of consumable statuses.

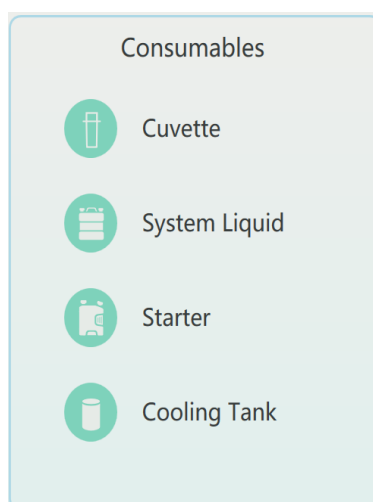


Fig. 5.7-2 Consumables

5.7.1.1 Cuvettes

Click the <Cuvette> tab in the [Status] interface to open the [Cuvette] interface, as shown in the following figure. This is used to display in real time the cuvette quantity in the current cuvette box and the consumption of cuvette boxes in the storage area in the cuvette storage and the recovery area. It is also used to set the cuvette quantity in the current cuvette box.

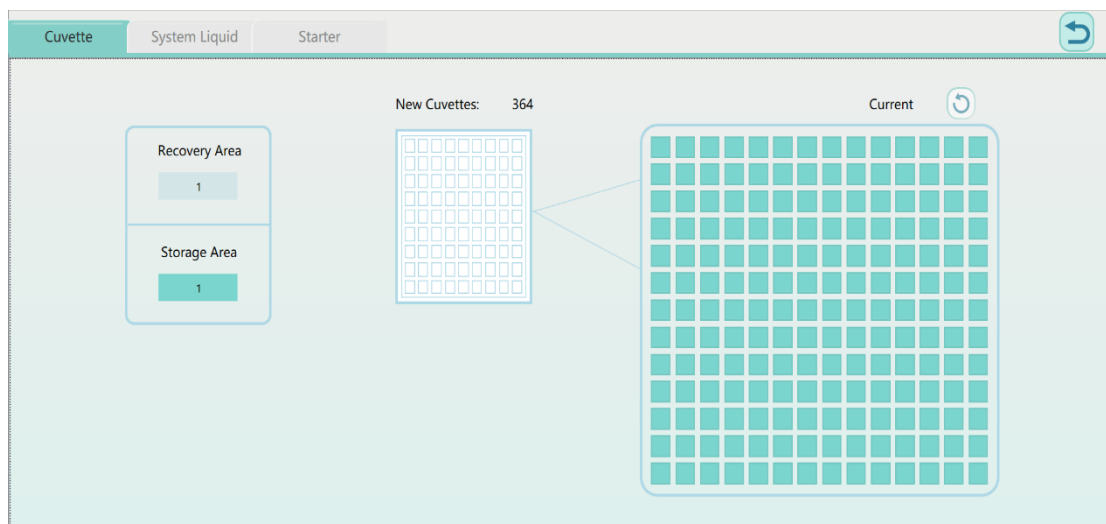


Fig. 5.7-3 Cuvettes

1. Explanation of displayed Cuvette information in the [Status] interface:

If the number of cuvettes > 72, the icon is displayed in green; if the number of cuvettes is 1–72, the icon is displayed in yellow; and if the number of cuvettes is 0, the icon is displayed in red.

2. The storage area statuses include the following:

Status	Description
	Cuvette boxes present
	No cuvette boxes

3. The recovery area status includes the following details:

Status	Description
	Cuvette boxes present
	No cuvette boxes

4. The cuvette statuses include the following:

Status	Description
	Cuvettes present
	No cuvettes

5. Reset the cuvette quantity:

In the current cuvette usage area on the right side of the [Cuvette] interface, select the initial gripping position of cuvettes and click the  button. The current total number of cuvettes then changes to the

corresponding value, and the number of cuvettes is reset.



Note:

1. The reset cuvette quantity should be the same as the actual cuvette quantity in the current cuvette box.
 2. This function is used only when cuvettes are manually added or a single cuvette is missing. If the initial gripping position of cuvettes is reset incorrectly, an error may occur when the analyzer grips cuvettes.
-

5.7.1.2 System liquid and starter

Click the <System Liquid> tab in the [Status] interface to open the [System Liquid] interface, as shown in the following figure. This is used to view the system liquid total volume in real time.

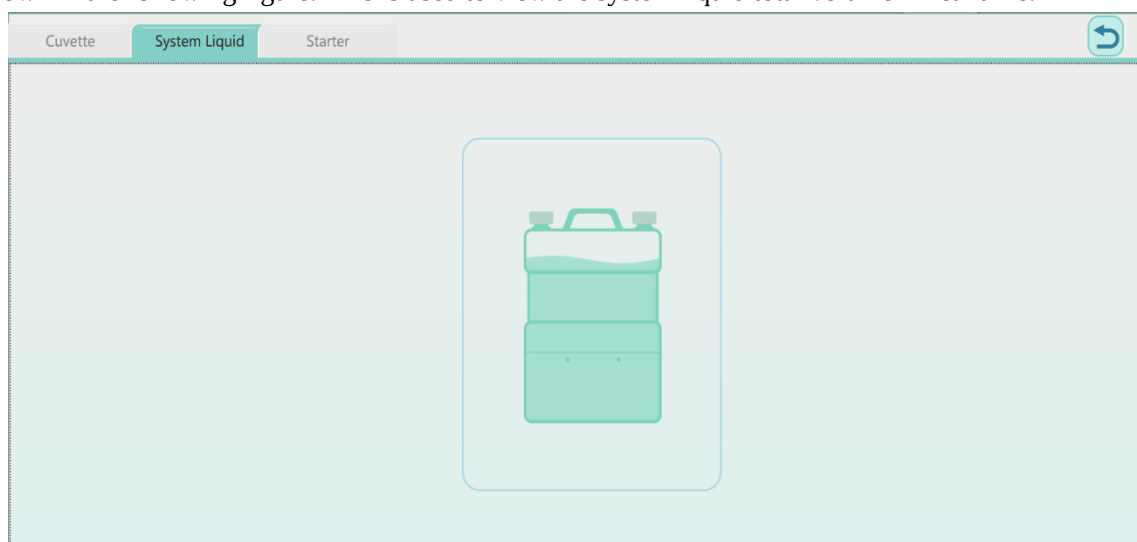


Fig. 5.7-4 System liquid

- When the total remaining volume > 20% of the total loadable volume, the icon is displayed in green.
- When the total remaining volume < 20% of the total loadable volume, the icon is displayed in yellow.
- When the waste liquid level is 0, the icon is displayed in red.

In the [Status] interface, you can click the <Starter> tab to go to the [Starter] interface, as shown in the following figure. This is used to view the total volume of starters 1 and 2 in real time.

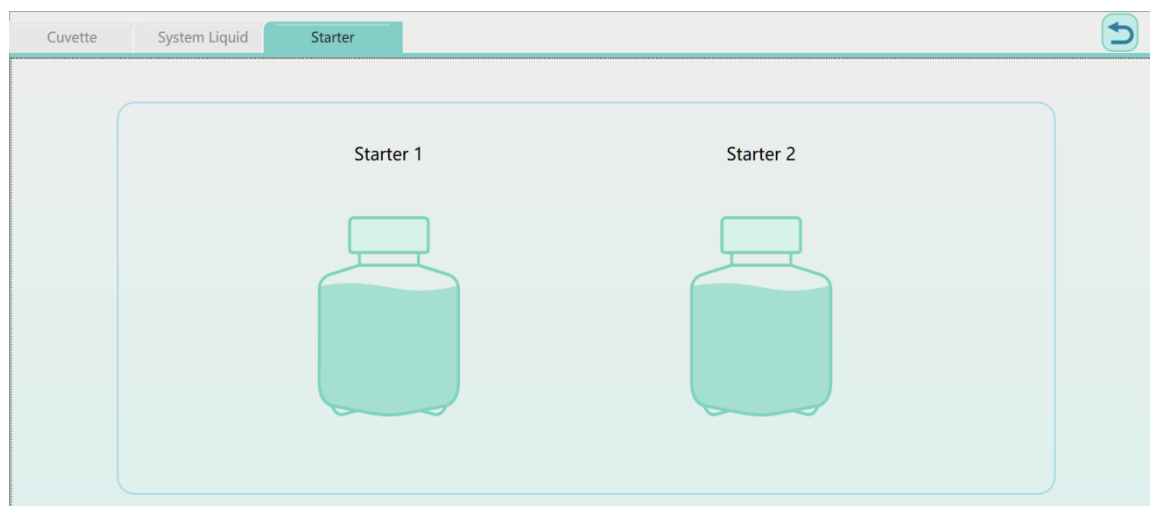


Fig. 5.7-5 Starter

- When the total remaining volume > 20% of the total loadable total, the icon is displayed in green.
- When the total remaining volume < 20% of the total loadable volume, the icon is displayed in yellow.
- When the total remaining volume is 0, the icon is displayed in red.

5.7.1.3 Cooling tank

In the consumable information area in the **[Status]** interface, you can view the status of the cooling tank. When the liquid level of the cooling tank is normal, the icon is displayed in green; when the liquid level of the cooling tank is abnormal, the icon is displayed in red.

5.7.2 Environmental parameters (temperature, voltage and humidity)

Click the environmental parameter message area of the **[Status]** interface to view messages on the ambient temperature and humidity of the analyzer, as well as the voltage of each analyzer component. The following is an introduction to the environmental parameters.

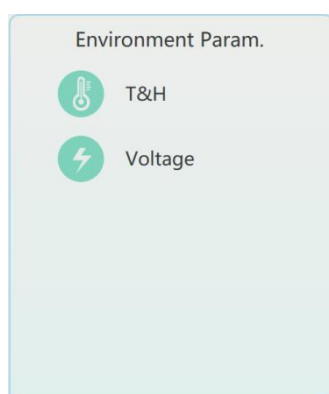


Fig. 5.7-6Environmental parameters

In the [Status] interface, you can click the <T&H> tab to go to the [T&H] interface, as shown in the following figure. This is used to view the temperature or ambient temperature of each analyzer component in real time; the unit of temperature is configurable. The value displayed inside the icon shows the real-time status. When the status value is within the normal range, the icon is displayed in green; when the status value is outside the normal range, the icon is displayed in red. The voltage and humidity interfaces are similar to the temperature interface.

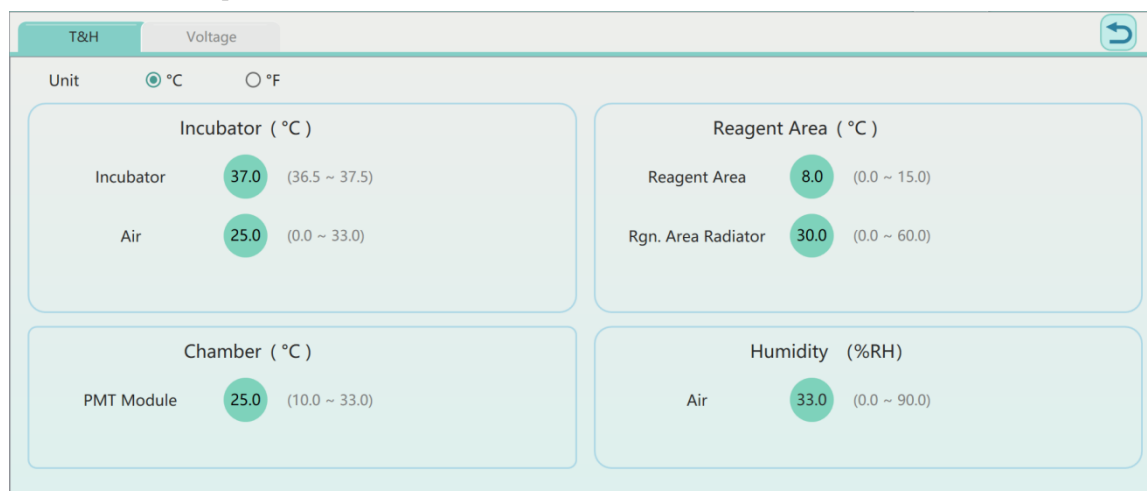


Fig. 5.7-7 T&H

5.7.3 Waste

Click the waste message area of the [Status] interface to view status messages for the analyzer's waste liquid and solid waste, as shown in the following figure. The following is an introduction of waste.

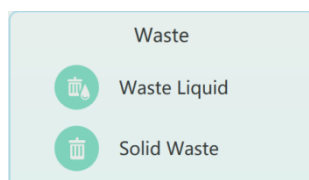


Fig. 5.7-8Waste

5.7.3.1 Waste liquid

Click the <Waste Liquid> tab in the [Status] interface to access the [Waste Liquid] interface, as shown in the following figure. This is used to view the status of the liquid level of the analyzer's light waste liquid and thick waste liquid in real time.



Fig. 5.7-9 Waste liquid

When the liquid level is normal, the icon is displayed in green; when the liquid level is full, the icon is displayed in red;

5.7.3.2 Solid waste

Click the **<Solid Waste>** button on the **[Status]** interface as shown in the following figure. It is used to view the real-time statuses of the cuvette dustbin and empty cuvette boxes.



Figure 5.7-10 Solid waste

1. When the number of waste cuvettes is less than 400, the icon is displayed in green; when the number of waste cuvettes is 400–520, the icon is displayed in yellow; when the number of waste cuvettes is greater than 520, the icon is displayed in red. You can click the **<Reset>** button to zero the corresponding waste status. Then seal the filled waste bag and dispose of it properly.
2. When the number of empty cuvette boxes is 0, the icon is displayed in green. When the number of empty cuvette boxes is 1, the icon is displayed in yellow.



Warning:

1. After emptying the waste, be sure to reset the corresponding waste status. Otherwise, the warning message will remain.

2. Since residual material in cuvettes may cause infection, they must be placed in a waste bag. Dispose of waste in accordance with local laws and regulations.

5.7.4 Analyzer status

In the [Status] interface, you can click the analyzer status information area to view the disk statuses of the analyzer, as shown in the following figure. The following is an introduction of analyzer statuses.

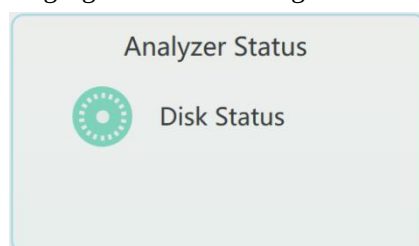


Fig. 5.7-11 Analyzer statuses

5.7.4.1 Disk status

Click the <Disk Status> button in the [Status] interface to open the [Disk Status] interface, as shown in the following figure. This is used to view cuvette position and status messages for each analyzer disk in real time.

In the center of the [Disk Status] interface, the three disks displayed are the sample disk, the chamber, and the washer in the order of left to right. The square disk is the incubator. The current statuses of cuvettes on the disk are differentiated by color. A legend is displayed on the left of the interface. The message list on the right shows messages on the test currently being run on the cuvette at the selected position.

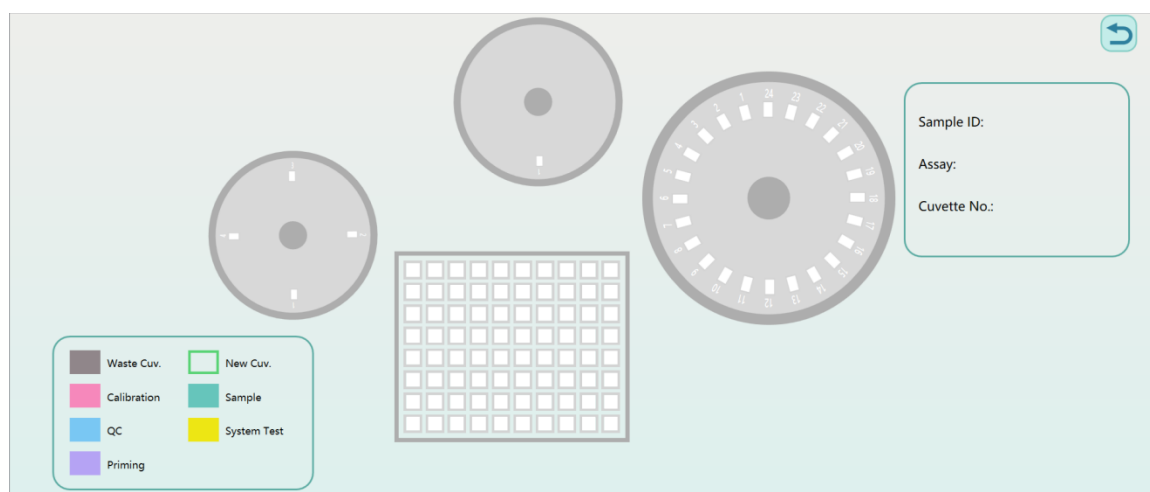


Fig. 5.7-12 Disk statuses

5.7.5 Shaker status

In the shaker status area in the [Status] interface, you can view the shaker statuses of the analyzer, as

shown in the following figure. When the shaker speed is normal, the icon is displayed in green. When the shaker speed is abnormal, the icon is displayed in red.

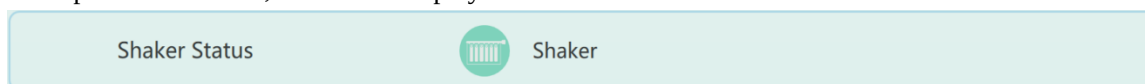


Fig. 5.7-13 Shaker statuses

5.8 [Set] function description

Click the <Set> button on the main menu to access the [Set] interface. [Set] functions are used to define various parameters, set system values, and maintain analyzer operations. This section mainly introduces each function of the [Set] interface one by one.

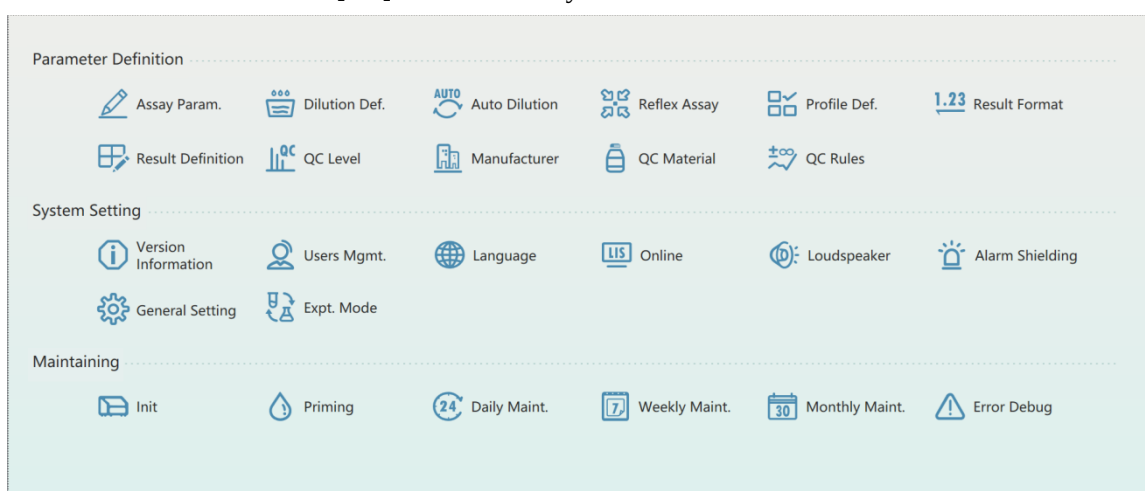


Fig. 5.8-1 Settings

5.8.1 Parameter definitions

The first column of the [Set] interface is [Parameter Definition], which can be used to set assay parameters, dilution definitions, auto dilutions, reflex assays, profile definitions, result formats, result definitions, QC levels, QC manufacturers, QC materials and QC rules. The following is an introduction to the various parts of the parameter definitions.

5.8.1.1 Assay parameters

Click the <Assay Param.> button to go to the [Assay Parameters] interface, as shown in the following figure. Define the relevant parameters of each test assay.

Click the corresponding button in the [Select Assay] area on the left to select the desired test assay. The assay details appear in the [Assay Info] area.

1. Modifying assay info:

Select the test assay in the [Select Assay] box, modify some of the data of the test assay in the [Assay Info] box, and click the <Save> button to finish modifying.

2. Delete assay:

In the **[Select Assay]** box, select the test assay to be deleted and click the **<Delete>** button to finish deleting.

Fig. 5.8-2 Assay parameters

Click the **<Master Curve>** button to go to the **[Master Curve]** interface, as shown in the following figure. The master curve is the basis for calibration and each test assay has a master curve labeled with an ID and a lot-no.

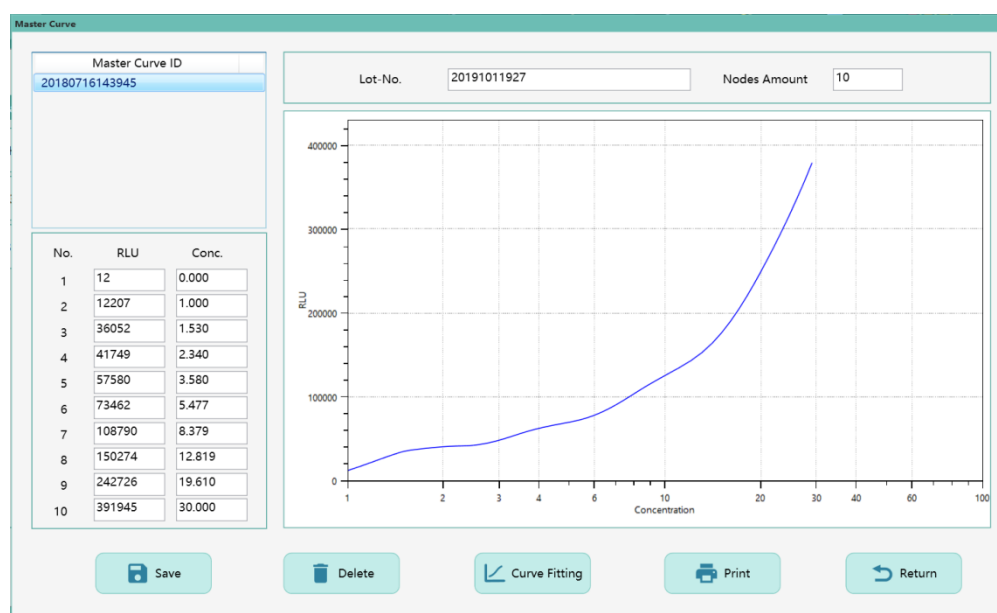


Fig. 5.8-3 Master curve

1. Displaying a master curve

Select a test assay in the **[Select Assay]** area to display the master curve ID of the corresponding assay in the **[Master Curve ID]** area. Click a master curve ID in the list to show the details of the master curve.

2. Master curve fitting

Click a master curve in the **[Master Curve ID]** area, modify the data of the master curve, click the

- <Curve Fitting> button to re-fit and show the new master curve. Then click the <Save> button to finish the curve fitting and save the data.
3. **Deleting a master curve**
Select a master curve and then click the <Delete> button. Click the <OK> button to delete the master curve.
 4. **Printing a master curve**
Select a master curve and click the <Print> button to print the corresponding assay master curve and data.

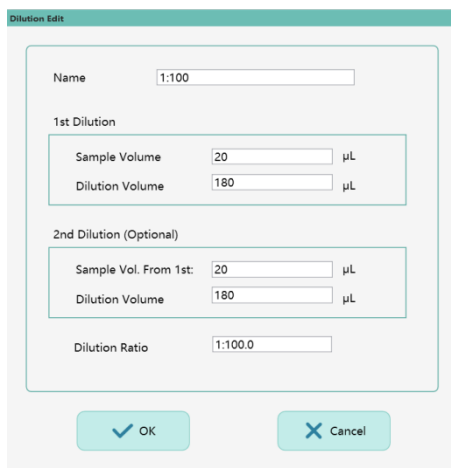
5.8.1.2 Dilution definitions

Click the <**Dilution Def.**> button to go to the [Dilution Definition] interface, as shown in the following figure. You can define a maximum of 10 dilution ratios for a test assay that uses sample dilution liquid. The dilution ratio can be customized.



Fig. 5.8-4 Dilution definitions

1. **Add a dilution ratio:**
 - Select a test assay in the [Select Assay] area, then select the dilution ratio to be added in the [Dilution Ratio Selection] area, to add the corresponding dilution ratio to the [Dilution Ratio] area.
 - If you need to customize the dilution ratio, select a test assay in the [Select Assay] area, click the <Add> button to open the [Dilution Edit] interface, edit the information about the primary and secondary dilutions, then click the <OK> button to finish the addition procedure.



The 'Dilution Edit' dialog box contains the following fields:

- Name:** 1:100
- 1st Dilution:**
 - Sample Volume: 20 μL
 - Dilution Volume: 180 μL
- 2nd Dilution (Optional):**
 - Sample Vol. From 1st: 20 μL
 - Dilution Volume: 180 μL
- Dilution Ratio:** 1:100.0

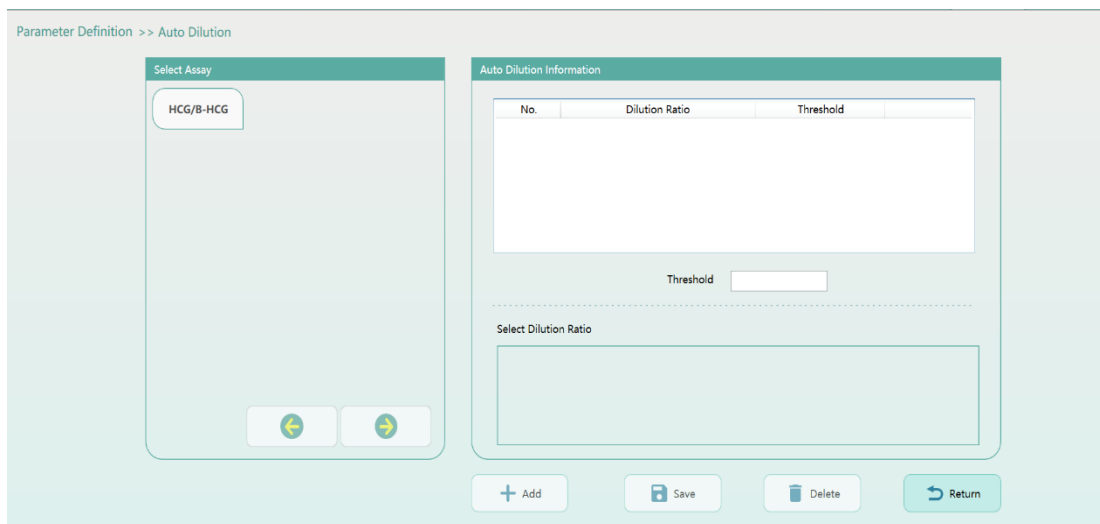
At the bottom are 'OK' and 'Cancel' buttons.

Fig. 5.8-5 Dilution edit

2. **Modify dilution ratio information:**
Select a test assay in the [Select Assay] area, select a dilution ratio in the [Dilution Ratio] area, and click the <Edit> button to open the [Dilution Edit] interface to modify the dilution definition.
3. **Delete a dilution ratio:**
Select a dilution ratio in the [Dilution Ratio] area and click the <Delete> button to delete the dilution ratio.

5.8.1.3 Auto dilution

Click the <Auto Dilution> button to go to the [Auto Dilution] interface, as shown in the following figure. Auto dilution means that when the concentration in the test result exceeds the auto dilution concentration, the diluted sample is automatically registered in the sample area and the analyzer automatically dilutes the liquid and finishes the test.



The 'Auto Dilution' interface is titled 'Parameter Definition >> Auto Dilution'. It is divided into two main sections:

- Select Assay:** A box containing 'HCG/B-HCG' with left and right arrow buttons below it.
- Auto Dilution Information:**
 - A table with columns: No., Dilution Ratio, Threshold.
 - A 'Threshold' input field.
 - A 'Select Dilution Ratio' section with a large empty box.

At the bottom are buttons: '+ Add', 'Save', 'Delete', and 'Return'.

Fig. 5.8-6 Auto dilution

1. **Add auto dilution:**
Select a test assay in the [Select Assay] area and select a dilution ratio in the [Select Dilution Ratio] area (If the assay has no dilutions defined, this area is empty. For defining dilutions, refer to "5.8.1.2 Dilution definitions."). Click the <Add> button, the new dilution ratio is displayed in the list at the top and

the addition is finished.

2. Modifying the auto dilution threshold:

Select a test assay in the **[Select Assay]** area, select the automatic dilution ratio in the list, enter the critical value, and click the **<Save>** button to finish modifying the automatic dilution threshold.

3. Delete auto dilution:

Select a test assay in the **[Select Assay]** area, select the automatic dilution ratio to be deleted in the list, and click the **<Delete>** button to complete the delete operation.

5.8.1.4 Reflex assays

Click the **<Reflex Assay>** button to access the **[Reflex Assay]** interface, as shown in the following figure. A reflex assay is when the test result of the assay in question is within the reflex detection range and an associated test assay is started up.

Fig. 5.8-7 Reflex assay

1. Add a reflex assay:

For example, if you need to start the reflex assay B when the test result of assay A is between 50 and 150, in the **[Select Assay]** area, select test assay A and select the reflex assay B in the **[Select Assay to Add]** area. For assay B, enter the high and low values (for example, high value is 150, and low value is 50), click the **<Add>** button, and add the corresponding information to the reflex assay information list.

2. Modifying a reflex assay:

Select a test assay in the **<Select Assay>** area, select the reflex assay to be modified in the reflex assay information list, modify the high value and low value input boxes, and click the **<Save>** button to finish modifying.

3. Deleting a reflex assay:

Select a test assay in the <Select Assay> area, select the reflex assay to be deleted in the reflex assay information list, and click the <Delete> button to finish deleting.

5.8.1.5 Profile definitions

Click the <Profile Def.> button to go to the **[Profile Definition]** interface, as shown in the following figure. The user can create a profile for a centralized test assay, which will appear in the **[Profile Selection Area]** of the work list. A one-click operation can be used to assign several test assays to one sample, thereby facilitating sample registration.

1. Add a profile:

Select the required assay (multiple selections allowed) in the **[Select Assay]** area. In the **[Profile Info]** box, enter the profile name and click the <Add> button to complete the addition.

2. Modify a profile:

Select a profile in the **[Profile List]**, modify the information in the profile info area on the left, and click the <Save> button to complete the modification.

3. Delete a profile:

Select a profile in the **[Profile List]** and click the <Delete> button to finish deleting.

Fig. 5.8-8 Profile definition

5.8.1.6 Result format

Click the <Result Format> button to go to the **[Result Format]** interface, as shown in the following figure. The result format defines the number of decimal places in the numerical range.

Enter the lower end of the numerical range and the corresponding decimal places. Enter the upper and lower limits of the numerical range to automatically change the upper limit of the next high range. Click the <Save> button to finish setting the result format parameters.

Parameter Definition >> Result Format

Result Format Information

Numerical Range		Decimal Places
0	< 1	3
1	< 10	2
10	< 100	1
100	< Max	0

Save

Return

Fig. 5.8-9 Result format

5.8.1.7 Results definition

Click the **<Result Definition>** button to open the **[Result Definition]** interface as shown in the following figure. You can check the corresponding definition option to set the information to display in the result. The result setting options are as follows:

- Concentration is not displayed for test results with Pipettor errors.
- The results show the flag beyond measuring range.
- Print detailed result when printing sample result, Cali. result and QC result.

Parameter Definition >> Result Definition

Result Definition

☐ Concentration is not displayed for test results with Pipettor errors

☐ The result show the flag beyond measuring range

☐ Print detailed result when printing sample result, Cali. result and QC result

Save

Return

Fig. 5.8-10 Results definition

In the **[Result Definition]** area, select the option to run and click the **<Save>** button to finish the modification.

5.8.1.8 QC level

Click the **<QC Level>** button to open the **[QC Level]** interface as shown in the following figure.

Users can define a QC level flag, enter the fields (letters, numbers, Chinese, symbols) in the corresponding level input box, and click the **<Save>** button to finish defining a QC level identifier.

Parameter Definition >> QC Level

QC Level Flag

Low Level 1

Mid Level 2

High Level 3

Save Return

Fig. 5.8-11 QC level

5.8.1.9 QC manufacturers

Click the **<Manufacturer>** button to access the **[QC Manufacturer]** interface, as shown in the following figure.

Parameter Definition >> QC Manufacturer

Manufacturer Set

No.	Manufacturer
1	snibe

Manufacturer Name

+ Add Save Delete Return

Fig. 5.8-12 QC manufacturers

1. Add a QC manufacturer

Enter the manufacturer name in the "Manufacturer Name" input box and click the <Add> button to finish.

2. Modifying a QC manufacturer

In the QC Manufacturer list, select the manufacturer that needs to be modified, modify the manufacturer name in the "Manufacturer Name" input box below, and click the <Save> button to finish.

3. Delete QC manufacturer

In the QC Manufacturer list, select the manufacturer that needs to be deleted and click the <Delete> button to finish.

5.8.1.10 QC material

Click the <QC Material> button to go to the [QC Material] interface, as shown in the following figure.

The screenshot displays the 'QC Material' interface. At the top, there's a breadcrumb 'Parameter Definition >> QC Material'. Below this, on the left, is a table listing QC materials with columns: QC Name, Lot-No., Level, and Expiry Date. The first entry is '朗道', '000001', '1', and '2021-08-03'. To the right of the table are input fields for 'QC Name', 'Lot-No.', 'Level' (a dropdown menu), 'Manufacturer' (a dropdown menu), 'Expiry Date' (a date picker), and 'Barcode'. Further right is a 'QC Assay' section with two vertical arrows for switching between assays. Below these is the 'QC Assay Parameters' section, which includes fields for 'Assay Abbr.', 'Repetitions', 'Target', 'Expected' (with radio buttons for 'Expected' and 'Interval'), 'Unit', 'User-defined', and 'Auto QC' (a checkbox). At the bottom, there's a row of buttons: 'Add QC Material', 'Add Assay', 'Save', 'Delete QC Material', 'Delete Assay', 'QC List', and 'Return'.

Fig. 5.8-13 QC material

1. QC material operations:

- Click the <Add QC Material> button to activate the [QC Material Messages] area, enter the relevant information about the QC material, and click the <Save> button to finish adding the QC material.
- In the QC material list, select the QC material that you want to delete and click the <Delete QC Material> button to finish deleting the QC material.

2. Operations for QC assays

- In the QC material list, select a QC material, click the <Add Assay> button to activate the [QC Assays] area, and select the assay to be added in the [QC Assay] area (multiple selection allowed). The [QC Assay Messages] area displays the assays just added in order and you can use the up and down page buttons on the left to switch between the assays. Fill in the assay QC

- info as required and click the **<Save>** button to finish adding the QC assay.
 - In the QC material list, select the QC material belonging to the QC assay that you want to delete and select the QC item to be deleted in the **[QC Assay]** area. Click the **<Delete Assay>** button to finish deleting the QC assay.
3. QC inventory:
- Click the **<QC List>** button to bring up the **[QC List]** interface. The QC inventory list will display all QC items by **[QC Material-QC Assay]**.

5.8.1.11 QC rules

Click the **<QC Rules>** button to access the **[QC Rules]** interface as shown in the following figure. The software supports Westgard Rules QC and the user can select the QC rules for the assay as needed.

In the **[Select Assay]** area, select the assay, check the QC rule in the **[Westgard Rules]** area, and click the **<Save>** button to complete the QC rule definition.

Fig. 5.8-14 QC rules

5.8.2 System settings

The second column in the **[Set]** interface is **[System Setting]**, including settings of Users Mgmt., Language, Online, Loudspeaker, Alarm Shielding, General Setting, Expt. Mode, and Version Info. The following is an introduction to the various parts of the system setup.

5.8.2.1 Version information

Click the **<Version Information>** button to access the **[Version Information]** interface as shown in the following figure. This is used to display version info for the host computer software and the lower-level program.

Fig. 5.8-15 Version information

5.8.2.2 User management

Click the **<Users Mgmt.>** button to access the **[User Management]** interface, as shown in the figure below. The software provides user management functions that are used by authorized system administrators to define the permissions of different users. Users can be added and usernames, passwords and permissions can be edited.

Fig. 5.8-16 User management

1. Add a user:

Enter the new user information in the **[User Information]** area and click the **<Add>** button to add the new user.

2. Modify a user:

In the **[User List]** area, select the user to be modified, directly modify the information in the **[User Information]** area, and click the **<Save>** button to finish modifying the user data.

3. Deleting users:

In the **[User List]** area, select the user to be deleted and click the **<Delete>** button to finish deleting the user.

5.8.2.3 Language

Click the <Language> button to go to the [Language] interface, as shown in the following figure. After the language is selected, the software will display text in the corresponding language.

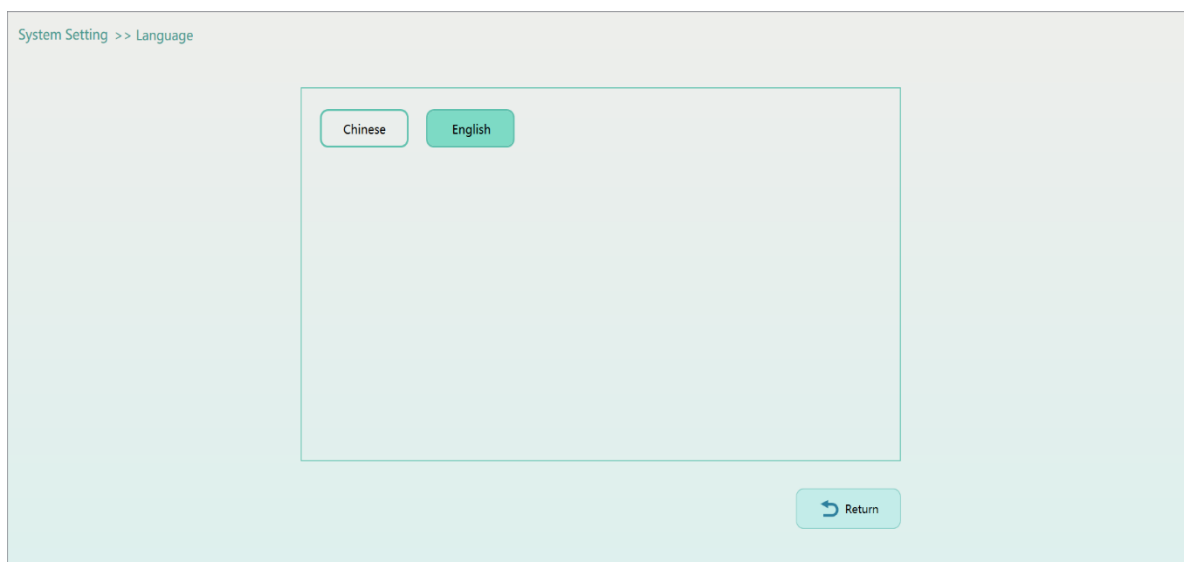


Fig. 5.8-17 Language

5.8.2.4 Online

In the [System Setting] area, click the <Online> button to go to the [Online] interface, as shown in the following figure.

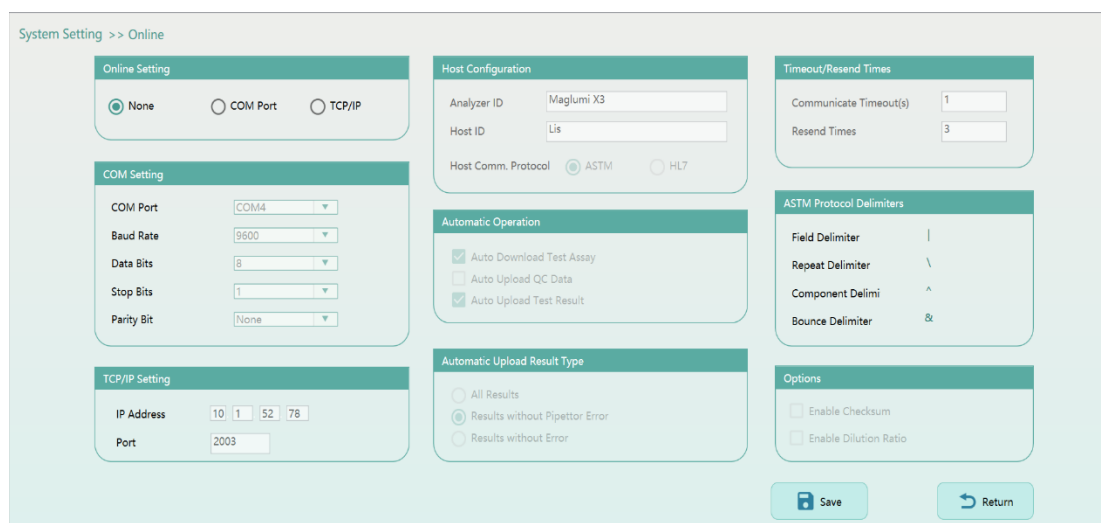


Fig. 5.8-18Online

The software provides two-way communication with a hospital LIS server. You can obtain the test assay information for a sample from the hospital LIS server by manually entering the sample ID or using a barcode, and send test results to the hospital LIS server.

1. Online settings:
 - Select "None" to not use online features.

- Select "COM Port" networking and select the corresponding COM port number and baud rate in the [COM Setting] area to finish.
- Select "TCP/IP" networking and enter the corresponding IP address and port in the [TCP/IP Setting] area to complete the setup.

2. Host configuration:

- Analyzer ID: Sets the unique channel no. of the analyzer;
- Host ID: Sets the unique channel no. of the LIS server;
- Host Comm. Protocol: Selects the communication protocol between the analyzer and the LIS server: "ASTM" or "HL7".

3. Other online settings:

You can define host testing, automatic operations, automatic uploads of results, timeout/resend times, whether to enable checksum, and whether to enable dilution ratio as needed. Click <Save> after all the settings are finished to complete the online settings. After successful connection, the LIS status indicator in the operation bar in the lower part of the interface turns green.

5.8.2.5 Loudspeaker

Click the <Loudspeaker> button to access the [Loudspeaker] interface, as shown in the following figure. This is used to set whether the computer or analyzer loudspeakers will sound when warnings or errors occur, and to select the type of sound.

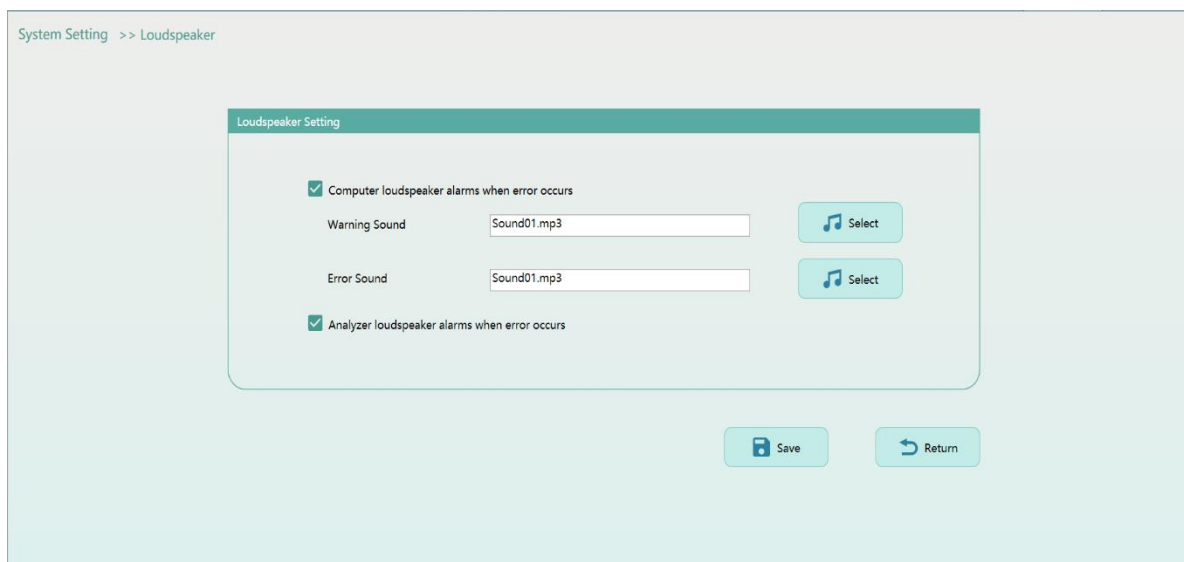


Fig. 5.8-19 Loudspeaker

Click the <Select> button in the [Loudspeaker Setting] area to go to the [Sound Selection] interface. Then select a sound file in the sound list in the [Sound Selection] interface. Click the <Play> button to listen to the sound and click the <Stop> button to stop listening. Click the <OK> button to finish selecting the sound type.

Click the <Save> button in the [Loudspeaker Setting] interface to finish saving the configuration.

5.8.2.6 Alarm shielding

Click the **<Alarm Shielding>** button to access the **[Alarm Shielding]** interface as shown in the following figure. This is used to set alarm options that need to be shielded during tests.

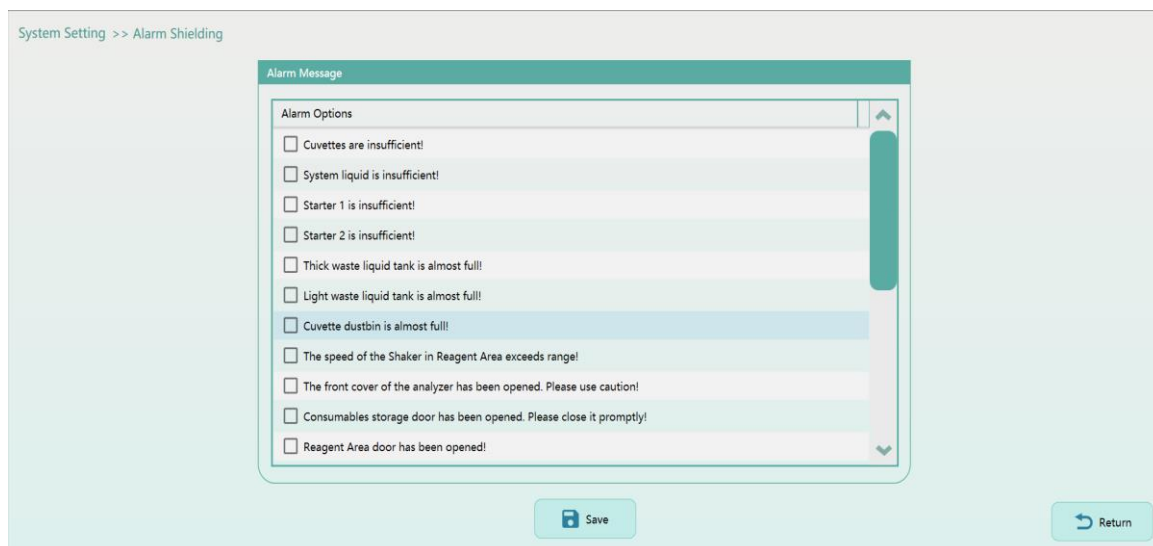


Fig. 5.8-20 Alarm shielding

5.8.2.7 General settings

Click the **<General Setting>** button to access the **[General Setting]** interface, as shown in the following figure. Sets whether a print pop-up is displayed when a screenshot setting is made and whether to enable the urgent mode.

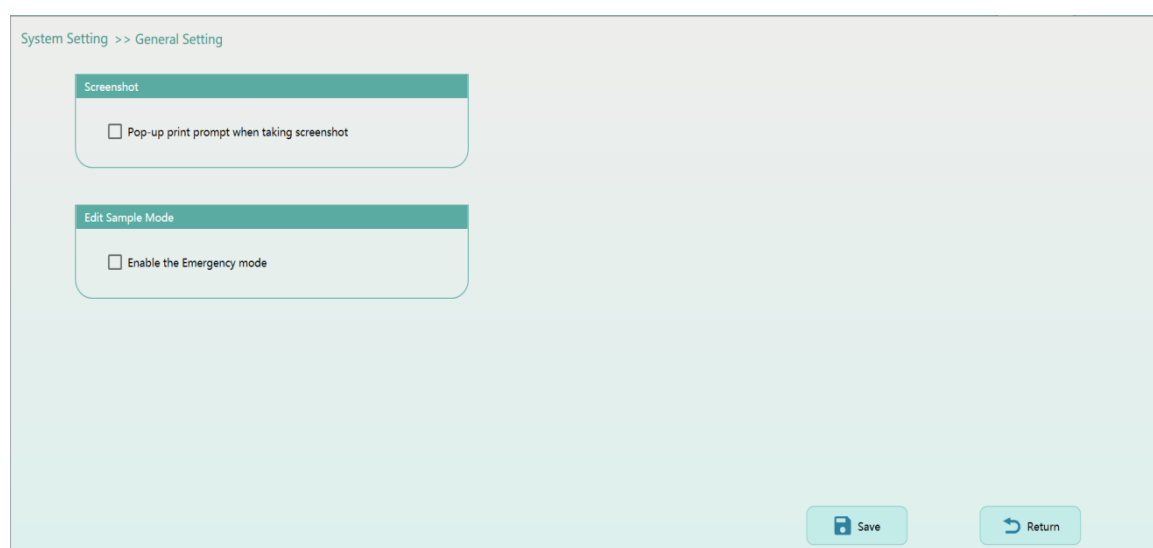
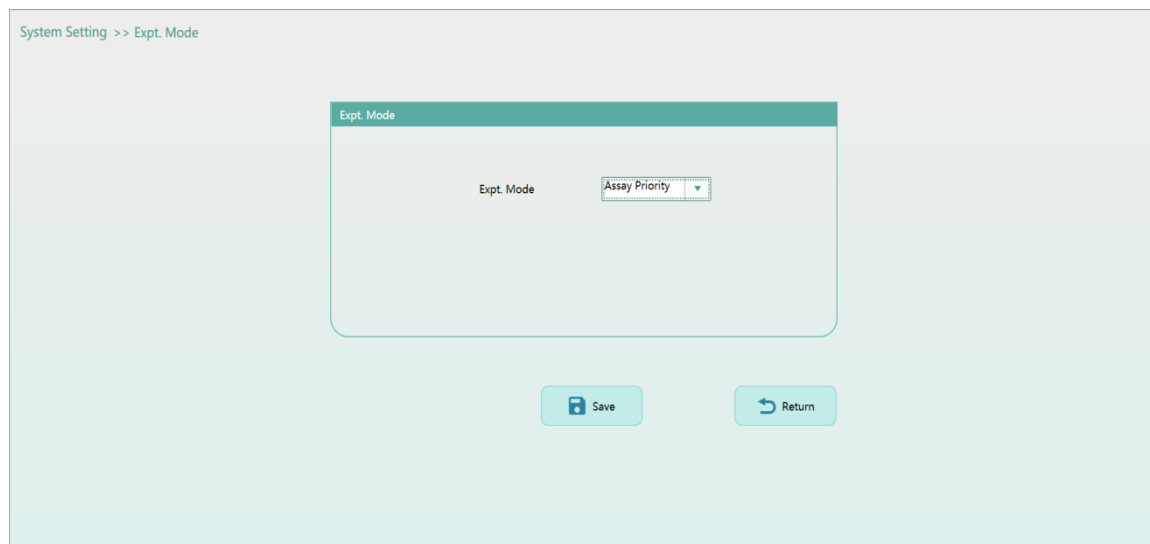


Fig. 5.8-21 General settings

After the user enables the urgent mode, they can manually edit the sample ID for test registration in the event of a code scan exception.

5.8.2.8 Expt. Mode

Click the <Expt. Mode> button to go to the [Expt. Mode] interface, as shown in the following figure. It is used to set the priority sequence of the tests in the system.



5.8-22 Expt. mode

1. Expt. Mode

Users can select the Expt. mode in the drop-down list: Assay Priority, Sample Rack Priority. This controls the testing sequence of sample and QC tests.

- Assay Priority: The tests in the same lot are sorted by assay category.
- Sample Rack Priority: The tests in the same lot are sorted by sample rack of the sample.

2. Whether reagent cross-contamination assays enjoy priority for testing by assay

If the user has selected the "Sample rack first" option, the "Whether reagent cross-contamination assays enjoy priority for testing by assay" option will be displayed in the interface.

Check this option, and all cross-contamination assays will be sorted in the "Assay Priority" mode for testing, and the remaining tests will be carried out in the "Sample Rack Priority" mode.

5.8.3 System maintenance

The third column of the [Set] interface is [System Maintenance], where you can view version info and run initialization, priming, daily maintenance, weekly maintenance, monthly maintenance and component debugging. The following is an introduction to the various parts of system maintenance.

5.8.3.1 Initialization

Click the <Init> button to access the [Initialization] interface. You can send a specific command to the analyzer and initialize the analyzer, as shown in the following figure.

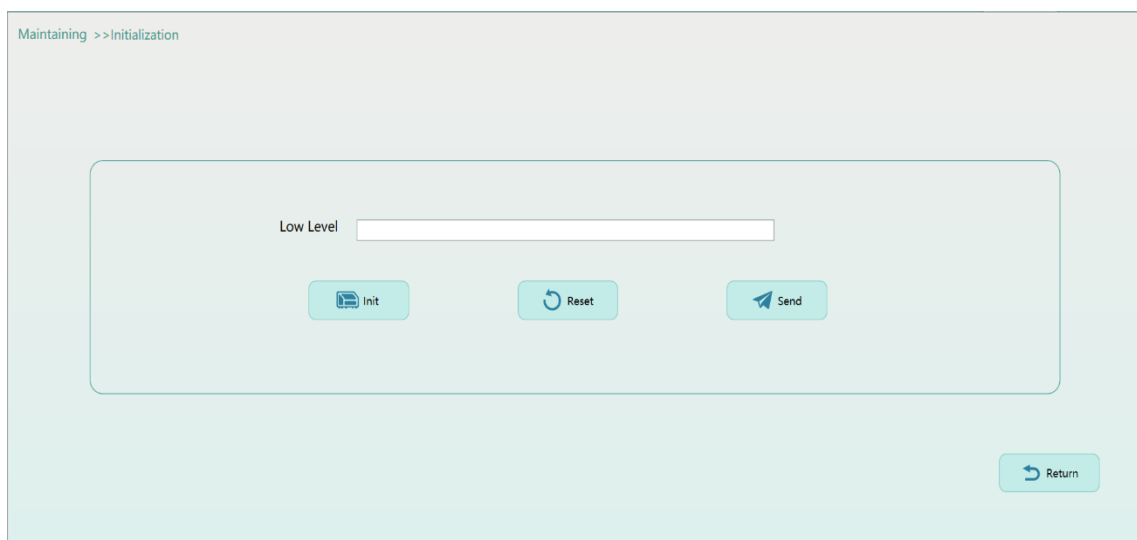


Fig. 5.8-23 Initialization

1、 Sending a command

Enter the specified command in the [Low Level] input box and then click the <Send> button to send the command.

2、 Initialization

Click <Init>: Initializes the analyzer.

Click <Reset>: Resets the analyzer. This operation does not clear the cuvettes inside the analyzer.

5.8.3.2 Priming

Click the <Priming> button to go to the [Priming] interface, as shown in the following figure.

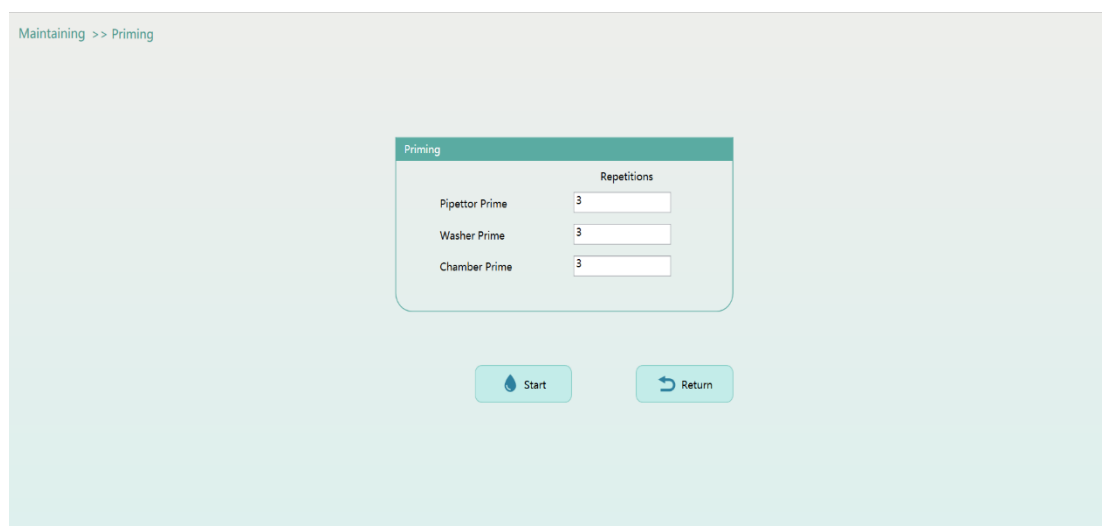


Fig. 5.8-24 Priming

The priming function washes the pipe system of the analyzer. Select the parts to be primed and the number of primings. Click the <Start> button to finish priming the relevant pipe system of the part.



Note:

You must perform priming before the analyzer starts testing. If the system liquid is replaced during a test, you must wash the pipe system immediately.

5.8.3.3 Daily, monthly and weekly maintenance

Click the <Daily Maint.> button to go to the [Daily Maintenance] interface, as shown in the following figure.

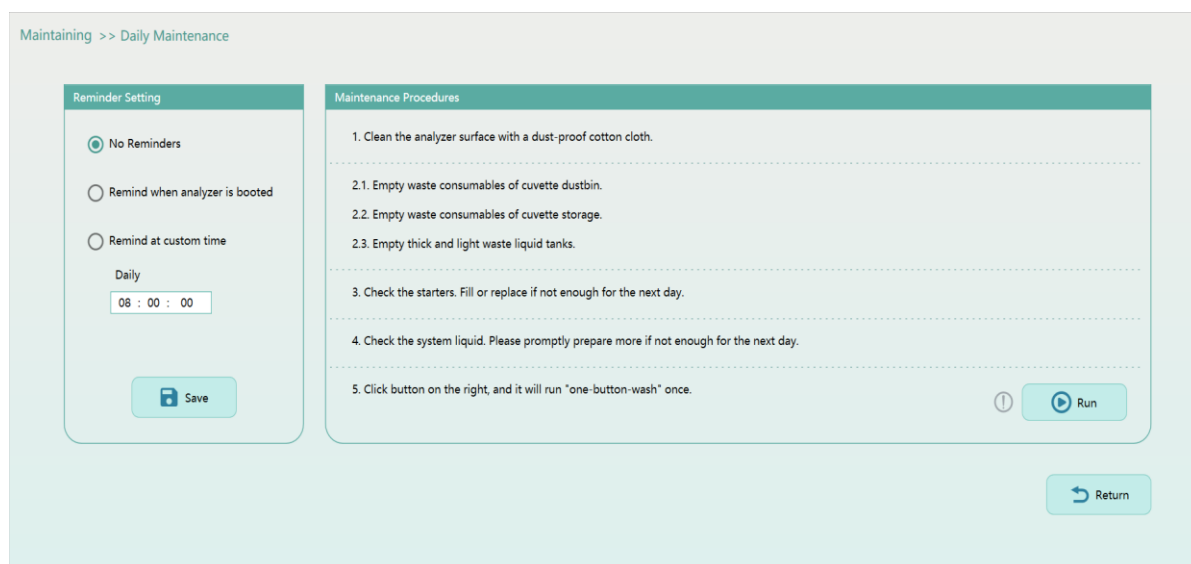


Fig. 5.8-25 Daily maintenance

1. Reminder settings:

In the **[Reminder Setting]** area, select the reminder type. These include: no reminders, reminder when analyzer is booted, remind at custom time. If you select remind at custom time, you need to enter the scheduled daily time for the reminder in the input box. Click the <Save> button to finish.

2. Maintenance steps:

Run maintenance operations in accordance with the maintenance procedures outlined in the interface prompts. When you need to run a one-click clean, click the <Run> button in the corresponding procedure to finish the cleaning operation in one go.

The weekly maintenance and monthly maintenance functions are similar and won't be described in detail here. See "Chapter 6 System maintenance" for detailed steps of daily, monthly and weekly maintenance.

5.8.3.4 Component debugging

Click the <Error Debug> button, or when the analyzer fails and reports an error, to go to the [Debug] interface, as shown in the following figure.

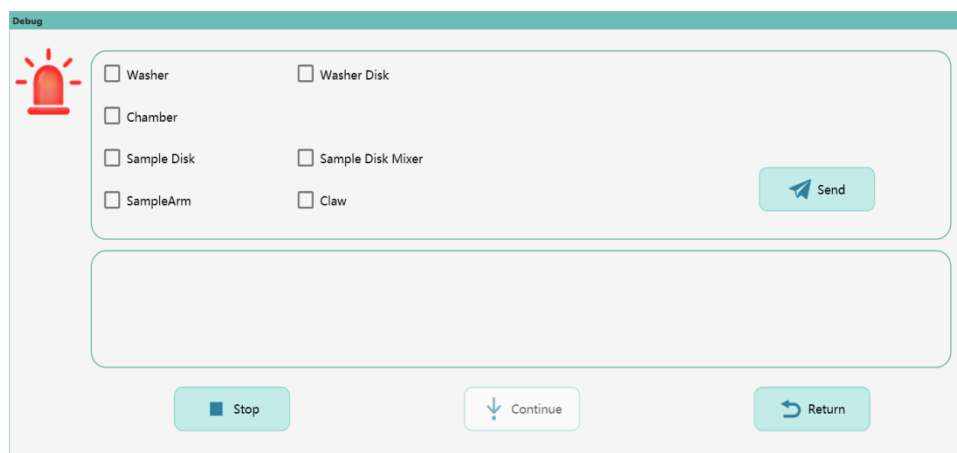


Fig. 5.8-26 Component debugging

Select one or more components to initialize separately in each component's initialization area above the dialog box and then click the <**Send**> button to initialize the selected components.

Click the <**Stop**> button to stop all tests. Click the <**Continue**> button to continue tests that can continue to run after the component is finished initializing.

5.9 [Home] function description

Click the <**Home**> button in the main menu to access the [**Home**] menu and related function interface, as shown in the following figure. The interface is divided into two parts. The central part shows statuses in real time and the two sides display the various statuses and indicator lights of the analyzer in real time. The following is an introduction to the contents of each part of Home.

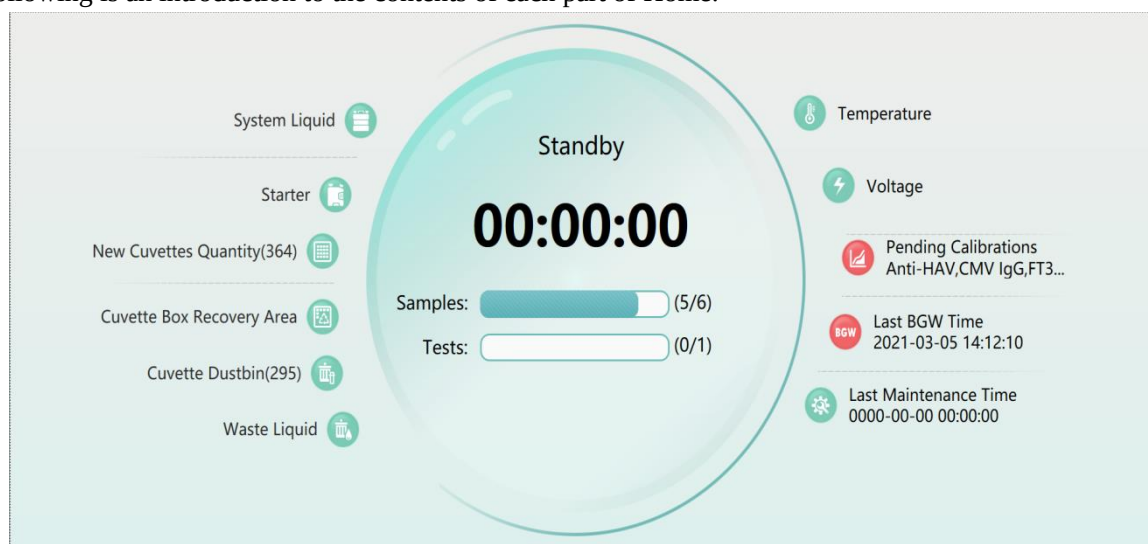


Fig. 5.9-1 Home

5.9.1 Test statuses

The central part of Home is the status display area. From top to bottom sequentially: analyzer status, remaining time for current testing, samples and tests.

1. Analyzer status:

Analyzer statuses include "Connecting", "Init", "Standby", "Testing", "Maintaining", "Pause Cycle", "Continue", "Stop", "Analyzer Fault", and "System Test".

2. Remaining time:

The software updates the remaining time for the current test in real time. If a new test is loaded when a test is still in progress, the remaining time indicates the remaining time before all the tests are completed.

3. Samples:

Progress bar for samples: (number of samples finished / total number of samples)

4. Tests:

Progress bar for tests: (number of tests finished / total number of tests)

5.9.2 Remaining statuses and indicator lights

Each analyzer status and indicator light is displayed on both sides of Home. The status on the left is consumables and waste. When the status value is normal, the icon is displayed in green; when the status reaches the prompt value, the icon is displayed in yellow; when the status is warning, the icon is displayed in red. To view the specific situation, click the corresponding icon to enter the corresponding detailed interface to view.

From top to bottom, the right side displays: temperature, voltage, calibration, BGW testing and maintenance.

1. Temperature and voltage

When the temperature and voltage values are in the normal range, the corresponding icon is displayed in green; when the temperature and voltage values are outside the normal range, the corresponding icon is displayed in red.

2. Pending calibration

When there is no new calibration to be processed, the icon is displayed in green; when there is a new calibration needing to be processed, the icon is displayed in red and the reagent assay to have its calibration confirmed is displayed below.

3. Last BGW Time

Shows the time of the most recent BGW test. When a BGW test has finished that day, the icon is displayed in green; when a BGW test has not finished on the day, the icon is displayed in red.

4. Last maintenance time

Displays the last maintenance operation time.

Chapter 6 System maintenance

Please conduct operations and routine maintenance in strict accordance with the Operating Instructions to guarantee the reliable performance, good work status and lifetime of the system.

For the problems occurring during operation that cannot be solved and the maintenance and repair problems not involved in the section, please contact the service department of Shenzhen New Industries Biomedical Engineering Co., Ltd.

Table 6-1 Parts information

Number	Component name	Position	Cycle	Remark (replacement method)
1	Washer waste liquid peristaltic pump hose	In the center of the front right plate of the analyzer	Replaced every year or every 73,440 tests, or when damaged	Contact the technical service engineer to replace it
2	Thick and thin waste liquid tube and connector	On the right side of the analyzer	Replaced regularly or when they are damaged and impede waste liquid discharge	Contact the technical service engineer to replace it
3	Pipettor	SampleArm	Replaced from time to time, or when it is damaged and becomes bent	Contact the technical service engineer to replace it

6.1 Tools

6.1.1 Accessories (built-in)

Internal hexagonal wrench: Removes the panel and some mechanism components of the analyzer.

Cross screwdriver: Removes the analyzer plates and installs the mechanism components of the analyzer.

Positioning tool: Calibrates the position of each component.

6.1.2 Materials you need to prepare

Clean cotton cloth: Cleans each component.

Cotton swab: Cleans pipettor.

Waste liquid tank or waste liquid discharge pipe: Used when you have special requirements for the

discharge of waste liquid.

Diagonal pliers (or scissors): Cuts cable ties off pipes.

Cable ties: Fastens pipes tight.

6.1.3 Cuvettes

Cuvette carries the sample and reagent required for the reaction. All cuvettes must be purchased from the Shenzhen New Industries Biomedical Engineering Co., Ltd. and may not be replaced by cuvettes supplied by other manufacturers. Our company does not assume any responsibilities for any inaccurate measurements because of the use of a cuvette not produced by our company.

6.1.4 System liquid

The system liquid is used for cleaning all parts of the analyzer. All system liquid must be purchased from Shenzhen New Industry Biomedical Engineering Co., Ltd. If other system liquids are used instead, the cuvettes, reagent probe and pipes may not be cleaned properly, or the mixture in the cuvette being tested will not be cleaned thoroughly, affecting the accuracy and precision of the test results. SNIBE will assume no responsibility for inaccurate results due to the cuvettes not being cleaned using SNIBE system liquid.

6.2 Daily maintenance

Necessary materials:

Clean waste bag, clean cotton cloth

Maintenance process:

1. Use the clean cotton cloth to clean the surface of the analyzer.
2. Empty the waste bag of cuvettes and replace the bag with a clean one; empty the waste liquid tank.
3. Check the volume of starter and supplement or replace the starter if the volume is insufficient for the next day.
4. Check the volume of system liquid and prepare more in time if the volume is insufficient for the next day.
5. Check the quantity of cuvettes and replenish more if the quantity is insufficient for the next day.
6. Prime the SampleArm, the washer, and the chamber 10 times each.

6.3 Weekly maintenance

Necessary materials:

Cotton swabs, alcohol

Maintenance process:

Use a cotton swab dipped in alcohol to clean the outer wall of the pipettor, the washing probe of the washer, the outer wall of the waste liquid probe, and the outer wall of the waste liquid probe of the chamber, then prime the SampleArm, the washer, and the chamber 10 times each.

6.4 Monthly maintenance

Necessary materials:

160 mL 84 disinfectant, pure water, System Tubing Cleaning Solution, Light Check

Maintenance process:

1. Replace the starter with pure water at 40 °C and prime the chamber 30 times, then replace the pure water with starter to prime the chamber for another 30 times.
2. Use System Tubing Cleaning Solution to run the Pipe Wash function once.
3. Empty the system liquid tank and waste liquid tank and wash them with 80mL 84 disinfectant + 2L pure water and then empty them. Then use pure water (2L each time) to wash the system liquid tank and waste liquid tank five times. Finally, fill the system liquid tank with system liquid to prime the SampleArm and washer 30 times each.
4. Place the Light Check at the 4th position of the 20th channel of the reagent area and perform five BGW tests and 10 LC tests for the SampleArm. View and record the results.
5. Requirements for system tests and results:

1) BGW	100-1200	$CV \leq 10\%$
2) LC	450000-650000	$CV \leq 3\%$

See the operating instructions for the LC test solution reagent for the specific expected values.

6.5 Long-term non-use

If you do not use the analyzer for more than five days, please carry out the following procedures:

Replace Starter 1 and Starter 2 with pure water.

Replace the system liquid with pure water.

Click the <Priming> button in the [Set] interface, and prime the SampleArm 30 times, the washer 30 times, and the chamber 30 times, to fill the pipes with pure water.

When you want to use the analyzer again, please follow the following procedure:

1. First remove the starter bottle and the system liquid tank. Click the <Start> button in the [Priming]

interface and prime the SampleArm 30 times, the washer 30 times, and the chamber 30 times. Then empty the pipes.

2. Place Starter 1 and Starter 2.
3. Place system liquid.
4. Click the <Start> button in the [Priming] interface to prime the SampleArm 30 times, the washer 30 times, and the chamber 30 times. Then confirm that the BGW meets the requirements before starting tests

Chapter 7 Common faults and troubleshooting

7.1 Fault classification

Faults are classified into the two major types of alarm fault and non-alarm fault. Alarm faults are graded as warnings, sample loading errors, and halt errors based on the severity. The actions that the analyzer will take after sending different alarm information are shown in the following table.

Table 7.1-1 Alarmed faults

Alarmed faults	Actions of the analyzer
Alarm at warning grade	The actions of sample loading and measurement will not be stopped and the analyzer will continue analysis testing.
Alarm at grade of stopping sample loading	Stop injection of new sample and continue the analysis of sample added in the cuvettes.
Alarm at halt grade	The analyzer will stop action immediately and the test result is not reliable.

7.2 Non-alarm-type faults and solutions

Some abnormal data not presented in the alarm information can be spotted by measuring the QC solution, reviewing, and checking the analyzer status.

Table 7.2-1 Data abnormalities without alarms and their solutions

Problem symptoms	Possible cause	To check	Solution
Abnormal background result CV or RLU out of range	Either the system liquid has been stored for too long, or a problem with the water quality system or configuration error has caused the system liquid to fail.	Observe whether the system liquid is too viscous, whether there is too much yellow dirt in the system liquid tank, or whether the conductivity of the pure water used for preparing the system liquid is less than 1 $\mu\text{S}/\text{cm}$.	Clean the system liquid tank following the monthly maintenance method, use pure water with conductivity of less than 1 $\mu\text{S}/\text{cm}$, and reprepare the system liquid according to the Operating Instructions.

Common faults and troubleshooting

	An abnormal amount of system liquid injected in the washer	Check whether the amount of system liquid injected by the three washing needles in the washer is consistent, whether the injection pipe is kinked or leaking, and whether the solenoid valve fixing screws are loose.	Contact technical service staff
	Malfunction of the washer waste liquid peristaltic pump, or the washer waste liquid needle is blocked.	Check if the liquid in the waste liquid pipe is flowing too slowly.	Contact technical service staff
	Chamber or translucent window is dirty	Check for liquid or crystals.	Contact technical service staff
	Abnormal amount of starter injected by starter pump	Check if the amount of starter is insufficient, if the starter pump makes an abnormal sound, and if the starter pipe is retracted or bent.	Contact technical service staff
	Contamination of cuvette	Check whether the cuvettes have not been used for a long time and check for dust or other impurities.	Replace the cuvettes and clean the cuvette storage
	PMT failure	Check whether the dark noise is normal.	Contact technical service staff
	The analyzer has been running for a long time and the pipes have not been properly maintained.	Check if the pipes are dirty or aged.	Use system tubing cleaning solution to run the Pipe Wash function once.
	Starter has expired	Check if the starter fluid has expired, or whether the starter fluid storage environment meets the requirements.	Change the starter.

Problem symptoms	Possible cause	To check	Solution
High CV value in LC test results	Water quality system problem	Check whether the conductivity of the pure water used for preparing the LC solution is less than 1 $\mu\text{S}/\text{cm}$.	Use pure water with a conductivity of less than 1 $\mu\text{S}/\text{cm}$ for preparing the solution.
	Problem with the SampleArm position	Check whether the position of the SampleArm is accurate or divergent.	Contact technical service staff
	The pipetting pipe has air leakage	Check whether there are any bubbles or water droplets on the probe tip after the pipettor is cleaned, whether there are air bubbles in the SampleArm pipe, and whether there is crystallization due to leakage at any pipe joints (solenoid valve, plunger pump, etc.)	Check fixing rings at valve interfaces for leaks, check whether the system liquid has been used up, check whether the SampleArm is at the right position for washing, and then select the SampleArm to prime in the Priming interface. Contact technical service staff if the leakage continues.
	SampleArm plunger pump, wash pump or valve failure	Check the alarm log for failure to initialize plunger pump, check whether the inner membrane of the washer pump is damaged or liquid spurts out.	Contact technical service staff
	Liquid level detection error, LC test solution not added	Observe whether any LC test results were almost 0.	Contact technical service staff
	The pipettor is dirty or damaged	Check the outer wall of the probe for crystallization, or for severe damage to the coating.	Wipe the pipettor. If it is damaged, contact technical service staff for replacement.
	Chamber or translucent window is dirty	Check for liquid or crystals.	Contact technical service staff

Common faults and troubleshooting

	Abnormal injection of starter by starter pump	Check if the amount of starter is insufficient, check if the starter pump makes an abnormal sound, check if the starter pipe is retracted or bent, and check for crystallization at the pipe outlet.	Contact technical service staff
	Contamination of cuvette	Check whether the cuvettes have not been used for a long time and check for dust or other impurities.	Replace the cuvettes and clean the cuvette storage
	PMT failure	Check whether the dark noise is normal.	Contact technical service staff
	The analyzer has been running for a long time and the pipes have not been properly maintained.	Check if the pipes are dirty or aged.	Use system tubing cleaning solution to run the Pipe Wash function once.

Problem symptoms	Possible cause	To check	Solution
High CV value in test results	Either the system liquid has been stored for too long, or a problem with the water quality system or configuration error has caused the system liquid to fail.	Observe whether the system liquid is too viscous, whether there is too much yellow dirt in the system liquid tank, or whether the conductivity of the pure water used for preparing the system liquid is less than 1 $\mu\text{S}/\text{cm}$.	Clean the system liquid tank following the monthly maintenance method, use pure water with conductivity of less than 1 $\mu\text{S}/\text{cm}$, and reprepare the system liquid according to the Operating Instructions.
	The pipe of the SampleArm has leakage	Check whether there are any bubbles or water droplets on the probe tip after the pipettor is cleaned, whether there are air bubbles in the SampleArm pipe, and whether there is crystallization due to leakage at any pipe joints (solenoid valve, plunger pump, etc.)	Check fixing rings at valve interfaces for leaks, check whether the system liquid has been used up, check whether the SampleArm is at the right position for washing, and then select the SampleArm to prime in the Priming interface. Contact technical service staff if the leakage continues.

	SampleArm plunger pump, wash pump or valve failure	Check the alarm log for failure to initialize plunger pump, check whether the inner membrane of the washer pump is damaged or liquid spurts out.	Contact technical service staff
	Liquid level detection error, sample or reagent not added	Check a large number of recent test results for extremely high or low RLU values.	Contact technical service staff
	The pipettor is dirty or damaged	Check the outer wall of the probe for crystallization, or for severe damage to the coating.	Wipe the pipettor. If it is damaged, contact technical service staff for replacement.
	An abnormal amount of system liquid injected in the washer	Check whether the amount of system liquid injected by the three washing needles in the washer is consistent, whether the injection pipe is kinked or leaking, and whether the solenoid valve fixing screws are loose.	Contact technical service staff
	Malfunction of the washer waste liquid peristaltic pump, or the washer waste liquid needle is blocked.	Check if the flexible pipe of the waste liquid peristaltic pump of the washer is due for replacement and check if the liquid in the waste liquid pipe is flowing too slowly.	Check the test counter to see if it is time for replacement, contact the technical service personnel when you need to replace or check the calibration.
	Dirty or damaged washer probe or improper cleaning position	Check the outer wall of the probe for crystallization or other deposits and whether the outer wall coating has peeled off or rusted. Check if the injection position of the washer lifting mechanism is too low and the suction level is too high.	Wipe the washing needle or contact the technical service staff.
	Chamber or translucent window is dirty	Check for liquid or crystals.	Contact technical service staff

Common faults and troubleshooting

	Abnormal injection of starter by starter pump	Check if the amount of starter is insufficient, check if the starter pump makes an abnormal sound, check if the starter pipe is retracted or bent, and check for crystallization at the pipe outlet.	Contact technical service staff
	Contamination of cuvette	Check whether the cuvettes have not been used for a long time and check for dust or other impurities.	Replace the cuvettes and clean the cuvette storage
	PMT failure	Check whether the dark noise is normal.	Contact technical service staff
	The analyzer has been running for a long time and the pipes have not been properly maintained.	Check if the pipes are dirty or aged.	Use system tubing cleaning solution to run the Pipe Wash function once.
	There are air bubbles or particulate impurities in the sample or reagent	Visual inspection	Remove the air bubbles.
	Serum insufficiently centrifuged	Visual inspection	Standardize the time and speed for serum centrifugation.
	The sample test tube is not inserted into the bottom of the sample rack or is crooked	Visual inspection	Correctly place the sample test tube in the sample rack.

Problem symptoms	Possible cause	To check	Solution
Cross-contamination	Either the system liquid has been stored for too long, or a problem with the water quality system or configuration error has caused the system liquid to fail.	Observe whether the system liquid is too viscous, whether there is too much yellow dirt in the system liquid tank, or whether the conductivity of the pure water used for preparing the system liquid is less than 1 $\mu\text{S}/\text{cm}$.	Clean the system liquid tank following the monthly maintenance method, use pure water with conductivity of less than 1 $\mu\text{S}/\text{cm}$, and reprepare the system liquid according to the Operating Instructions.

	An abnormal amount of system liquid injected in the washer	Check whether the amount of system liquid injected by the three washing needles in the washer is consistent, whether the injection pipe is kinked or leaking, and whether the solenoid valve fixing screws are loose.	Contact technical service staff
	Malfunction of the washer waste liquid peristaltic pump, or the washer waste liquid needle is blocked.	Check if the flexible pipe of the waste liquid peristaltic pump of the washer is due for replacement and check if the liquid in the waste liquid pipe is flowing too slowly.	Check the test counter to see if it is time for replacement, contact the technical service personnel when you need to replace or check the calibration.
	Dirty or damaged washer probe or improper cleaning position	Check the outer wall of the probe for crystallization or other deposits and whether the outer wall coating has peeled off or rusted. Check if the injection position of the washer lifting mechanism is too low and the suction level is too high.	Wipe the washing needle or contact the technical service staff.
	The pipettor is dirty or damaged	Check the outer wall of the probe for crystallization, or for severe damage to the coating.	Wipe the pipettor. If it is damaged, contact technical service staff for replacement.
	SampleArm plunger pump, wash pump or valve failure	Check the alarm log for failure to initialize plunger pump, check whether the inner membrane of the washer pump is damaged or liquid spurts out.	Contact technical service staff

Problem symptoms	Possible cause	To check	Solution
Too low RLU value in test results(even 0)	Liquid level detection error, sample or reagent not added	Check a large number of recent test results for extremely high or low RLU values.	Contact technical service staff

	Sample size too small	Visual inspection	Supplement samples. If the quantity of samples is sufficient, contact the technical service staff.
	Abnormal injection of starter by starter pump	Check if the amount of starter is insufficient, check if the starter pump makes an abnormal sound, check if the starter pipe is retracted or bent, and check for crystallization at the pipe outlet.	Contact technical service staff
	PMT failure	Check whether the dark noise is normal.	Contact technical service staff
	Regent not correctly calibrated or unverified calibration	Check reagent calibration for rotation or other conditions.	Recalibrate or click to confirm.
	Too high concentration of sample	Check whether the assay uses the competition method.	Dilution
Too high RLU value in test results	Sample concentration exceeds the measuring range	None	Dilution
	Either the system liquid has been stored for too long, or a problem with the water quality system or configuration error has caused the system liquid to fail.	Observe whether the system liquid is too viscous, whether there is too much yellow dirt in the system liquid tank, or whether the conductivity of the pure water used for preparing the system liquid is less than 1 $\mu\text{s}/\text{cm}$.	Clean the system liquid tank following the monthly maintenance method, use pure water with conductivity of less than 1 $\mu\text{s}/\text{cm}$, and reprepare the system liquid according to the Operating Instructions.
	An abnormal amount of system liquid injected in the washer	Check whether the amount of system liquid injected by the three washing needles in the washer is consistent, whether the injection pipe is kinked or leaking, and whether the solenoid valve fixing screws are loose.	Contact technical service staff

	Malfunction of the washer waste liquid peristaltic pump, or the washer waste liquid needle is blocked.	Check if the flexible pipe of the waste liquid peristaltic pump of the washer is due for replacement and check if the liquid in the waste liquid pipe is flowing too slowly.	Check the test counter to see if it is time for replacement, contact the technical service personnel when you need to replace or check the calibration.
	Dirty or damaged washer probe or improper cleaning position	Check the outer wall of the probe for crystallization or other deposits and whether the outer wall coating has peeled off or rusted. Check if the injection position of the washer lifting mechanism is too low and the suction level is too high.	Wipe the washing needle or contact the technical service staff.
	Contamination of cuvette	Check whether the cuvettes have not been used for a long time and check for dust or other impurities.	Replace the cuvettes and clean the cuvette storage
	Light leakage of chamber	Observe whether the RLU values in the same batch of test results are abnormally high.	Contact technical service staff
	PMT failure	Check whether the dark noise is normal.	Contact technical service staff
	Serum insufficiently centrifuged	Visual inspection	Standardize the time and speed for serum centrifugation.

7.3 Alarm-type faults and solutions

When a fault serious enough to raise an alarm occurs, please carry out a simple inspection of the analyzer and follow the prompts in the dialog boxes brought up by the software. Contact the technical service personnel or an authorized dealer if recovery is not possible or other faults occur

Appendix A Performance indexes

1.1 Accuracy and repeatability of sample pipetting

Pipette 10 μ L, 200 μ L and 450 μ L samples for tests, respectively (using the same pipettor for sample and reagent transfers), and the results should meet the requirements in Table 1.

Table 1 Requirements on pipetting accuracy and repeatability

Nominal pipetted volume (v)/ μ L	Requirements	
	Offset	Coefficient of variation (CV)/%
10	Within $\pm 1\mu$ L	≤ 5
200	Within $\pm 5\%$	≤ 2
450	Within $\pm 5\%$	≤ 2

1.2 Temperature control accuracy and fluctuation of the reaction area

The temperature offset of the reaction area should be within the 37 $^{\circ}\text{C} \pm 0.3$ range, with the fluctuation not greater than 0.2 degrees centigrade.

1.3 LC test components

1.3.1 Analyzer noise

After the analyzer is powered on and enters the stable operating status, the noise per calculation should be within the 100—1,200 range, with the CV $\leq 10\%$.

1.3.2 RLU linearity

In the luminescence reagent method, when the order of magnitude of the RLU is not less than 3, the linear correlation coefficient (r) should be equal to or greater than 0.99.

1.3.3 RLU repeatability

In the luminescence reagent method, the coefficient of variation (CV, %) should not exceed 5%.

1.3.4 RLU stability

In the luminescence reagent method, the RLU variation should not exceed $\pm 10\%$.

1.4 Carryover

Carryover rate should be $\leq 10^{-5}$.

1.5 The intra-lot precision of clinical assays

The coefficient of variation (CV, %) should not exceed 5%.

Appendix B Network interfaces

1.1 Protocol description

The software uses the ASTM E1394 and HL7 protocols to implement communication.

All the communication information of the software with the LIS system is stored at logs\data\online.

1.2 ASTM protocol

1.2.1 Control code description

Table 1.2-1 Control code description

Character	Description	Corresponding ASCII
<ENQ>	Inquiry	0x05
<ACK>	Confirm response	0x06
<STX>	Beginning of document	0x02
<ETX>	End of document	0x03
<CR>	Carriage return	0x0D
<EOT>	End of transmission	0x04

1.2.2 Basic communication format description

<ENQ><STX>content<ETX><EOT>

For example:

<ENQ><STX>TEST<ETX><EOT>

Table 1.2-2 Basic communication format specification

Character	Description	Corresponding ASCII
<ENQ>	Inquiry	0x05
<STX>	Beginning of document	0x02
T	Letter T	0x54
E	Letter E	0x45
S	Letter S	0x53
T	Letter T	0x54
<ETX>	End of document	0x03
<EOT>	End of transmission	0x04

1.2.3 Separator description

The ASTM E1394 protocol defines four separators to distinguish the communication content. Refer to Section A.4 of E1394 for details. The following table shows the implication of each separator:

Table 1.2-3 Separator specification

Separator	Description	Corresponding ASCII
<CR>	Record separator	0x0D
	Field separator	0x7C
\	Repeating separator	0x5C
^	Component separator	0x5E
&	Special separator	0x26

1.2.4 Message type description

Table 1.2-4 Message type description

Message label	Description
H	Message header record
P	Patient information record
O	Test assay record
R	Result record
Q	Inquiry information record
L	Message termination record

1.2.4.1 Message header record (H)

Note:

The content in this section corresponds to Section 7 of ASTM E1394.

Message header records should be put in the front of all transmission records to describe some basic information of the protocol and transmission.

Example:

H|\^&||PSWD|MAGLUMI X3||||LIS||P|E1394-97|20180717<CR>

Table 1.2-5 Message header record

Field	E1394	ASTM field name	Content	Maximum length	Whether required
1	7.1.1	Message type	H	1	Yes
2	7.1.2	Separator definition	\^&	4	Yes
4	7.1.4	Password	PSWD	20	No
5	7.1.5	Transmitter name	MAGLUMI X3	20	Yes
10	7.1.10	Receiver name	LIS	20	Yes

12	7.1.14	Processing method	P	1	Yes
13	7.1.13	Protocol version number	E1394-97	10	Yes
14	7.1.14	Date	YYYYMMD D	14	Yes

Note:



- 1) The protocol includes a total of 14 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software. The "|" separator is also needed to separate different fields if values are not required for some of these fields.
- 2) The <CR> separator at the end of a message record is required. This separator must be added at the end of each message record to indicate the end of each message record.

1.2.4.2 Patient information record (P)

Note:

The content in this section corresponds to Section 8 of ASTM E1394.

Every patient's related information should be described in this record.

Example:

P|1|||ABC|||F<CR>

Table 1.2-6 Patient information record

Field	E1394	ASTM field name	Content	Maximum length	Whether Required
1	8.1.1	Message type	P	1	Yes
2	8.1.2	Serial number	1	6	Yes
6	8.1.6	Patient name	ABC	30	No
9	8.1.9	Sex	M, F, U	1	No

Note:



- 1) Since only the first and second fields are required, patient information records in the software are generally written in the P|1 <CR> format.
- 2) The protocol includes a total of 35 fields. Not all the fields required by

this software are listed in the table. Other fields can be added during the actual communication process and are automatically recognized by the software.

- 3) The <CR> separator at the end of a message record is required. This separator must be added at the end of each message record to indicate the end of each message record.

1.2.4.3 Test assay record (O)

Note:

The content in this section corresponds to Section 9 of ASTM E1394. Information for each test assay should be described in this record.

Example:

O|1|1234567||^ALT^1:50|R<CR>

Table 1.2-7 Test assay record

Field	E1394	ASTM field name	Content	Maximum length	Whether required
1	9.1.1	Message type	O	1	Yes
2	9.1.2	Serial number	1	6	Yes
3	9.1.3	Sample ID	1234567	22	Yes
5	5.4	9.1.5	Test assay	30	Yes
	5.6		Dilution ratio	30	No
6	9.1.6	Priority	S, R	1	Yes

Note:

- 1) In the sixth field "Priority", S refers to STAT and R refers to routine. R is generally used.
- 2) The protocol includes a total of 31 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.
- 3) The <CR> separator at the end of a message record is required. This separator must be added at the end of each message record to indicate the end of each message record.



1.2.4.4 Result record (R)

Note:

The content in this section corresponds to Section 10 of ASTM E1394.

All information about each test result should be described in this record.

Example:

R|1|^^^ALT^^1:50|115.3|pg/mL|0 to 200|N|||||20180326172956<CR>

Table 1.2-8 Result record

Field		E1394	ASTM field name	Content	Maximum length	Whether required
1		10.1.1	Message type	R	1	Yes
2		10.1.2	Serial number	1	5	Yes
3	3.4	10.1.3	Test item	ALT	10	Yes
	3.6	10.1.3	Dilution ratio	1:50	10	No
4		10.1.4	Result	115.3	12	No
5		10.1.5	Unit	pg/mL	10	No
6		10.1.6	Reference range	0 to 200	30	No
7		10.1.7	Result flag	L,H,N,< ,>	1	No
13		10.1.13	Test completion time	YYYYMMDD DHHMMSS	14	No

Note:

- 1) In the seventh field "Result Flag", L refers to lower than normal range, H refers to higher than normal range, and N refers to normal. < refers to lower than the measuring range, and > refers to higher than the measuring range.
- 2) The protocol includes a total of 14 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.
- 3) The <CR> separator at the end of a message record is required. This separator must be added at the end of each message record to indicate the end of each message record.

1.2.4.5 Request information record (Q)

Note:

The content in this section corresponds to Section 12 of ASTM E1394.
It is used to request assay information corresponding to samples from LIS.

Example:

Q|1|^1234567||ALL|||||||O<CR>

Table 1.2-9 Inquiry information record

Field	E1394	ASTM field name	Content	Maximum length	Whether required
1	12.1.1	Message type	Q	1	Yes
2	12.1.2	Serial number	1	6	Yes
3	12.1.3	Sample ID	^1234567	22	Yes
5	12.1.5	All test assays	All	10	Yes
13	10.1.13	Request information status	O	1	Yes

Note:

- 1) The protocol includes a total of 13 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.
 - 2) The <CR> separator at the end of a message record is required. This separator must be added at the end of each message record to indicate the end of each message record.
-

1.2.4.6 Message termination record (L)

Note:

The content in this section corresponds to Section 13 of ASTM E1394.
It is used in the last transmission record, indicating that transmission is completed.

Example:

L|1|N<CR>

Table 1.2-10 Message termination record

Field	E1394	ASTM field name	Content	Maximum length	Whether required
1	13.1.1	Message type	L	1	Yes
2	13.1.2	Serial number	1	6	Yes
3	13.1.3	End code	N	1	Yes



Note:

The <CR> separator at the end of a message record is required. This separator must be added at the end of each message record to indicate the end of each message record.

1.2.5 Example description

1.2.5.1 Query test assay

When you insert a sample rack into the sample area and the analyzer scans the sample barcode, the software will request the test assay information corresponding to the sample from the LIS system by using the following message.

Message content:

```
--><ENQ>
<--<ACK>
--><STX>
<--<ACK>
-->H|^&||PSWD| MAGLUMI X3 ||||Lis||P|E1394-97|20100323<CR>
-->Q1|^1234567||ALL||||||O<CR>
-->L|1|N<CR>
<--<ACK>
--><ETX>
<--<ACK>
--><EOT>
<--<ACK>
```

Table 1.2-11 Character implication

Character	Description	Corresponding ASCII
-->	Software sending	
<--	Software receiving	

Where:

```
H|\^&||PSWD|MAGLUMI X3 ||||Lis||P|E1394-97|20100323<CR>
Q|1|^1234567||ALL||||||O<CR>
L|1|N<CR>
```

The above is a message sent by this software via the ASTM E1394 communication protocol to the LIS system to request the corresponding test assay of the sample. The above text requests the corresponding test assay of the 1234567 sample. It is a counterpart of Appendix B Section 1.2.5.2 which describes how to return the test assay information. For analysis of the corresponding fields, refer to Appendix B Section 1.2.4.



Note:

- 1) The <ACK> in the above transmitted content is sent by the LIS system, which requires sending the corresponding <ACK> command in the written position. Otherwise, the software will regard that the LIS system as disconnected.

1.2.5.2 Returning test assay

After the LIS system receives the test assay request from the software, it returns the test assay information.

Message content:

```
<--<ENQ>
--><ACK>
<--<STX>
--><ACK>
<--H|\^&||PSWD| MAGLUMI X3 ||||Lis||P|E1394-97|20180319<CR>
<--P|1<CR>
<--O|1|1234567||^FT3\^FT4|R<CR>
<--L|1|N<CR>
<--<ETX>
--><ACK>
<--<EOT>
--><ACK>
```

Table 1.2-11 Character meaning

Character	Description	Corresponding ASCII
-->	Software sending	
<--	Software receiving	

Where:

```
<--H|\^&||PSWD| MAGLUMI X3 ||||Lis||P|E1394-97|20180319<CR>
<--P|1<CR>
<--O|1|1234567||^FT3\^FT4|R<CR>
<--L|1|N<CR>
```

The above is a message sent by the LIS system via the ASTM E1394 communication protocol to return the corresponding test assay of the sample. The above text returns the corresponding test assays FT3 and FT4 of the 1234567 sample. It is a counterpart of Appendix B Section 1.2.5.1 which describes how to request the test assay information. For analysis of the corresponding fields, refer to Appendix B Section 1.2.4.



Note:

The <ACK> in the above transmitted content is sent by the analyzer, which requires sending the corresponding <ACK> command in the written position. Otherwise, the LIS system will regard that the software as disconnected.

1.2.5.3 Send result

Message content:

```
--><ENQ>
<--<ACK>
--><STX>
<--<ACK>
-->H|\^&||PSWD| MAGLUMI X3 ||||Lis||P|E1394-97|20180326<CR>
-->P|1<CR>
-->O|1|1234567||^FT3 \^FT4<CR>
-->R|1|^FT3|123|pg/mL|0-200|N||||20180326172956<CR>
-->R|2|^FT4|25.1|pg/mL|0-50|N||||20180326172956<CR>
-->L|1|N<CR>
--><ETX>
<--<ACK>
--><EOT>
<--<ACK>
```

Table 1.2-11 Character meaning

Character	Description	Corresponding ASCII
-->	Software sending	
<--	Software receiving	

Where:

```
H|\^&|PSWD| MAGLUMI X3 ||||Lis|P|E1394-97|20180326<CR>
P|1<CR>
O|1|1234567||^FT3 \^FT4<CR>
R|1|^ FT3|123|pg/mL|0-200|N|||||20180326172956<CR>
R|2|^ FT4|25.1|pg/mL|0-50|N|||||20180326172956<CR>
```

The above is a message sent by this software via the ASTM E1394 communication protocol to send the test result to the LIS system. The above text is a message sent by the analyzer to the LIS system to transmit the FT3 and FT4 test results of the reagent corresponding to the 1234567 sample. For analysis of the corresponding fields, refer to Appendix B Section 1.2.4

1.3 HL7 protocol

1.3.1 Basic communication format description

Format

<SB>CONTENT<EB><CR>

Example

<SB>TEXT <EB><CR>

Table 1.3-1 Basic communication format description

Character	Description	Corresponding ASCII
<SB>	Start character	0x0B
T	Letter T	0x54
E	Letter E	0x45
S	Letter S	0x53
T	Letter T	0x54
<EB>	End character	0x1C
<CR>	Carriage return character	0x0D

1.3.2 Separator description

Table 1.3-2 Separator description

Separators	Description	Corresponding ASCII
	Field delimiter	0x7C
~	Repeating separator	0x7E
^	Component delimiter	0x5E
&	Sub-component delimiter	0x26
\	Escape delimiter	0x5C

1.3.3 Message type description

Table 1.3-3 Message type description

Message label	Description
OML^O33	Used to return test assays
TSREQ	Used to inquire about test assays
OUL^R22	Used to send test results
RSREQ	Used to inquire about test results

1.3.3.1 Return test assay (OML^O33)

This message is used when the LIS system takes the initiative to send orders to the software, or when the LIS system responds to the order queries from the software.

Table 1.3-4 OML^O33 message structure

Message segment	Meaning	Remark
MSH	Message header segment	
[PID]	Patient identification segment	Optional
SPM	Specimen segment	
ORC	Order common segment	
{OBR}	Order test segment	Repeatable item

**Note:**

1. The protocol includes a total of 7 message segments. The list only includes the message segments required by this software. Other message segments can be added during the actual communication process and automatically recognized by the software.

1.3.3.2 Query test assay (TSREQ)

Note:

This message is used when the software queries the LIS system about the test assay.

Table 1.3-5 TSREQ message structure

Message segment	Meaning	Remark
MSH	Message header segment	
QPD	Query parameter definition	
RCP	Response control parameter	

1.3.3.3 Transmission result (OUL^R22)

Note:

This message is used when the software takes the initiative to send test results to the LIS system, or when the software returns query results of the LIS system.

Table 1.3-6 OUL^R22 message structure

Message segment	Meaning	Remark
MSH	Message header segment	
SPM	Specimen segment	
ORC	Order common segment	
{OBR	Assay order segment	Repeatable item
OBX	Observation result segment	Repeatable item
[NTE]}	Comment segment	Optional, repeatable

**Note:**

1. The protocol includes a total of 11 message segments. The list only includes the message segments required by this software. Other message segments can be added during the actual communication process and automatically recognized by the software.

1.3.3.4 Query result (RSREQ)

Note:

This message is used when the LIS system queries the test result from the software.

Table 1.3-7 RSREQ message

Message segment	Meaning	Remark
MSH	Message header segment	
QPD	Query parameter definition	
RCP	Response control parameter	

1.3.4 Message segment description

Note:

The protocol includes a total of 13 message segments. The list only includes the message segments required by this software. Other message segments can be added during the actual communication process and automatically recognized by the software.

1.3.4.1 Header segment (MSH)

Format:

MSH|^~\&|Sender||Receiver||Message Time||Message Type|Verification Code|P|2.5|||NE|NE||UTF-8

Example:

MSH|^~&|MAGLUMI||LIS||20151025105016||TSREQ|013329845|P|2.5||NE|NE||UTF-8

Table 1.3-8 MSH message segment description

Position	Field	Content	Maximum length	Required	Remark
1	Field separator		1	Yes	Fixed
2	Encoding characters	^~\&	4	Yes	Fixed
3	Sending app	MAGLUMI	180	Yes	
5	Receiving app	LIS	180	Yes	
7	Message time	20151025105016	14	Yes	yyyyMMddHHmmSS
9	Message type	TSREQ	7	Yes	OML^O33, OUL^R22, RSREQ or TSREQ
10	Message control ID/verification code	013329845	20	Yes	Unique identifier
11	Processing ID	P	3	Yes	Fixed
12	Version ID	2.5	8	Yes	Fixed
15	Message acknowledgment	NE	2	Yes	Fixed
16	App acknowledgment	NE	2	Yes	Fixed
18	Character set	UTF-8	16	Yes	Fixed



Note:

The protocol includes a total of 18 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.2 Patient identification segment (PID)

Format:

PID|1||Patient ID||Patient Name||yyyyMMdd(Date of Birth)|Gender (Enumeration)|||Age^Age Unit

Example:

PID|1||10248734||Zhang Jianguo||20110101|1|||8^1

Table 1.3-9 PID message segment description

Position	Field	Content	Maximum length	Required	Remark
1	No.	1	1	Yes	Fixed
3	Patient ID	10248734	20		
5	Patient name	Zhang Jianguo	50	Yes	
7	Date/time of birth	20110101	8		yyyyMMdd
8	Sex	1	1	Yes	1:Male,2:Female,3:Unknown
38	38.1	Age	8	Yes	Positive integer
	38.2	Age unit	1	Yes	1:Year,2:Month,3:Day,4:Hour



Note:

The protocol includes a total of 38 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.3 Specimen segment (SPM)

Format:

SPM|No.|Sample ID ^ ^QC Material ID^Sample Rack ID^Sample Rack Position||Sample Type|||||Sample Role|||||

Example:

(sample)SPM|1|034581229010^^A001^2||0|||||P|||||

(QC)SPM|1|^#2020030301#^C01^4||0|||||Q|||||

Table 1.3-10 SPM message segment description

Position	Field	Content	Maximum length	Required	Remark
1	No.	1	1	Yes	Fixed
2	2.1	Sample ID	034581229010	Yes	
	2.3	QC material ID	#2020030301#	Yes	

	2.4	Sample rack ID	A001/C01	22		
	2.5	Sample rack positions	2/4	5		
4		Specimen type	1/6	1	Yes	0:Serum 1:Plasma 2:Urine 3:Hydrothorax and ascite 4:Cerebrospinal fluid 5:Other
11		Specimen role	P/Q	1	Yes	P:Patient sample Q:QC material

**Note:**

The protocol includes a total of 18 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.4 Order common segment (ORC)

Format:

ORC|Order Control|Priority|Order Time||

Example:

ORC|NW|0|201909131511433||

Table 1.3-11 ORC message segment description

Position	Field	Content	Maximum length	Required	Remark
1	Order control	NW	2	Yes	NW:Newly generated advice of the doctor RF:Second request for the order
7	Priority	0	1	Yes	0: General 1: Quick check
9	Order time	201909131511433	14	Yes	yyMMddHHmmSS

**Note:**

The protocol includes a total of 12 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.5 Order test segment (OBR)

Format:

OBR|No.|||Assay Abbr.^Dilution Ratio

Example: OBR|1|||FT3^1:10

Table 1.3-12 OBR message segment description

Position		Field	Content	Maximum length	Required	Remark
1		No.	1	4	Yes	Start from 1
4	4.1	Assay abbr.	FT3	10	Yes	Channel no. of reagent
	4.2	Dilution ratio	1:10	10		



Note:

- 1、 The protocol includes a total of 4 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.
- 2、 Set whether to send the dilution ratio in the user software.

1.3.4.6 Result segment (OBX)

Format:

OBX|No.||Assay Abbr.||Sample Result|Unit|Reference Range|Abnormal Flag|||F||Test Start Time||
||Module Name^Equipment Name|Test End Time|Recheck Flag|Messaging Type

Example:

OBX|1||FT4||20.9|mmol/L|8.9-17.2|H|||F||20191025104016|||Module1^Maglumi|20191025104816|0|0

Table 1.3-13 OBX message segment description

Position	Field	Content	Maximum length	Required	Remark
1	No.	1	4	Yes	Start from 1
3	Assay abbr.	FT4	10	Yes	Reagent name abbreviation
5.1	Quantitative result	20.9	10	Yes	
6	Unit	2	10		Reagent assay unit

7	Reference range	8.9-17.2	100		Reference value range of reagent assay results	
8	Abnormal flags	H	6		H: Higher than normal HH: Higher than measuring range L: Lower than normal LL: Lower than measuring range N: No flag	
11	Result status	F	1	Yes	Fixed, indicating the final result	
14	Time of test start	20191025104016	14	Yes	yyyyMMddHHmmSS	
18	18.1	Module name	Module1	20	Yes	
	18.2	Equipment name	Maglumi	20	Yes	
19	Time of test end	20191025104816	14	Yes	yyyyMMddHHmmSS	
20	Recheck flag	0	1	Yes	0: Regular result 1: Recheck result	
21	Messaging type	0	1	Yes	0: Automatic messaging 1: Manual messaging	



Note:

The protocol includes a total of 21 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.7 Comment segment (NTE)

Format:

NTE|No.

Example:

NTE|1

Table 1.3-14 NTE message segment description

Position	Field	Content	Maximum length	Required	Remark
1	No.	1	4	Yes	Start from 1



Note:

The protocol includes a total of 3 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.8 Query parameter segment (QPD)

Format:

QPD|Query Name||Sample ID^^Sample Rack ID^Sample Rack Position|Priority

Example:

QPD|RSREQ||123456^^N0001^2|0

Table 1.3-15 QPD message segment description

Position	Field	Content	Maximum length	Required	Remark
1	Message query name	RSREQ	7	Yes	RSREQ or TSREQ
3	3.1	Sample ID	123456	22	Yes
	3.3	Sample rack ID	N0001	5	
	3.4	Sample rack positions	2	22	
4	Priority	0		Yes	0: General 1: STAT



Note:

The protocol includes a total of 4 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.4.9 Response control segment (RCP)

Format:

RCP|Query Priority||Response Form

Example:

RCP|I||R|

Table 1.3-16 RCP message segment description

Position	Field	Content	Maximum length	Required	Remark
----------	-------	---------	----------------	----------	--------

1	Query priority	I	4	Yes	Fixed, indicating immediate queries
3	Response modality	R	10	Yes	Fixed, indicating real-time responses



Note:

The protocol includes a total of 3 fields. Not all fields this software requires are listed in the table. The rest of the fields can be added during the actual communication process and can be automatically recognized by the software.

1.3.5 Example description

1.3.5.1 Return test assay

Actual message:

```
<-- MSH|^~\&|Lis||Maglumi X3||20200311140350||OML^O33|2|P|2.5|||NE|NE||UTF-8
<-- PID|1||34||zhangsan||19960101|Male|||||||||||||||||^Years Old
<-- SPM|1|1234567^^|P
<-- ORC|NW||||N
<-- OBR|1||FT3^
<-- OBR|2||FT4^
```

Messaging direction description:

Character	Description
<--	Messages received by this software

The above is a message sent by the software via the HL7 communication protocol to return the test assay information. The above text returns the corresponding test assays FT3 and FT4 of the 1234567 sample. It is a counterpart of Appendix B Section 1.3.5.2 which describes test assay queries. For analysis of the corresponding fields, refer to Appendix B Section 1.3.4.

1.3.5.2 Query test assay

Actual message:

```
--> MSH|^~\&|Maglumi X3||Lis||20200311140350||TSREQ|2|P|2.5|||NE|NE||UTF-8
--> QPD|TSREQ||1234567^^A006^2|0
--> RCP|I||R|
```

Messaging direction description:

Character	Description
-->	Messages sent by this software

The above is a message sent by this software via the HL7 communication protocol to the LIS system to request the corresponding test assay of the sample. The above text requests the corresponding test assay of the 1234567 sample. For analysis of the corresponding fields, refer to Appendix B Section 1.3.4.



Note:

After this software sends this content, the LIS system will return the test assay information. See the example in Appendix B Section 1.3.5.1.

1.3.5.3 Messaging result

Actual message:

```
--> MSH|^~\&|Maglumi X3||Lis||20200311140933||OUL^R22|4|P|2.5||NE|NE||UTF-8
--> SPM|1|1234567^^^A006^2||0|||||P|||||
--> ORC|NW|||||0||20200311140350||
--> OBR|1||FT4
--> OBX|1||FT4||9.2|pg/mL|8.90 - 45.00|HH||F||20200311140702|||Module1^Maglumi|20200311140933|0|0
--> NTE|1
```

Messaging direction description:

Character	Description
-->	Messages sent by this software

The above is a message sent by this software via the HL7 communication protocol to send the test result to the LIS system. The above text is a message sent by the analyzer to the LIS system to transmit the FT4 test result of the reagent corresponding to the 1234567 sample. For analysis of the corresponding fields, refer to Appendix B Section 1.3.4.

1.3.5.4 Query result

Actual message:

```
<-- MSH|^~\&|Lis||Maglumi X3||20200311142249||RSREQ|5|P|2.5|||NE|NE||UTF-8  
<-- QPD|RSREQ||1234567^|N  
<-- RCP|I||R|
```

Messaging direction description:

Character	Description
<--	Messages received by this software

The above is a message sent by the LIS system via the HL7 communication protocol to the software to request the test result. The above text requests the assay test result corresponding to the 1234567 sample. For analysis of the corresponding fields, refer to Appendix B Section 1.3.4.

**Note:**

After this software receives this content, it will return the assay test result. See the example in Appendix B Section 1.3.5.3.